Saviours AI

ICSE CLASS 10 · 2027 SYLLABUS

SIDE KICK

Mathematics

Twenty-one chapters written fresh for the 2027 syllabus: formulas, worked examples, HOT questions and practice sets.



EVERY TOPPER HAS A SIDEKICK

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Hey. I'm your Sidekick.

I live inside this book. Wherever things get tricky, I pop up with a tip, a trap to dodge, or a question the board loves to ask. Read the chapter, then try my HOT questions. Deal?



Aligned to the official ICSE 2027 syllabus. If it's in the syllabus, it's in here.

SIDEKICK TIP
Shortcuts, mnemonics and exam hacks.

FUN FACT
Things that make the concept stick.

REAL LIFE
Where this shows up outside the exam hall.

KEY POINT
Definitions and laws you must write exactly.

HOT QUESTION
Higher-order thinking. Try before you peek.

✦ NEW
Added in this edition to match the 2027 syllabus.

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01

CHAPTER · MATHEMATICS

GST

Every bill you see in a shop carries a tax line. This chapter shows you how that line is worked out, how to run it backwards and how a dealer settles with the government.



IN THIS CHAPTER

- Discount and taxable value
- CGST and SGST
- Inverse GST
- Tax payable
- Comparing offers

What GST Is and Why You Pay It

Goods and Services Tax (GST) is a single tax charged on the sale of goods and services in India. Earlier, a product could carry several different taxes at different stages. GST replaced most of them with one tax, collected at every step of the supply chain but finally borne by the **consumer**.

- **CGST** (Central GST) goes to the Central Government and **SGST** (State GST) goes to the State Government. They are charged when a sale happens **inside one State**.
- Both are charged at the **same rate**, each being exactly half of the GST rate. A 12% GST means 6% CGST + 6% SGST.
- The rate depends on the item. Typical slabs seen in questions are 5%, 12%, 18% and 28%, and the question always **tells you the rate**.

KEY POINT

GST is always calculated on the **selling price after discount**, never on the marked price. The bill amount is the discounted price **plus** the tax.

The Vocabulary

TERM	MEANING
List price / Marked price (MP)	The price printed or tagged on the article, before any discount and before GST
Discount	Reduction on the marked price. Discount = MP × discount% / 100
Selling price (SP) / Taxable value	MP − Discount. This is the amount on which GST is charged
Cost price (CP)	What the shopkeeper paid for the article (also called basic price)
Input tax	GST the shopkeeper paid while buying goods
Output tax	GST the shopkeeper collected from customers while selling
Tax payable to government	Output tax − Input tax

Formulae

$$\text{Discount} = \text{MP} \times \frac{d}{100} \qquad \text{SP} = \text{MP} - \text{Discount}$$

$$\text{GST} = \text{SP} \times \frac{r}{100} \qquad \text{Bill amount} = \text{SP} + \text{GST} = \text{SP} \times \frac{100 + r}{100}$$

$$\text{CGST} = \text{SGST} = \frac{\text{GST}}{2} \qquad \text{Inverse: SP} = \text{Bill} \times \frac{100}{100 + r}$$

SIDEKICK TIP

Treat GST like a percentage increase. A bill with 18% GST is just **1.18 × SP**. To go backwards, divide by 1.18. This single idea solves every inverse question.

Tax Payable by a Dealer

A dealer buys goods, adds his profit and sells them. He pays GST on his purchases and collects GST on his sales. Government wants only the tax on the **value he added**, so he deposits the difference.

1. Output tax
GST charged on the price at which he sells.

2. Input tax
GST he paid on the price at which he bought.

3. Deposit
Tax payable = output tax − input tax. This equals GST on his profit margin.

4. Split
Half of it is CGST and half is SGST when sold within the State.

Worked Example 1

Q. A school bag has a marked price of ₹ 2,400. The shop allows a discount of 25% and GST is charged at 12% on the discounted price. Find the amount on the bill.

- Discount = 25% of 2400 = ₹ 600, so SP = 2400 − 600 = ₹ 1,800.
- GST = 12% of 1800 = ₹ 216.
- Bill = 1800 + 216 = ₹ 2,016.

KEY POINT
Answer: ₹ 2,016 (CGST ₹ 108 and SGST ₹ 108).

Worked Example 2

Q. The bill for a sofa, including 18% GST, is ₹ 35,400. Find the price before tax and the CGST.

- Bill = 1.18 × SP, so SP = 35400 ÷ 1.18 = ₹ 30,000.
- GST = 35400 − 30000 = ₹ 5,400.
- CGST = half of 5400 = ₹ 2,700.

KEY POINT
Answer: Price before tax ₹ 30,000; CGST ₹ 2,700.

Worked Example 3

Q. A shopkeeper buys a table for ₹ 5,000 and sells it for ₹ 6,000. GST on both transactions is 12%. How much GST must he deposit, and how much of it is SGST?

- Output tax = 12% of 6000 = ₹ 720.
- Input tax = 12% of 5000 = ₹ 600.
- Payable = 720 − 600 = ₹ 120. (Check: 12% of his margin 1000 = 120.)
- SGST = 120 ÷ 2 = ₹ 60.

KEY POINT
Answer: He deposits ₹ 120, of which ₹ 60 is SGST.

Worked Example 4

Q. A laptop is marked at ₹ 48,000. Shop A gives 10% discount and charges 18% GST. Shop B gives 5% discount and charges 12% GST on the same laptop. Which shop is cheaper and by how much?

- Shop A: SP = 48000 × 0.90 = 43,200. Bill = 43200 × 1.18 = ₹ 50,976.
- Shop B: SP = 48000 × 0.95 = 45,600. Bill = 45600 × 1.12 = ₹ 51,072.
- Difference = 51072 − 50976 = ₹ 96.

KEY POINT
Answer: Shop A is cheaper by ₹ 96.

Worked Example 5

Q. A customer paid ₹ 4,956 for a jacket. This included 18% GST charged on the price after a 30% discount. Find the marked price.

- SP after discount = 4956 ÷ 1.18 = ₹ 4,200.
- SP is 70% of MP, so MP = 4200 ÷ 0.70 = ₹ 6,000.

KEY POINT

Answer: Marked price = ₹ 6,000.

Worked Example 6

Q. A bill has three items: a pen set ₹ 1,500 (GST 5%), a bag ₹ 2,000 (GST 12%) and a lamp ₹ 800 (GST 18%). Find the total tax and the total bill.

- GST: 5% of 1500 = 75; 12% of 2000 = 240; 18% of 800 = 144.
- Total GST = 75 + 240 + 144 = ₹ 459. Taxable total = 4,300.
- Bill = 4300 + 459 = ₹ 4,759.

KEY POINT

Answer: Total GST ₹ 459; bill ₹ 4,759 (CGST = SGST = ₹ 229.50).

Worked Example 7

Q. A dealer marks a camera 40% above its cost price of ₹ 9,000. He sells it at a 10% discount, charging 5% GST. Find (i) the customer's price (ii) the dealer's profit percent, before tax.

- MP = 9000 × 1.40 = ₹ 12,600.
- SP = 12600 × 0.90 = ₹ 11,340.
- Customer's price = 11340 × 1.05 = ₹ 11,907.
- Profit = 11340 − 9000 = 2,340. Profit % = 2340 ÷ 9000 × 100 = 26%.

KEY POINT

Answer: ₹ 11,907 and 26%.

Worked Example 8

Q. The price of an article is ₹ 2,500 before tax and the bill is ₹ 2,800. Find the rate of GST.

- GST = 2800 − 2500 = ₹ 300.
- Rate = 300 ÷ 2500 × 100 = 12%.

KEY POINT

Answer: GST rate = 12% (6% CGST + 6% SGST).

BOARD ALERT

Common errors. (1) Charging GST on the **marked price** instead of the discounted price. (2) Subtracting 18% from the bill to get the base price. The correct method is to **divide by 1.18**. (3) Writing CGST as the full GST. Each is half.

REAL LIFE

Look at any shop bill: it lists the taxable value, then CGST and SGST in two equal lines. Check if they really are equal. They always are for a sale within one State.

HOT QUESTION

A trader sells an article for ₹ 11,200 including 12% GST, having bought it for ₹ 8,400 including 12% GST. Without finding either base price, how can you tell the GST he deposits is ₹ 300?

Mind Map Recap: GST

REVISE IN ONE LOOK

SP = MP − discount. GST is charged on SP only. **Bill = SP × (100 + r)/100.**

Inverse: base = bill × 100/(100 + r). Never subtract r% from the bill.

CGST = SGST = half of GST (same State sale).

Tax payable by dealer = output tax − input tax = GST on his margin.

Compare two offers by finding **both final bills**, not by comparing the percentages.

Practice Set

Try each one on paper first. Answers are at the end of the set.

1. An LED TV has marked price ₹ 24,000. After a discount of 20% it is sold with 28% GST. Find the bill amount.
2. A watch is billed at ₹ 3,360 including 12% GST. Find its price before tax and the CGST.
3. Rohan buys a printer with a marked price of ₹ 7,500 at a 12% discount. GST at 18% is charged. Find the total he pays and the SGST.
4. The bill for a coat is ₹ 7,140 including 5% GST on the discounted price. If the discount was 15%, find the marked price.
5. A retailer buys goods for ₹ 20,000 and sells them for ₹ 26,000, GST being 18% on both. Find the GST he pays to the government.
6. A shopkeeper sells a chair for ₹ 1,650 (before tax) and charges 12% GST. How much does the buyer pay in all, and what is the CGST?
7. The price of a fan is ₹ 1,800 before tax and the bill is ₹ 2,124. Find the GST rate.
8. Two stores sell the same sports shoe marked at ₹ 5,000. Store P: 20% off, 12% GST. Store Q: 10% off, 5% GST. Find the cheaper bill.
9. A trader marks goods 25% above cost ₹ 6,400 and sells at a 5% discount. If GST is 12%, find what the customer pays.
10. The total bill for three items is ₹ 5,900. All carry 18% GST. Find the total price before tax and the total GST.
11. A tour company charges ₹ 18,000 for a package, with 5% GST. Find the CGST and the final charge.
12. Input tax paid by a dealer is ₹ 840 and output tax collected is ₹ 1,320. Find the amounts of CGST and SGST he deposits.

Answers

1. SP = 19,200; GST = 5,376; bill = **₹ 24,576**.

2. Before tax ₹ 3,000; CGST = **₹ 180**.

3. SP = 6,600; GST = 1,188; total **₹ 7,788**; SGST = **₹ 594**.

4. SP = 7140 ÷ 1.05 = 6,800; MP = 6800 ÷ 0.85 = **₹ 8,000**.

5. Output 4,680, input 3,600; pays **₹ 1,080**.

6. GST = 198; pays **₹ 1,848**; CGST = **₹ 99**.

7. GST = 324; rate = 324/1800 = **18%**.

8. P: 4000 × 1.12 = 4,480. Q: 4500 × 1.05 = 4,725. **Store P** is cheaper by ₹ 245.

9. MP = 8000; SP = 7,600; pays 7600 × 1.12 = **₹ 8,512**.

10. Before tax = 5900 ÷ 1.18 = **₹ 5,000**; total GST = **₹ 900**.

11. GST = 900; CGST = **₹ 450**; final **₹ 18,900**.

12. Payable = 1320 − 840 = 480; CGST = **₹ 240**, SGST = **₹ 240**.

02

CHAPTER · MATHEMATICS

Banking: Recurring Deposits

Saving a little every month beats waiting for a big lump sum. See how a Recurring Deposit earns interest on each instalment and how to find the maturity value.



IN THIS CHAPTER

- RD formula
- Interest and maturity value
- Finding r, P and n
- Quadratic in n

What a Recurring Deposit Is

In a **Recurring Deposit (RD)** account you put away a fixed amount **every month** for a chosen number of months. The bank pays interest on every instalment for the time it stays in the bank, and hands you everything back on the **maturity date**.

- **P** = the monthly instalment, **n** = number of months, **r** = yearly rate of interest.
- Interest is simple interest, worked out on each instalment for the months it actually stayed in the account.
- Total money deposited = **P × n**. Maturity value (MV) = deposits + interest.

Where the Formula Comes From

The first instalment stays for **n** months, the second for **n − 1**, and so on until the last, which stays for just **1** month. Add the months: n + (n−1) + … + 1 = **n(n+1)/2**. Each rupee of P earns simple interest at r% per year for these months, so the interest is P × (months) × r/(12 × 100).

$$\text{Interest } I = P \times \frac{n(n+1)}{2} \times \frac{r}{12 \times 100}$$

$$\text{Maturity value } MV = P \times n + I$$

KEY POINT

Divide by 12 because the rate is **per year** but the time in this formula is counted in **months**. Forgetting the 12 is the commonest slip.

SIDEKICK TIP

Time saver: compute n(n+1)/2 first. For 24 months it is 300; for 36 months it is 666; for 12 months it is 78. These come up again and again.

MONTHS N	N(N+1)/2	MONTHS N	N(N+1)/2
6	21	24	300
10	55	30	465
12	78	36	666
18	171	48	1176

Worked Example 1

Q. Meera deposits ₹ 800 every month in an RD for 2 years at 7.5% per annum. Find the interest and the maturity value.

- n = 24, so n(n+1)/2 = 300.
- I = 800 × 300 × 7.5 / 1200 = ₹ 1,500.
- Deposits = 800 × 24 = ₹ 19,200. MV = 19200 + 1500 = ₹ 20,700.

KEY POINT

Answer: Interest ₹ 1,500; maturity value ₹ 20,700.

Worked Example 2

Q. A man deposits ₹ 750 per month for 3 years at 6.8% per annum. Find the amount he receives at maturity.

- n = 36, n(n+1)/2 = 666.
- I = 750 × 666 × 6.8 / 1200 = ₹ 2,830.50.
- Deposits = 750 × 36 = 27,000. MV = 27000 + 2830.50 = ₹ 29,830.50.

KEY POINT

Answer: ₹ 29,830.50

Worked Example 3

Q. Amit deposited ₹ 600 per month for 20 months. He received ₹ 12,682.50 at maturity. Find the rate of interest.

- Deposits = 600 × 20 = 12,000, so I = 12682.50 − 12000 = ₹ 682.50.
- n(n+1)/2 = 210. I = 600 × 210 × r / 1200 = 105 r.
- 105 r = 682.50, so r = 6.5.

KEY POINT
Answer: Rate = 6.5% per annum.

Worked Example 4

Q. An RD runs for 1 year at 8% per annum and the maturity value is ₹ 12,520. Find the monthly instalment.

- n = 12, n(n+1)/2 = 78. I = P × 78 × 8 / 1200 = 0.52 P.
- MV = 12 P + 0.52 P = 12.52 P.
- 12.52 P = 12520, so P = 1000.

KEY POINT
Answer: Monthly instalment = ₹ 1,000.

Worked Example 5

Q. Sana deposits ₹ 500 every month at 6% per annum and earns ₹ 750 as interest. For how many months did she deposit?

- I = 500 × n(n+1)/2 × 6/1200 = 1.25 n(n+1).
- 1.25 n(n+1) = 750, so n(n+1) = 600, i.e. $n^2 + n - 600 = 0$.
- (n + 25)(n − 24) = 0, so n = 24 (months cannot be negative).

KEY POINT
Answer: 24 months.

Worked Example 6

Q. Ritu opens an RD of ₹ 1,200 per month for 18 months at 9% per annum. How much more is the maturity value than the total money she put in?

- n(n+1)/2 = 171.
- I = 1200 × 171 × 9 / 1200 = 171 × 9 = ₹ 1,539.

KEY POINT
Answer: She gains ₹ 1,539 over her deposits of ₹ 21,600; MV = ₹ 23,139.

Worked Example 7

Q. Two friends open 2-year RDs at 6% per annum. Karan deposits ₹ 1,000 per month and Lata deposits ₹ 900 per month. Find the difference in their maturity values.

- Interest factor for both: 300 × 6/1200 = 1.5 per rupee of monthly instalment.
- Karan: MV = 24000 + 1500 = ₹ 25,500.
- Lata: MV = 21600 + 1350 = ₹ 22,950.
- Difference = 25500 − 22950 = 2,550.

KEY POINT
Answer: ₹ 2,550.

BOARD ALERT
Common errors. (1) Using n in years instead of months. (2) Using P × n × r / 100 as for a normal deposit: an RD has its own formula. (3) Giving only the interest when the question asks for the **maturity value** (interest + deposits).

REAL LIFE

Banks use RDs for people who cannot save a lump sum but can set aside a little each month, for example for school fees or a festival fund.

HOT QUESTION

Priya's RD interest is exactly one-tenth of her total deposits after 24 months. Taking P as the instalment, find the annual rate (no need to know P).

Mind Map Recap: Banking

REVISE IN ONE LOOK

I = P × n(n+1)/2 × r/1200, with n in months.

MV = P × n + I. Deposits are P × n.

To find r, P or n: write I (or MV) as an equation and solve. A quadratic in n comes up when interest and P are given.

Unit check: r is per year, n is months, so keep the 12 in the denominator.

Practice Set

Try each one on paper first. Answers are at the end of the set.

1. Find the interest and maturity value of an RD of ₹ 1,500 per month for 2 years at 7% per annum.
2. Neha deposits ₹ 400 per month for 15 months at 8% per annum. Find the maturity value.
3. Find the monthly instalment of an RD at 6% per annum for 3 years if the interest earned is ₹ 3,330.
4. An RD of ₹ 2,000 per month for 12 months gives ₹ 24,780 at maturity. Find the rate of interest.
5. For how many months must ₹ 1,000 be deposited monthly at 6% per annum to earn interest of ₹ 1,265? (Form a quadratic in n.)
6. A person deposits ₹ 250 monthly for 4 years at 5.4% per annum. Find the interest.
7. Raj gets ₹ 40,400 on maturity of an RD of ₹ 1,600 per month for 2 years. Find the rate of interest.
8. Compare an RD of ₹ 500 per month for 2 years at 6% and one of ₹ 500 per month for 3 years at 6%. Find the difference in interest.
9. Mala opens an RD of ₹ 700 per month at 9% per annum for 1 year. Find the maturity value.
10. The maturity value of an RD for 2 years at 5% per annum is ₹ 30,300. Find the monthly instalment.
11. Tina invests ₹ 3,000 per month for 10 months at 6.6% per annum. Find the interest earned.
12. An RD at 7.2% per annum for 20 months earns ₹ 2,520 as interest. Find the monthly instalment.

Answers

1. n(n+1)/2 = 300; I = 1500 × 300 × 7/1200 = **₹ 2,625**; MV = 36,000 + 2,625 = **₹ 38,625**.

2. n(n+1)/2 = 120; I = 400 × 120 × 8/1200 = 320; MV = 6000 + 320 = **₹ 6,320**.

3. I = P × 666 × 6/1200 = 3.33 P; 3.33 P = 3330, **P = ₹ 1,000**.

4. Deposits = 24,000, so I = 780; 2000 × 78 × r/1200 = 130 r; r = **6%**.

5. I = 1000 × n(n+1)/2 × 6/1200 = 5 × n(n+1)/2. So n(n+1)/2 = 253, n(n+1) = 506 = 22 × 23, **n = 22** months.

6. n = 48; n(n+1)/2 = 1176; I = 250 × 1176 × 5.4/1200 = **₹ 1,323**.

7. Deposits = 38,400, so I = 2,000. 1600 × 300 × r/1200 = 400 r, hence **r = 5%**.

8. 2 yr: I = 500 × 300 × 6/1200 = 750. 3 yr: I = 500 × 666 × 6/1200 = 1,665. Difference **₹ 915**.

9. I = 700 × 78 × 9/1200 = 409.50; MV = 8400 + 409.50 = **₹ 8,809.50**.

10. MV = 24P + P × 300 × 5/1200 = 25.25 P; P = 30300/25.25 = **₹ 1,200**.

11. n(n+1)/2 = 55; I = 3000 × 55 × 6.6/1200 = **₹ 907.50**.

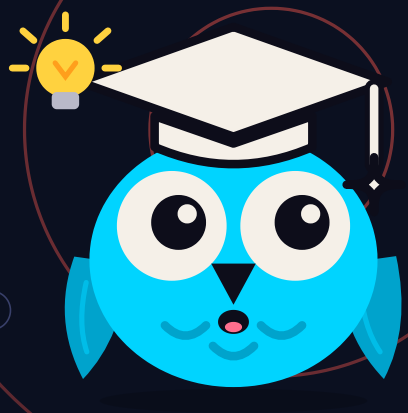
12. I = P × 210 × 7.2/1200 = 1.26 P; P = 2520/1.26 = **₹ 2,000**.

03

CHAPTER · MATHEMATICS

Shares and Dividends

Shares let you own a slice of a company. Learn the difference between face value and market value, and how to find the income and the return on an investment.



IN THIS CHAPTER

- Face and market value
- Premium and discount
- Income and return
- Comparing investments

Owning a Piece of a Company

A company raises money by dividing its capital into equal parts called **shares**. Anyone who buys shares becomes a part-owner and receives a share of the profit, called the **dividend**.

TERM	MEANING
Face value (FV) / Nominal value	The value printed on the share certificate. Dividend is always paid on this
Market value (MV)	The price at which the share is actually bought or sold on the stock exchange today
At par	MV = FV
At premium	MV is **more** than FV. Premium = MV − FV
At discount	MV is **less** than FV. Discount = FV − MV
Dividend	Profit paid to a shareholder, quoted as a percentage **of the face value**
Return (yield)	Income as a percentage of the money you actually invested



BOARD ALERT

Dividend % is on the **face value**. Investment is at the **market value**. Mixing these two is the number one mistake in this chapter.

Formulae

$$\text{Number of shares } N = \frac{\text{Total investment}}{\text{Market value of one share}}$$

$$\text{Income (dividend)} = N \times \frac{\text{rate of dividend}}{100} \times FV$$

$$\text{Return \%} = \frac{\text{Income}}{\text{Investment}} \times 100$$



SIDEKICK TIP

For one share: Return % = (dividend per share ÷ MV) × 100. You can compare two companies without knowing how much is invested.

Brokerage and fractional shares are not part of the syllabus, so the number of shares will always be a whole number in questions.

Worked Example 1

Q. Ravi buys 150 shares of face value ₹ 100 that pay 8% dividend. The market value is ₹ 125. Find his investment, income and return.

- Investment = 150 × 125 = ₹ 18,750.
- Dividend on one share = 8% of 100 = ₹ 8. Income = 150 × 8 = ₹ 1,200.
- Return = 1200 ÷ 18750 × 100 = 6.4%.



KEY POINT

Answer: Investment ₹ 18,750; income ₹ 1,200; return 6.4%.

Worked Example 2

Q. Anita invests ₹ 45,000 in ₹ 50 shares quoted at a premium of ₹ 10. The dividend is 12%. Find the number of shares, her income and the return.

- MV = 50 + 10 = ₹ 60. N = 45000 ÷ 60 = 750 shares.
- Dividend per share = 12% of 50 = ₹ 6. Income = 750 × 6 = ₹ 4,500.
- Return = 4500 ÷ 45000 × 100 = 10%.

KEY POINT

Answer: 750 shares; income ₹ 4,500; return 10%.

Worked Example 3

Q. ₹ 10 shares are available at ₹ 8 each. Imran invests ₹ 12,000 and the company pays 15% dividend. Find his income and return.

- Shares are at a discount of ₹ 2, so MV = 8. N = 12000 ÷ 8 = 1,500.
- Dividend per share = 15% of 10 = ₹ 1.50. Income = 1500 × 1.5 = ₹ 2,250.
- Return = 2250 ÷ 12000 × 100 = 18.75%.

KEY POINT

Answer: Income ₹ 2,250; return 18.75%.

Worked Example 4

Q. Sunita owns 400 shares of face value ₹ 75 and receives ₹ 3,600 as dividend. Find the rate of dividend.

- Total face value = 400 × 75 = ₹ 30,000.
- Rate = 3600 ÷ 30000 × 100 = 12%.

KEY POINT

Answer: Rate of dividend = 12%.

Worked Example 5

Q. Which is better? A: ₹ 100 share at ₹ 150 paying 12%. B: ₹ 50 share at ₹ 40 paying 6%. If ₹ 30,000 is invested, find the income from each.

- A: dividend per share 12, return = 12/150 = 8%.
- B: dividend per share 3, return = 3/40 = 7.5%.
- Income from A = 8% of 30000 = ₹ 2,400. Income from B = 7.5% of 30000 = ₹ 2,250.

KEY POINT

Answer: A is better: it earns ₹ 150 more.

Worked Example 6

Q. Mohan sells 300 shares of ₹ 100 (market value ₹ 140, dividend 8%) and puts all the money into ₹ 10 shares priced at ₹ 12 that pay 15%. Find the change in his income.

- Sale proceeds = 300 × 140 = ₹ 42,000. Old income = 300 × 8 = ₹ 2,400.
- New shares: 42000 ÷ 12 = 3,500. Income = 3500 × 1.50 = ₹ 5,250.
- Change = 5250 − 2400 = ₹ 2,850.

KEY POINT

Answer: Income rises by ₹ 2,850.

Worked Example 7

Q. A man invests ₹ 27,500 in ₹ 50 shares at a premium of 10%. If he earns a return of 10%, find the rate of dividend.

- Premium is 10% of 50 = 5, so MV = 55. N = 27500 ÷ 55 = 500.
- Income = 10% of 27500 = ₹ 2,750, so income per share = 5.50.
- Rate = 5.50 ÷ 50 × 100 = 11%.

KEY POINT

Answer: Rate of dividend = 11%.

Worked Example 8

Q. Deepa gets a dividend of ₹ 1,800 which is a return of 9% on her investment. How much did she invest, and how many ₹ 100 shares did she buy at ₹ 125?

- Investment = 1800 ÷ 0.09 = ₹ 20,000.

- N = 20000 ÷ 125 = 160 shares.

KEY POINT

Answer: Investment ₹ 20,000; 160 shares (dividend rate = 1800/(160 × 100) = 11.25%).

REAL LIFE

Share prices change every minute but the dividend stays tied to the face value. That is why a share bought at a high price may give a smaller percentage return even from a profitable company.

HOT QUESTION

Two companies pay the same dividend on ₹ 100 shares. One trades at ₹ 200, the other at ₹ 125. Compare the returns without knowing the dividend, and say which is better for a buyer.

Mind Map Recap: Shares

REVISE IN ONE LOOK

N = investment ÷ MV. Use MV to count shares.

Income = N × rate% × FV. Use FV for dividend.

Return % = income/investment × 100 = (dividend per share / MV) × 100.

Premium: MV above FV. Discount: MV below FV. Par: equal.

To compare investments, compare the **return %**.

Practice Set

Try each one on paper first. Answers are at the end of the set.

1. Find the income and return on 200 shares of face value ₹ 50 paying 14% dividend, bought at ₹ 70 each.
2. Kiran invests ₹ 36,000 in ₹ 100 shares at a premium of ₹ 20. The dividend is 9%. Find the number of shares, income and return.
3. ₹ 10 shares are on sale at a discount of ₹ 2. A person buys shares worth ₹ 40,000 and the dividend is 12%. Find the annual income and the return.
4. A person has 600 shares of ₹ 35 each and receives ₹ 2,520 as dividend. Find the rate of dividend.
5. How many ₹ 100 shares paying 7.5% dividend must a person own to receive ₹ 4,500 a year?
6. Compare A: ₹ 100 share at ₹ 150, dividend 12% with B: ₹ 50 share at ₹ 40, dividend 6%. Which gives the higher return, and what is the difference in returns?
7. A man invests ₹ 27,500 in shares of face value ₹ 50 quoted at 10% premium. Find the number of shares and, at 11% dividend, his income.
8. Latha sells 250 shares of ₹ 100 at ₹ 120 and buys ₹ 20 shares at ₹ 25 with the proceeds. Find the number of new shares.
9. A dividend of ₹ 1,800 gives a return of 9%. Find the investment.
10. A person buys 250 shares of ₹ 40 at a 25% premium and gets a dividend of 7.5%. Find the investment, income and return.
11. Investment of ₹ 15,000 in ₹ 100 shares at ₹ 125 gives ₹ 1,440 as income. Find the rate of dividend and the return.
12. A company declares 8% dividend. A shareholder holding ₹ 25 shares receives ₹ 1,600. How many shares does he hold, and what is their total face value?

Answers

1. Investment 14,000; income 200 × 7 = **₹ 1,400**; return **10%**.

2. MV = 120; N = **300**; income 300 × 9 = **₹ 2,700**; return **7.5%**.

3. MV = 8; N = 5,000; income 5000 × 1.20 = **₹ 6,000**; return **15%**.

4. Face value total = 21,000; rate = **12%**.

5. Dividend per share = 7.5; N = 4500/7.5 = **600**.

6. A: 8%; B: 7.5%. **A** is better by **0.5%**.

7. MV = 55; N = **500**; income = 500 × 5.50 = **₹ 2,750**.

8. Proceeds 30,000; new shares = **1,200**.

9. Investment = 1800/0.09 = **₹ 20,000**.

10. MV = 50; investment = **₹ 12,500**; income = 250 × 3 = **₹ 750**; return **6%**.

11. N = 120; rate = 1440/(120 × 100) = **12%**; return = **9.6%**.

12. Dividend per share = 2; N = **800**; face value total = **₹ 20,000**.

04

CHAPTER · MATHEMATICS

Linear Inequations

An inequation gives you a whole range of answers instead of one. Learn to solve it, write it in set notation and draw it on the number line.



IN THIS CHAPTER

- Inequality rules
- N, W, Z, R
- Double inequalities
- Number line
- Common solutions

Inequalities and Their Symbols

An equation says two things are **equal**. A linear inequation says one side is **bigger or smaller**. Instead of one answer you usually get a whole **range** of values that work, called the **solution set**.

SYMBOL	READ AS	CIRCLE ON NUMBER LINE
$x < a$	x is less than a	Open circle at a, shade to the left
$x > a$	x is greater than a	Open circle at a, shade to the right
$x \leq a$	x is less than or equal to a	Filled circle at a, shade to the left
$x \geq a$	x is greater than or equal to a	Filled circle at a, shade to the right

The Replacement Set Decides the Answer

The same inequation gives different solution sets for different domains. Always check what x is allowed to be.

SET	SYMBOL	MEMBERS
Natural numbers	N	1, 2, 3, 4, ...
Whole numbers	W	0, 1, 2, 3, ...
Integers	Z	... −2, −1, 0, 1, 2, ...
Real numbers	R	Every number on the number line, including fractions and surds

Rules for Solving

1. Add or subtract
Add or subtract the same number on both sides. The sign stays the same.

2. Multiply or divide by a positive
Sign stays the same.

3. Multiply or divide by a negative
The sign reverses: < becomes > and ≤ becomes ≥.

4. Remove fractions
Multiply every term by the LCM of the denominators (a positive number).

BOARD ALERT
Flip the sign **only** when you multiply or divide by a negative number. Moving a term across the sign changes its sign but does **not** flip the inequality.

Writing the Solution

- **Roster form (list):** {1, 2, 3} for N or Z restricted domains.
- **Set-builder form:** $\{x : x \in R, x > 2\}$.
- **Number line:** mark the boundary, then shade the solution. For N, W, Z mark only the **dots** that belong, not a continuous line.

SIDEKICK TIP
A double inequality such as $-2 < x \leq 4$ is solved in one go: do the same operation on **all three parts**. Always keep the smaller number on the left when writing the answer.

Worked Example 1

Q. Solve $5x - 3 < 2x + 9$, where $x \in N$.

- $5x - 2x < 9 + 3$, so $3x < 12$ and $x < 4$.
- Natural numbers below 4: 1, 2, 3.

KEY POINT
Answer: Solution set = {1, 2, 3}.

Worked Example 2

Q. Solve $-4x + 5 \leq 13$, $x \in R$, and show the solution on a number line.

- $-4x \leq 8$.
- Divide by -4 and reverse the sign: $x \geq -2$.

KEY POINT
Answer: $\{x : x \in R, x \geq -2\}$. Filled circle at -2, shaded to the right.



Worked Example 3

Q. Solve $-3 < 2x + 1 \leq 9$, $x \in Z$.

- Subtract 1 from all parts: $-4 < 2x \leq 8$.
- Divide by 2: $-2 < x \leq 4$.
- Integers greater than -2 and up to 4.

KEY POINT
Answer: Solution set = $\{-1, 0, 1, 2, 3, 4\}$.



Worked Example 4

Q. Solve $7 - 2x > -5$, $x \in W$.

- $-2x > -12$.
- Divide by -2, reversing: $x < 6$.
- Whole numbers less than 6 start at 0.

KEY POINT
Answer: Solution set = {0, 1, 2, 3, 4, 5}.

Worked Example 5

Q. Solve $\frac{x}{3} - 2 \geq \frac{x}{2} - 3$, $x \in R$.

- Multiply by 6: $2x - 12 \geq 3x - 18$.
- $2x - 3x \geq -18 + 12$, so $-x \geq -6$.
- Multiply by -1 and reverse: $x \leq 6$.

KEY POINT
Answer: $\{x : x \in R, x \leq 6\}$.

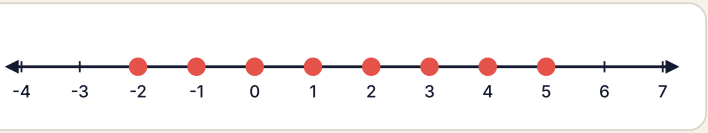
Worked Example 6

Q. Find the solution set common to $2x - 5 < 7$ and $3x + 2 \geq -4$, where $x \in Z$. Show it on a number line.

- First: $2x < 12$, so $x < 6$.

- Second: $3x \geq -6$, so $x \geq -2$.
- Common part: $-2 \leq x < 6$, integers only.

KEY POINT
Answer: $\{-2, -1, 0, 1, 2, 3, 4, 5\}$.

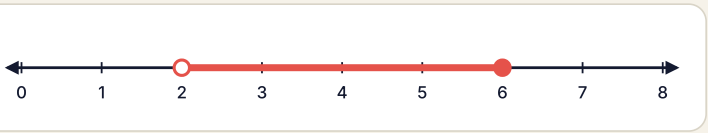


Worked Example 7

Q. Solve $2x + 3 > 7$ and $5 - x \geq -1$ for $x \in R$ and show on a number line.

- First: $2x > 4$, so $x > 2$.
- Second: $-x \geq -6$, so $x \leq 6$.
- Together: $2 < x \leq 6$.

KEY POINT
Answer: $\{x : x \in R, 2 < x \leq 6\}$. Open circle at 2, filled circle at 6, line shaded between.



Worked Example 8

Q. Find the largest integer x that satisfies $4 - 3x > -11$.

- $-3x > -15$, so $x < 5$.
- x is strictly less than 5, so 5 is not allowed.

KEY POINT
Answer: Largest integer = 4.

BOARD ALERT
Common errors. (1) Not reversing the sign after dividing by a negative. (2) Using a filled circle for < or >. (3) Writing the solution of an N or W question as a continuous line instead of separate dots or a list. (4) Forgetting to apply the operation to **every** part of a double inequality.

REAL LIFE
Inequations describe limits: "at least 75% attendance" is a ≥ condition, "fewer than 40 students per bench" is a < condition.

HOT QUESTION
If $x \in R$ and $-3 < x \leq 5$, what is the largest integer value of $2x - 7$? Then say how many integer values $2x - 7$ can take in all.

Mind Map Recap: Linear Inequations

REVISE IN ONE LOOK

Solve like an equation, but **flip the sign when multiplying or dividing by a negative**.

Domain matters: N (1, 2, ...), W (0, 1, ...), Z, R.

Number line: open circle for < >, filled circle for ≤ ≥. Dots only for N, W, Z.

Double inequality: operate on all three parts.

Two conditions together: shade the **overlap** (intersection).

Practice Set

Try each one on paper first. Answers are at the end of the set.

1. Solve $4x - 5 < 11$, $x \in N$.
2. Solve $3 - 2x \leq 9$, $x \in R$, and show the solution on a number line.
3. Solve $-5 < 3x + 4 \leq 16$, $x \in Z$.
4. Solve $2(x - 3) \geq 5x + 9$, $x \in R$. Write the greatest integer in the solution.
5. Solve $\frac{x + 1}{2} - \frac{x - 2}{3} < 2$, $x \in W$.
6. Solve $2x - 1 > 3$ and $7 - x \geq 1$, $x \in R$. Show the answer on a number line.
7. Find the greatest integer x such that $7x - 3 \leq 25$.
8. Solve $-4 \leq \frac{x}{2} - 3 < 1$, $x \in Z$.
9. Solve $3x - 2 \geq 4x + 1$, $x \in R$.
10. Solve $1 \leq 3 - 2x < 9$, $x \in R$.
11. Find the integers that satisfy both $2x + 1 > -3$ and $4x - 3 < 9$.
12. Find the smallest integer x satisfying $5x - 4 > 2x + 9$.

Answers

1. $x < 4$: **{1, 2, 3}**.

2. $-2x \leq 6$ gives $x \geq -3$. **{x : x ∈ R, x ≥ −3}**; filled circle at −3, shade right.

3. $-9 < 3x \leq 12$ gives $-3 < x \leq 4$: **{−2, −1, 0, 1, 2, 3, 4}**.

4. $2x - 6 \geq 5x + 9$, $-3x \geq 15$, **x ≤ −5**; greatest integer **−5**.

5. $3(x + 1) - 2(x - 2) < 12$, $x + 7 < 12$, $x < 5$: **{0, 1, 2, 3, 4}**.

6. $x > 2$ and $x \leq 6$: **2 < x ≤ 6**; open circle at 2, filled at 6.

7. $7x \leq 28$, $x \leq 4$: **4**.

8. Add 3: $-1 \leq x/2 < 4$, so $-2 \leq x < 8$: **{−2, −1, 0, 1, 2, 3, 4, 5, 6, 7}**.

9. $-x \geq 3$, so **x ≤ −3**.

10. $-2 \leq -2x < 6$, so **−3 < x ≤ 1**.

11. $x > -2$ and $x < 3$: **{−1, 0, 1, 2}**.

12. $3x > 13$, $x > 4.33$: smallest integer **5**.

05

CHAPTER · MATHEMATICS

Quadratic Equations

Two roots, one formula and a discriminant that tells you the story before you solve. Factorise, use the formula and turn word problems into equations.



IN THIS CHAPTER

- Factorisation
- Formula
- Nature of roots
- Word problems

What Makes an Equation Quadratic

A quadratic equation in one variable has the standard form **$ax^2 + bx + c = 0$** with $a \neq 0$. The highest power of x is 2, so it can have at most **two roots**. A root is a value of x that makes the equation true.

SIDEKICK TIP

Always bring everything to one side first so the equation reads $ax^2 + bx + c = 0$. Many equations that look different (with brackets or fractions) turn into this form after simplifying.

Method 1: Factorisation

1. Standard form
Write $ax^2 + bx + c = 0$.

2. Split the middle term
Find two numbers whose product is $a \times c$ and whose sum is b.

3. Factorise by grouping
This gives two brackets, $(px + q)(rx + s) = 0$.

4. Zero product rule
If a product is zero, one of the factors is zero. Put each bracket equal to 0.

Method 2: The Quadratic Formula

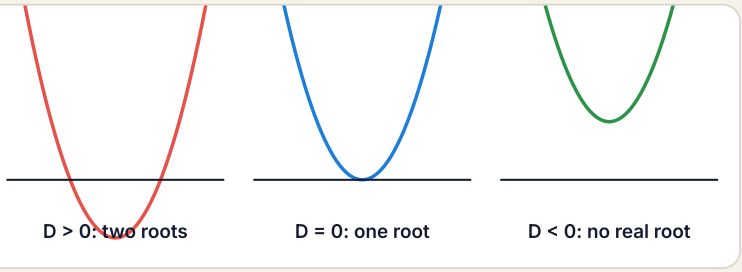
When the factors are not easy to see, use the formula. It always works.

$$x = \frac{-b \pm \sqrt{(b^2 - 4ac)}}{2a}$$

Nature of the Roots

The expression under the root is the **discriminant**, $D = b^2 - 4ac$. Its sign tells you what kind of roots exist, without solving.

DISCRIMINANT	NATURE OF ROOTS	GRAPH TOUCHES X-AXIS
$D > 0$	Two distinct real roots	At two points
$D = 0$	Two equal real roots, $x = -b/2a$	At one point
$D < 0$	No real roots	Does not touch the axis



KEY POINT

Equal roots is the condition $D = 0$. This is how you find an unknown constant k in an equation that is said to have “equal roots” or “real roots” ($D \geq 0$).

Solving Word Problems

1. Choose the unknown
Let x be the quantity you are asked about, with its unit.

2. Form the equation
Translate each sentence into algebra. Typical themes: consecutive numbers, areas, ages, speed-distance-time, work.

3. Solve
Factorise or use the formula.

4. Reject what makes no sense
Lengths, speeds and counts cannot be negative. Check the answer in the story.

Worked Example 1

Q. Solve $x^2 - 5x - 14 = 0$ by factorisation.

- Need two numbers with product −14 and sum −5: −7 and 2.
- $x^2 - 7x + 2x - 14 = x(x - 7) + 2(x - 7) = (x - 7)(x + 2)$.
- $x - 7 = 0$ or $x + 2 = 0$.

KEY POINT
Answer: $x = 7$ or $x = -2$.

Worked Example 2

Q. Solve $6x^2 + x - 12 = 0$.

- $a \times c = -72$. Numbers with product −72 and sum 1: 9 and −8.
- $6x^2 + 9x - 8x - 12 = 3x(2x + 3) - 4(2x + 3) = (2x + 3)(3x - 4)$.
- $2x + 3 = 0$ or $3x - 4 = 0$.

KEY POINT
Answer: $x = -\frac{3}{2}$ or $x = \frac{4}{3}$.

Worked Example 3

Q. Solve $2x^2 - 7x + 4 = 0$, giving answers correct to two decimal places.

- $a = 2$, $b = -7$, $c = 4$. $D = 49 - 32 = 17$, and $\sqrt{17} = 4.1231$.
- $x = (7 \pm 4.1231) / 4$.
- $x = 11.1231/4 = 2.78$ or $x = 2.8769/4 = 0.72$.

KEY POINT
Answer: $x \approx 2.78$ or 0.72.

Worked Example 4

Q. Find k if $3x^2 - 2kx + 12 = 0$ has equal roots.

- $a = 3$, $b = -2k$, $c = 12$. Equal roots need $D = 0$.
- $(-2k)^2 - 4 \times 3 \times 12 = 4k^2 - 144 = 0$.
- $k^2 = 36$.

KEY POINT
Answer: $k = 6$ or $k = -6$.

Worked Example 5

Q. Solve $\frac{x}{x + 2} + \frac{x + 2}{x} = \frac{34}{15}$.

- Let $y = x/(x + 2)$. Then $y + 1/y = 34/15$, so $15y^2 - 34y + 15 = 0$.
- $D = 1156 - 900 = 256$, so $y = (34 \pm 16)/30 = 5/3$ or $3/5$.
- $y = 5/3$: $3x = 5x + 10$, so $x = -5$. $y = 3/5$: $5x = 3x + 6$, so $x = 3$.

KEY POINT

Answer: x = 3 or x = −5.

Worked Example 6

Q. The sum of the squares of two consecutive positive integers is 313. Find them.

- Let the integers be n and n + 1. $n^2 + (n + 1)^2 = 313$.
- $2n^2 + 2n - 312 = 0$, so $n^2 + n - 156 = 0$.
- $(n + 13)(n - 12) = 0$. Positive: n = 12.

KEY POINT

Answer: 12 and 13 (check: 144 + 169 = 313).

Worked Example 7

Q. A train travels 240 km at a uniform speed. If the speed had been 10 km/h more, it would have taken 2 hours less. Find the speed.

- Let speed be x km/h. $240/x - 240/(x + 10) = 2$.
- $240(x + 10) - 240x = 2x(x + 10)$, so $2400 = 2x^2 + 20x$.
- $x^2 + 10x - 1200 = 0$, $(x + 40)(x - 30) = 0$. Speed is positive.

KEY POINT

Answer: 30 km/h (times 8 h and 6 h).

Worked Example 8

Q. The length of a rectangle is 3 m more than twice its breadth and its area is 90 m². Find its dimensions.

- Breadth b, length 2b + 3. $b(2b + 3) = 90$.
- $2b^2 + 3b - 90 = 0$. $D = 9 + 720 = 729 = 27^2$.
- $b = (-3 + 27)/4 = 6$ (reject −7.5).

KEY POINT

Answer: Breadth 6 m, length 15 m.

Worked Example 9

Q. Two pipes fill a tank in 6 hours together. The smaller pipe alone takes 5 hours more than the larger pipe. How long does each take alone?

- Larger pipe takes x hours, smaller x + 5. $1/x + 1/(x + 5) = 1/6$.
- $6(2x + 5) = x(x + 5)$, so $x^2 - 7x - 30 = 0$.
- $(x - 10)(x + 3) = 0$, so x = 10.

KEY POINT

Answer: Larger pipe 10 hours, smaller pipe 15 hours.

BOARD ALERT

Common errors. (1) Dividing both sides by x and losing the root x = 0. (2) Mis-signing b in the formula: it is **−b**, so if b = −7 you write +7. (3) Giving a negative length or speed as an answer. (4) Leaving the equation unsimplified: first clear fractions and brackets.

REAL LIFE

Throw a ball upwards and its height after t seconds is a quadratic in t. Setting the height to a value and solving gives the two moments at which it passes that height: once going up, once coming down.

HOT QUESTION

If the roots of $x^2 + px + 12 = 0$ are real, what are the possible integer values of p? Then take the smallest positive such p and solve the equation.

Mind Map Recap: Quadratic Equations

REVISE IN ONE LOOK

Standard form **$ax^2 + bx + c = 0$**, $a \neq 0$.

Factorise: product ac, sum b. Or **formula**: $x = (-b \pm \sqrt{D})/2a$.

$D = b^2 - 4ac$: $D > 0$ two real roots; $D = 0$ equal roots; $D < 0$ none.

Word problems: define x, form equation, solve, reject impossible roots, check.

Speed-time: distance/speed gives time; the difference in times forms the equation.

Practice Set

Try each one on paper first. Answers are at the end of the set.

1. Solve $x^2 - 9x + 20 = 0$.
2. Solve $3x^2 - 14x - 5 = 0$ by factorisation.
3. Solve $2x^2 + 5x - 3 = 0$ using the formula.
4. Solve $x^2 - 6x + 4 = 0$, correct to two decimal places.
5. Find k if $kx^2 + 8x + 2 = 0$ has equal roots.
6. Without solving, state the nature of the roots of $2x^2 - 3x + 5 = 0$.
7. The product of two consecutive positive odd numbers is 143. Find them.
8. A rectangle has perimeter 34 cm and area 60 cm^2. Find its sides.
9. An aeroplane flies 600 km at a steady speed. At 50 km/h faster it would take 1 hour less. Find the original speed.
10. The sum of the squares of two consecutive positive even numbers is 340. Find the numbers.
11. Solve $\frac{1}{x} + \frac{1}{x+3} = \frac{7}{10}$.
12. If $x = 3$ is a root of $2x^2 - 5x + m = 0$, find m and the other root.

Answers

1. $(x - 4)(x - 5) = 0$: **4 or 5**.

2. $(3x + 1)(x - 5) = 0$: **$x = 5$ or $-1/3$**.

3. $D = 25 + 24 = 49$; $x = (-5 \pm 7)/4$: **1/2 or −3**.

4. $D = 36 - 16 = 20$; $x = 3 \pm \sqrt{5}$: **5.24 or 0.76**.

5. $D = 64 - 8k = 0$: **$k = 8$**.

6. $D = 9 - 40 = -31 < 0$: **no real roots**.

7. $n(n + 2) = 143$, $n^2 + 2n - 143 = 0$, $(n + 13)(n - 11) = 0$: **11 and 13**.

8. $l + b = 17$, $x(17 - x) = 60$, $x^2 - 17x + 60 = 0$: **5 cm and 12 cm**.

9. $600/x - 600/(x + 50) = 1$ gives $x^2 + 50x - 30000 = 0$, $(x + 200)(x - 150) = 0$: **150 km/h**.

10. $n^2 + (n + 2)^2 = 340$ gives $n^2 + 2n - 168 = 0$, $n = 12$: **12 and 14**.

11. $10(2x + 3) = 7x(x + 3)$, $7x^2 + x - 30 = 0$, $(7x + 15)(x - 2) = 0$: **$x = 2$ or $-15/7$**.

12. $18 - 15 + m = 0$: **$m = -3$**; $2x^2 - 5x - 3 = (2x + 1)(x - 3)$: other root **$-1/2$**.

06

CHAPTER · MATHEMATICS

Ratio and Proportion

Ratios compare, proportions equate them. Learn the properties that turn messy-looking fractions into one-line answers.



IN THIS CHAPTER

- Mean and third proportional
- Continued proportion
- Componendo and dividendo
- Equal ratios

Ratio Refresher

A ratio a : b compares two quantities of the **same kind** and has no units. It can be written as a fraction a/b. Multiplying or dividing both terms by the same non-zero number does not change the ratio.

Proportion

Four quantities a, b, c, d are in **proportion** when a : b = c : d. We write a : b :: c : d.

$$a : b = c : d \iff a \times d = b \times c \quad \text{(product of extremes = product of means)}$$

- **Fourth proportional** to a, b, c is the number d with a : b = c : d, so d = bc/a.
- **Third proportional** to a and b is c with a : b = b : c, so c = b^2/a.
- **Mean proportional** between a and c is b with a : b = b : c, so b = √(ac).

Continued Proportion

a, b, c are in continued proportion when a : b = b : c. Then **b^2 = ac**. For a continued proportion the powerful trick is to write **a/b = b/c = k**, so b = ck and a = ck^2. Many "prove that" questions fall to this substitution.

The Four Properties

Suppose a/b = c/d. Then each of the following is also true.

PROPERTY	RESULT
Invertendo	b/a = d/c
Alternendo	a/c = b/d
Componendo	(a + b)/b = (c + d)/d
Dividendo	(a − b)/b = (c − d)/d
Componendo and dividendo	(a + b)/(a − b) = (c + d)/(c − d)



SIDEKICK TIP

Use componendo-dividendo when the equation looks like (something + something)/(something − something) = a number. Add the denominator to the numerator on both sides, and subtract it too, then divide the two results.



SIDEKICK TIP

For three or more equal ratios a/b = c/d = e/f, put them all equal to k. Then a = bk, c = dk, e = fk and the algebra simplifies.

Worked Example 1

Q. Find the mean proportional between 4 and 25, and the third proportional to 6 and 9.

- Mean proportional = √(4 × 25) = √100 = 10.
- Third proportional c: 6 : 9 = 9 : c, so c = 81/6 = 13.5.



KEY POINT

Answer: 10 and 13.5.

Worked Example 2

Q. Check whether 6, 9, 8, 12 are in proportion.

- Product of extremes = 6 × 12 = 72.

- Product of means = 9 × 8 = 72.

KEY POINT
Answer: Equal, so 6 : 9 = 8 : 12. They are in proportion.

Worked Example 3

Q. If (5x + 3y) : (5x − 3y) = 7 : 3, find x : y.

- Componendo-dividendo: (2 × 5x)/(2 × 3y) = (7 + 3)/(7 − 3) = 10/4.
- So 5x/3y = 5/2, which gives x/y = (5/2)(3/5) = 3/2.

KEY POINT
Answer: x : y = 3 : 2.

Worked Example 4

Q. If a : b = c : d, prove that (3a + 5b) : (3a − 5b) = (3c + 5d) : (3c − 5d).

- Let a/b = c/d = k, so a = bk and c = dk.
- LHS = (3bk + 5b)/(3bk − 5b) = (3k + 5)/(3k − 5).
- RHS = (3dk + 5d)/(3dk − 5d) = (3k + 5)/(3k − 5).

KEY POINT
Answer: LHS = RHS, hence proved.

Worked Example 5

Q. If a, b, c are in continued proportion, prove that $(a^2 + b^2)/(b^2 + c^2) = a/c$.

- Let a/b = b/c = k. Then b = ck and $a = ck^2$.
- $a^2 + b^2 = c^2k^4 + c^2k^2 = c^2k^2(k^2 + 1)$.
- $b^2 + c^2 = c^2k^2 + c^2 = c^2(k^2 + 1)$.
- Ratio = k^2 and $a/c = ck^2/c = k^2$.

KEY POINT
Answer: Both equal k^2, hence proved.

Worked Example 6

Q. What number must be added to each of 6, 15, 20, 43 so that the results are in proportion?

- (6 + x) : (15 + x) = (20 + x) : (43 + x).
- (6 + x)(43 + x) = (15 + x)(20 + x), so 258 + 49x = 300 + 35x.
- 14x = 42, x = 3.

KEY POINT
Answer: 3 (the numbers become 9, 18, 23, 46).

Worked Example 7

Q. Two numbers are in the ratio 5 : 7. If 8 is added to each, the ratio becomes 3 : 4. Find the numbers.

- Numbers 5k and 7k. (5k + 8)/(7k + 8) = 3/4.
- 20k + 32 = 21k + 24, so k = 8.

KEY POINT
Answer: 40 and 56 (check: 48/64 = 3/4).

Worked Example 8

Q. If x : y = 3 : 4 and y : z = 6 : 7, find x : y : z.

- Make the y-terms equal. x : y = 9 : 12 and y : z = 12 : 14.

- So x : y : z = 9 : 12 : 14.

KEY POINT

Answer: 9 : 12 : 14.

BOARD ALERT

Common errors. (1) Using componendo on only one side. Do the same on both sides. (2) Confusing mean proportional (square root) with third proportional (square over first). (3) Cancelling terms across a sum, e.g. (a + b)/b ≠ a + 1 unless you write it as a/b + 1 properly.

REAL LIFE

Mixing paint, scaling a recipe or reading a map scale are all proportion. If 3 cups of rice need 5 cups of water, 9 cups need 15: 3 : 5 = 9 : 15.

HOT QUESTION

If a : b = b : c and a, b, c are positive, show that a : c = a^2 : b^2. Use it to find a : c when a : b = 2 : 3.

Mind Map Recap: Ratio and Proportion

REVISE IN ONE LOOK

a : b = c : d ⇔ ad = bc.

Continued proportion: **b^2 = ac**; mean proportional = √(ac).

Properties: invertendo, alternendo, componendo, dividendo, **componendo-dividendo**.

For equal ratios, set each ratio = k and substitute.

“Add the same number” questions give a linear equation after cross-multiplying.

Practice Set

Try each one on paper first. Answers are at the end of the set.

1. Find the mean proportional between 9 and 16.
2. Find the third proportional to 8 and 12.
3. Find the fourth proportional to 4, 9 and 6.
4. Are 7, 12, 21, 36 in proportion?
5. Find x if (x + 2) : (3x − 1) = 1 : 2.
6. What must be added to each of 5, 9, 11 and 18 so that the numbers are in proportion?
7. If a : b = c : d, prove that (a + 3b) : (a − 3b) = (c + 3d) : (c − 3d).
8. If x : y = 3 : 5, find (2x + 3y) : (3x + 2y).
9. If a, b, c are in continued proportion, prove that a : c = a^2 : b^2.
10. Two numbers are in the ratio 8 : 11. When 6 is added to each, the ratio becomes 10 : 13. Find the numbers.
11. If 4, b, 25 are in continued proportion, find b and the ratio (4 + b) : (b + 25).
12. If x/3 = y/4 = z/5 and x + y + z = 36, find x, y and z.

Answers

1. √(9 × 16) = **12**.

2. 144/8 = **18**.

3. (9 × 6)/4 = **13.5**.

4. 7 × 36 = 252 = 12 × 21: **yes**.

5. 2(x + 2) = 3x − 1: **x = 5**.

6. (5 + x)(18 + x) = (9 + x)(11 + x): 90 + 23x = 99 + 20x: **3**.

7. Put a = bk, c = dk: both sides become (k + 3)/(k − 3).

8. x = 3k, y = 5k: (6 + 15)/(9 + 10) = **21 : 19**.

9. a = ck^2, b = ck: a/c = k^2 and a^2/b^2 = $(c^2k^4)/(c^2k^2)$ = k^2.

10. 13(8k + 6) = 10(11k + 6): k = 3: **24 and 33**.

11. b^2 = 100, **b = 10**; ratio 14 : 35 = **2 : 5**.

12. Common value k = 3: **x = 9, y = 12, z = 15**.

07

CHAPTER · MATHEMATICS

Factorisation of Polynomials

The remainder and factor theorems let you test a number in seconds instead of dividing for minutes. Use them to break cubics into factors.



IN THIS CHAPTER

- Remainder theorem
- Factor theorem
- Cubic factorisation
- Finding unknown constants

Polynomials in One Variable

A polynomial such as $f(x) = 2x^3 - 5x^2 + 3x - 7$ has terms with whole-number powers of x. The highest power is the **degree**. In your syllabus f(x) is not more than degree 3.

The Remainder Theorem

Dividing a polynomial by $(x - a)$ leaves a remainder. You do not need long division to find it.

KEY POINT

Remainder Theorem: when a polynomial f(x) is divided by $(x - a)$, the remainder is **f(a)**. For a divisor $(px + q)$, the remainder is $f(-q/p)$.

Why it works: division gives $f(x) = (x - a) \times Q(x) + R$. Put $x = a$ and the first term vanishes, leaving $f(a) = R$.

The Factor Theorem

KEY POINT

Factor Theorem: $(x - a)$ is a factor of f(x) if and only if **f(a) = 0**. In that case a is a root of $f(x) = 0$.

This is the remainder theorem with the remainder equal to zero. It is how you hunt for factors of a cubic.

Factorising a Cubic Completely

1. Guess a root
Try $x = \pm1, \pm2, \pm3$ ... (factors of the constant term). The first value giving $f(a) = 0$ gives the factor $(x - a)$.

2. Divide
Divide f(x) by $(x - a)$ by long division, or compare coefficients, to get a quadratic.

3. Factorise the quadratic
Split the middle term or use the formula.

4. Write the product
$f(x) = (x - a)$(factor)(factor).

Division in a nutshell. To divide $x^3 + 2x^2 - 5x - 6$ by $(x + 1)$: the first term is x^2, so subtract $x^3 + x^2$ to leave $x^2 - 5x$. Next term x, subtract $x^2 + x$ to leave $-6x - 6$. Next term -6, subtract $-6x - 6$ to leave 0. The quotient is $x^2 + x - 6$.

SIDEKICK TIP

To test candidate roots fast, remember: if the polynomial has integer coefficients and leading coefficient 1, any integer root is a **factor of the constant term**. For $2x^3 + ... + 6$, try $\pm1, \pm2, \pm3, \pm6$ and fractions like 1/2.

Worked Example 1

Q. Find the remainder when $f(x) = 2x^3 - 5x^2 + 3x - 7$ is divided by $(x - 2)$.

- Remainder $= f(2) = 2(8) - 5(4) + 3(2) - 7$.
- $= 16 - 20 + 6 - 7 = -5$.

KEY POINT

Answer: Remainder = −5.

Worked Example 2

Q. Find the remainder when $4x^3 + 3x^2 - x + 6$ is divided by $(2x + 1)$.

- $2x + 1 = 0$ gives $x = -1/2$.
- $f(-1/2) = 4(-1/8) + 3(1/4) + 1/2 + 6 = -0.5 + 0.75 + 0.5 + 6 = 6.75$.

KEY POINT

Answer: Remainder = 27/4 = 6.75.

Worked Example 3

Q. Show that $(x + 1)$ is a factor of $x^3 + 2x^2 - 5x - 6$ and factorise it completely.

- $f(-1) = -1 + 2 + 5 - 6 = 0$, so $(x + 1)$ is a factor.
- Dividing gives quotient $x^2 + x - 6 = (x + 3)(x - 2)$.

KEY POINT

Answer: $f(x) = (x + 1)(x + 3)(x - 2)$.

Worked Example 4

Q. If $(x + 2)$ is a factor of $x^3 + kx^2 - 4x - 12$, find k and then factorise.

- $f(-2) = -8 + 4k + 8 - 12 = 4k - 12 = 0$, so $k = 3$.
- $x^3 + 3x^2 - 4x - 12$ divided by $(x + 2)$ gives $x^2 + x - 6 = (x + 3)(x - 2)$.

KEY POINT

Answer: $k = 3$; $f(x) = (x + 2)(x + 3)(x - 2)$.

Worked Example 5

Q. The polynomial $x^3 + ax^2 + bx - 6$ is exactly divisible by $(x - 1)$ and $(x + 2)$. Find a and b, and the third factor.

- $f(1) = 1 + a + b - 6 = 0$, so $a + b = 5$.
- $f(-2) = -8 + 4a - 2b - 6 = 0$, so $2a - b = 7$.
- Add the two equations: $3a = 12$, $a = 4$, $b = 1$.
- $f(x) = x^3 + 4x^2 + x - 6 = (x - 1)(x + 2)(x + 3)$.

KEY POINT

Answer: $a = 4$, $b = 1$; third factor $(x + 3)$.

Worked Example 6

Q. Factorise $2x^3 - 3x^2 - 11x + 6$ completely and solve $f(x) = 0$.

- $f(3) = 54 - 27 - 33 + 6 = 0$, so $(x - 3)$ is a factor.
- Quotient: $2x^2 + 3x - 2 = (2x - 1)(x + 2)$.
- Roots from $(x - 3)(2x - 1)(x + 2) = 0$.

KEY POINT

Answer: $(x - 3)(2x - 1)(x + 2)$; $x = 3$, $1/2$ or -2.

Worked Example 7

Q. When $x^3 - ax^2 + 6x - a$ is divided by $(x - 1)$ the remainder is 5. Find a.

- $f(1) = 1 - a + 6 - a = 7 - 2a$.
- $7 - 2a = 5$ gives $a = 1$.

KEY POINT

Answer: $a = 1$.

Worked Example 8

Q. Polynomials $2x^3 + ax^2 + 3x - 5$ and $x^3 + x^2 - 2x + a$ leave the same remainder when divided by $(x - 2)$. Find a.

- First: $f(2) = 16 + 4a + 6 - 5 = 17 + 4a$.
- Second: $g(2) = 8 + 4 - 4 + a = 8 + a$.
- $17 + 4a = 8 + a$, so $3a = -9$.

KEY POINT

Answer: a = −3.

BOARD ALERT

Common errors. (1) Using f(a) for a divisor (x + a): the correct value is f(−a). (2) For (2x − 1) use x = 1/2, not 2. (3) Stopping at one factor. "Factorise completely" means break the quadratic too. (4) Sign slips in f(−2): put the number in brackets before squaring or cubing.

REAL LIFE

Engineers use the same idea to test whether a particular value is a "zero" of a model. If plugging it in gives 0, that value is a solution and the model splits into simpler pieces.

HOT QUESTION

If (x − 2) is a factor of $x^3 - 3x^2 + px + 2$ and the remainder on dividing the same cubic by (x + 1) is q, find p and q. Then find the quadratic factor and its discriminant.

Mind Map Recap: Factorisation

REVISE IN ONE LOOK

Remainder theorem: f(x) ÷ (x − a) leaves f(a). Divisor (px + q): f(−q/p).

Factor theorem: f(a) = 0 ⇔ (x − a) is a factor.

Cubic: find one root by trial → divide → factorise the quadratic → write all three factors.

Unknown constants: put f(a) = 0 or f(a) = remainder, solve the equations.

Same remainder for two polynomials: set f(a) = g(a).

Practice Set

Try each one on paper first. Answers are at the end of the set.

1. Find the remainder when $x^3 - 4x^2 + 6x - 5$ is divided by (x − 2).
2. Find the remainder when $3x^3 + x^2 - 5$ is divided by (x + 1).
3. Show that (x − 2) is a factor of $x^3 - 3x^2 - 4x + 12$ and factorise completely.
4. If (x + 3) is a factor of $x^3 + kx^2 - x - 12$, find k.
5. Find a if $x^3 + ax^2 - 5x + 4$ leaves remainder 10 when divided by (x + 1).
6. Find a and b if $x^3 + ax^2 + bx - 12$ is divisible by (x − 1) and (x + 3).
7. Factorise $2x^3 + x^2 - 13x + 6$ completely.
8. Solve $x^3 - 6x^2 + 11x - 6 = 0$.
9. Find the remainder when $4x^3 - 6x^2 + x + 5$ is divided by (2x − 1).
10. Without dividing, show that (x + 2) is a factor of $x^3 + x^2 - 4x - 4$ and factorise it.
11. Find a and b if $x^3 + ax^2 - bx + 10$ is divisible by $x^2 - 3x + 2$.
12. Find a if $2x^3 - 5x^2 + ax - 6$ leaves remainder 12 when divided by (x − 2).

Answers

1. f(2) = 8 − 16 + 12 − 5 = **−1**.

2. f(−1) = −3 + 1 − 5 = **−7**.

3. f(2) = 8 − 12 − 8 + 12 = 0. Quotient $x^2 - x - 6$: **(x − 2)(x − 3)(x + 2)**.

4. f(−3) = −27 + 9k + 3 − 12 = 9k − 36 = 0: **k = 4**.

5. f(−1) = −1 + a + 5 + 4 = a + 8 = 10: **a = 2**.

6. a + b = 11 and 3a − b = 13: **a = 6, b = 5**.

7. f(2) = 0. Quotient $2x^2 + 5x - 3$: **(x − 2)(2x − 1)(x + 3)**.

8. x = 1 is a root; quotient $x^2 - 5x + 6$: **x = 1, 2, 3**.

9. f(1/2) = 0.5 − 1.5 + 0.5 + 5 = **4.5**.

10. f(−2) = −8 + 4 + 8 − 4 = 0. Quotient $x^2 - x - 2$: **(x + 2)(x − 2)(x + 1)**.

11. f(1) = 0: a − b = −11. f(2) = 0: 2a − b = −9. **a = 2, b = 13**.

12. 16 − 20 + 2a − 6 = 12: **a = 11**.

08

CHAPTER · MATHEMATICS

Matrices

Numbers in a grid that add, scale and multiply by rules. Get the order right, know when an operation is possible and watch out: AB is not BA.



IN THIS CHAPTER

- Order and types
- Equal matrices
- Addition and subtraction
- Multiplication
- Matrix equations

What a Matrix Is

A matrix is a rectangular arrangement of numbers in rows and columns, enclosed in brackets. Each number is an **element**. A matrix with **m rows and n columns** has **order m × n** (rows first, always).

Example:

$\begin{vmatrix} 3 & -2 & 5 \\ 0 & 7 & 1 \end{vmatrix}$ has 2 rows and 3 columns, so its order is 2 × 3 and it has 6 elements.

Types of Matrices

TYPE	MEANING	EXAMPLE
Row matrix	Only one row, order 1 × n	$\begin{vmatrix} 4 & -1 & 6 \end{vmatrix}$
Column matrix	Only one column, order m × 1	$\begin{vmatrix} 2 \\ 9 \end{vmatrix}$
Square matrix	Number of rows = number of columns	2 × 2 or 3 × 3
Null (zero) matrix	Every element is 0. Adding it changes nothing	$\begin{vmatrix} 0 & 0 \\ 0 & 0 \end{vmatrix}$
Identity matrix I	Square, 1 on the main diagonal, 0 elsewhere. AI = IA = A	$\begin{vmatrix} 1 & 0 \\ 0 & 1 \end{vmatrix}$

Equal Matrices

Two matrices are equal only if they have the **same order** and **every pair of matching elements is equal**. This turns a matrix equation into ordinary equations: each position gives one equation.

Addition, Subtraction and Scalar Multiplication

- **A ± B** exists only when A and B have the **same order**. Add or subtract element by element.
- **kA** (k a number) multiplies **every** element of A by k.
- Addition is commutative: A + B = B + A. The null matrix O satisfies A + O = A.

Multiplication of Matrices

KEY POINT

Compatibility: AB can be found only if the **number of columns of A = number of rows of B**. If A is m × n and B is n × p, then AB is of order **m × p**.

Each element of AB is a **row of A times a column of B**: multiply matching entries and add. For 2 × 2 matrices:

$$\begin{vmatrix} a & b \\ c & d \end{vmatrix} \begin{vmatrix} p & q \\ r & s \end{vmatrix} = \begin{vmatrix} ap + br & aq + bs \\ cp + dr & cq + ds \end{vmatrix}$$

BOARD ALERT

In general **AB ≠ BA**. Even when both products exist they usually differ. Never swap the order. Also AB = O does not force A = O or B = O.

SIDEKICK TIP

Slide your left index finger along the row of A and your right index finger down the column of B, multiplying pairs as you go. It stops you from picking the wrong entries.

Worked Example 1

Q. State the order of each matrix and say whether it can be added to

$\begin{vmatrix} 1 & 2 \\ 3 & 4 \end{vmatrix}$: (i) $\begin{vmatrix} 5 & 0 & 1 \\ 2 & 2 & 2 \end{vmatrix}$ (ii) $\begin{vmatrix} 7 & 8 \\ 9 & 1 \end{vmatrix}$ (iii) $\begin{vmatrix} 3 \\ 6 \end{vmatrix}$.

- (i) Order 2 × 3. Not the same order as 2 × 2, so not addable.
- (ii) Order 2 × 2. Same order, so addable.
- (iii) Order 2 × 1 (a column matrix). Not addable.

KEY POINT

Answer: Only (ii) can be added.

Worked Example 2

Q. If A = $\begin{vmatrix} 2 & -1 \\ 3 & 4 \end{vmatrix}$ and B = $\begin{vmatrix} 5 & 0 \\ -2 & 1 \end{vmatrix}$, find 3A − 2B.

- 3A = $\begin{vmatrix} 6 & -3 \\ 9 & 12 \end{vmatrix}$ and 2B = $\begin{vmatrix} 10 & 0 \\ -4 & 2 \end{vmatrix}$.
- 3A − 2B = $\begin{vmatrix} -4 & -3 \\ 13 & 10 \end{vmatrix}$.

KEY POINT

Answer: 3A − 2B = $\begin{vmatrix} -4 & -3 \\ 13 & 10 \end{vmatrix}$.

Worked Example 3

Q. Find x and y if

$\begin{vmatrix} x & 2 \\ 3 & y \end{vmatrix} + \begin{vmatrix} 1 & -4 \\ 2 & 5 \end{vmatrix} = \begin{vmatrix} 6 & -2 \\ 5 & 8 \end{vmatrix}$.

- Add on the left: $\begin{vmatrix} x+1 & -2 \\ 5 & y+5 \end{vmatrix} = \begin{vmatrix} 6 & -2 \\ 5 & 8 \end{vmatrix}$.
- x + 1 = 6 so x = 5. y + 5 = 8 so y = 3.

KEY POINT

Answer: x = 5, y = 3.

Worked Example 4

Q. If P = $\begin{vmatrix} 2 & 3 \\ 1 & -1 \end{vmatrix}$ and Q = $\begin{vmatrix} 4 & 0 \\ -2 & 5 \end{vmatrix}$, find PQ and QP. Are they equal?

- PQ: first row (2 × 4 + 3 × (−2), 2 × 0 + 3 × 5) = (2, 15). Second row (1 × 4 + (−1)(−2), 1 × 0 + (−1)(5)) = (6, −5).
- PQ = $\begin{vmatrix} 2 & 15 \\ 6 & -5 \end{vmatrix}$.
- QP: first row (4 × 2 + 0, 4 × 3 + 0) = (8, 12). Second row (−2 × 2 + 5 × 1, −2 × 3 + 5 × (−1)) = (1, −11).
- QP = $\begin{vmatrix} 8 & 12 \\ 1 & -11 \end{vmatrix}$.

KEY POINT

Answer: PQ ≠ QP. Matrix multiplication is not commutative.

Worked Example 5

Q. Find the matrix X if 2X + $\begin{vmatrix} 1 & 3 \\ -2 & 0 \end{vmatrix} = \begin{vmatrix} 7 & 5 \\ 4 & -6 \end{vmatrix}$.

- $2X = \begin{vmatrix} 7 & 5 \\ 4 & -6 \end{vmatrix} - \begin{vmatrix} 1 & 3 \\ -2 & 0 \end{vmatrix} = \begin{vmatrix} 6 & 2 \\ 6 & -6 \end{vmatrix}$.
- Halve every element.

KEY POINT

Answer: X = $\begin{vmatrix} 3 & 1 \\ 3 & -3 \end{vmatrix}$.

Worked Example 6

Q. If A = $\begin{vmatrix} 1 & 2 \\ 3 & -1 \end{vmatrix}$, find A^2 and show that $A^2 = 7I$.

- $A^2 = A \times A = \begin{vmatrix} 7 & 0 \\ 0 & 7 \end{vmatrix}$: (1 + 6, 2 − 2; 3 − 3, 6 + 1).
- $7I = \begin{vmatrix} 7 & 0 \\ 0 & 7 \end{vmatrix}$.

KEY POINT

Answer: $A^2 = 7I$ (equal element by element).

Worked Example 7

Q. Find x and y if $\begin{vmatrix} 2 & 1 \\ 3 & 4 \end{vmatrix} \begin{vmatrix} x \\ y \end{vmatrix} = \begin{vmatrix} 7 \\ 18 \end{vmatrix}$.

- Order 2 × 2 times 2 × 1 gives 2 × 1. Multiply: 2x + y = 7 and 3x + 4y = 18.
- From the first, y = 7 − 2x. Substitute: 3x + 28 − 8x = 18, so x = 2.
- y = 7 − 4 = 3.

KEY POINT

Answer: x = 2, y = 3.

Worked Example 8

Q. Compute $\begin{vmatrix} 3 & -1 \end{vmatrix} \begin{vmatrix} 2 & 4 \\ 1 & 5 \end{vmatrix}$ and state the order of the product.

- Row (1 × 2) times matrix (2 × 2): inner orders match (2 and 2). Result is 1 × 2.
- First element: 3 × 2 + (−1)(1) = 5. Second: 3 × 4 + (−1)(5) = 7.

KEY POINT

Answer: [5 7], of order 1 × 2.

Worked Example 9

Q. If $A = \begin{vmatrix} 3 & 1 \\ -1 & 2 \end{vmatrix}$, find $A^2 - 5A + 7I$.

- $A^2 = \begin{vmatrix} 8 & 5 \\ -5 & 3 \end{vmatrix}$, $5A = \begin{vmatrix} 15 & 5 \\ -5 & 10 \end{vmatrix}$, $7I = \begin{vmatrix} 7 & 0 \\ 0 & 7 \end{vmatrix}$.
- $A^2 - 5A + 7I = \begin{vmatrix} 0 & 0 \\ 0 & 0 \end{vmatrix}$.

KEY POINT

Answer: The null matrix O.

BOARD ALERT

Common errors. (1) Writing the order as columns × rows. It is rows × columns. (2) Multiplying matching elements for a matrix product (that is only for scalars). (3) Trying to multiply when the inner orders differ. (4) Assuming AB = BA.

REAL LIFE

A school canteen records the number of plates sold of three items on two days in a 2 × 3 matrix. Multiplying by a column of prices gives the daily earnings in one step.

HOT QUESTION

If A is a 2 × 3 matrix and B is a matrix such that both AB and BA exist, what must the order of B be? Give one example of each matrix and compute AB.

Mind Map Recap: Matrices

REVISE IN ONE LOOK

Order = rows × columns. Row, column, square, null and identity matrices.

A ± B only for equal orders. **kA** scales every element.

AB exists when columns of A = rows of B; order m × p.

AB ≠ BA in general. AI = IA = A.

Matrix equation: equate matching elements to get ordinary equations.

Practice Set

Try each one on paper first. Answers are at the end of the set.

1. State the order of $\begin{vmatrix} 1 & 4 & 2 \\ 0 & 3 & 5 \end{vmatrix}$ and of $\begin{vmatrix} 6 \\ 7 \\ 8 \end{vmatrix}$.
2. If $A = \begin{vmatrix} 2 & -3 \\ 4 & 1 \end{vmatrix}$ and $B = \begin{vmatrix} -1 & 5 \\ 0 & 2 \end{vmatrix}$, find A + B and A − B.
3. With A and B as in Q2, find 2A + 3B.
4. Find x and y if $\begin{vmatrix} 2x & 3 \\ 1 & y-1 \end{vmatrix} = \begin{vmatrix} 8 & 3 \\ 1 & 5 \end{vmatrix}$.
5. If $R = \begin{vmatrix} 1 & 4 \\ -2 & 3 \end{vmatrix}$ and $S = \begin{vmatrix} 0 & 2 \\ 5 & -1 \end{vmatrix}$, find RS.
6. With R and S as in Q5, find SR and decide whether RS = SR.

7. Find the matrix X such that $X + \begin{vmatrix} 2 & -3 \\ 4 & 1 \end{vmatrix} = \begin{vmatrix} -1 & 5 \\ 0 & 2 \end{vmatrix}$.

8. If $T = \begin{vmatrix} 3 & 1 \\ 2 & 0 \end{vmatrix}$, find T^2.

9. Find x and y if $\begin{vmatrix} 3 & 2 \\ 1 & 4 \end{vmatrix} \begin{vmatrix} x \\ y \end{vmatrix} = \begin{vmatrix} 12 \\ 14 \end{vmatrix}$.

10. If $A = \begin{vmatrix} 1 & 2 \\ 0 & -1 \end{vmatrix}$, show that $A^2 = I$.

11. Find the order of AB if A is 3 × 2 and B is 2 × 4. Is BA defined?

12. Find the matrix X: $3X - \begin{vmatrix} 4 & 2 \\ -1 & 5 \end{vmatrix} = \begin{vmatrix} 2 & 4 \\ 8 & -2 \end{vmatrix}$.

Answers

1. 2 × 3 and 3 × 1.

2. $A + B = \begin{vmatrix} 1 & 2 \\ 4 & 3 \end{vmatrix}$; $A - B = \begin{vmatrix} 3 & -8 \\ 4 & -1 \end{vmatrix}$.

3. $\begin{vmatrix} 1 & 9 \\ 8 & 8 \end{vmatrix}$.

4. $2x = 8$, $y - 1 = 5$: **x = 4, y = 6**.

5. $RS = \begin{vmatrix} 20 & -2 \\ 15 & -7 \end{vmatrix}$.

6. $SR = \begin{vmatrix} -4 & 6 \\ 7 & 17 \end{vmatrix}$. **RS ≠ SR**.

7. $X = B - A = \begin{vmatrix} -3 & 8 \\ -4 & 1 \end{vmatrix}$.

8. $T^2 = \begin{vmatrix} 11 & 3 \\ 6 & 2 \end{vmatrix}$.

9. $3x + 2y = 12$ and $x + 4y = 14$ give **x = 2, y = 3**.

10. $A^2 = \begin{vmatrix} 1 & 0 \\ 0 & 1 \end{vmatrix} = I$.

11. AB is **3 × 4**. BA is not defined (columns of B = 4, rows of A = 3).

12. $3X = \begin{vmatrix} 6 & 6 \\ 7 & 3 \end{vmatrix}$, $X = \begin{vmatrix} 2 & 2 \\ 7/3 & 1 \end{vmatrix}$.

09

CHAPTER · MATHEMATICS

Arithmetic and Geometric Progressions

Patterns that grow by adding or by multiplying. Find any term, add up the first n terms and apply both ideas to savings, populations and everyday growth.



IN THIS CHAPTER

- AP formulae
- GP formulae
- Sum of n terms
- Tn from Sn
- Applications

Arithmetic Progression (AP)

A sequence in which each term is obtained by **adding the same number** to the previous term. That number is the **common difference d**. Example: 7, 11, 15, 19, ... has a = 7 and d = 4.

$$\text{General term: } T_n = a + (n-1)d \quad \text{Sum: } S_n = \frac{n}{2}[2a + (n-1)d] = \frac{n}{2}(a + l)$$

- l is the last term, so $l = a + (n-1)d$.
- Three numbers a, b, c are in AP exactly when **2b = a + c**.
- For three terms in AP, call them **a − d, a, a + d**: the sum is then 3a.



SIDEKICK TIP

$T_n = S_n - S_{n-1}$. If a question gives a formula for S_n, this finds the nth term. Also $T_1 = S_1$.

Geometric Progression (GP)

A sequence in which each term is the previous term **multiplied by the same number**, the **common ratio r**. Example: 3, 6, 12, 24, ... has a = 3 and r = 2. Find r by dividing any term by the one before it.

$$\text{General term: } T_n = ar^{n-1} \quad \text{Sum: } S_n = \frac{a(r^n - 1)}{r - 1} \ (r > 1) \ \text{ or } \ \frac{a(1 - r^n)}{1 - r} \ (r < 1)$$

- Three numbers a, b, c are in GP exactly when **$b^2 = ac$**.
- If r = 1 every term is the same and $S_n = na$.
- Sum to infinity is not in your syllabus; only the first n terms.

	AP	GP
Change from term to term	Add d	Multiply by r
n^{th} term	$a + (n-1)d$	ar^{n-1}
Sum of n terms	$(n/2)[2a + (n-1)d]$	$a(r^n - 1)/(r - 1)$
Test for three terms	$2b = a + c$	$b^2 = ac$
Growth looks like	Straight line	Doubling or halving (exponential)

Worked Example 1

Q. For the AP 7, 11, 15, ... find the 15th term and the term that equals 123.

- a = 7, d = 4. $T_{15} = 7 + 14 \times 4 = 63$.
- $7 + (n-1)4 = 123$, so $(n-1)4 = 116$, $n - 1 = 29$.



KEY POINT

Answer: $T_{15} = 63$ and 123 is the 30th term.

Worked Example 2

Q. Find the sum of the first 20 terms of 3, 8, 13, ...

- a = 3, d = 5, n = 20.
- $S_{20} = 10[2(3) + 19(5)] = 10(6 + 95) = 1010$.



KEY POINT

Answer: $S_{20} = 1010$.

Worked Example 3

Q. In an AP the 4th term is 14 and the 9th term is 34. Find a, d and the sum of 12 terms.

- $a + 3d = 14$ and $a + 8d = 34$. Subtract: $5d = 20$, $d = 4$, $a = 2$.
- $S_{12} = 6[4 + 11 \times 4] = 6 \times 48 = 288$.

KEY POINT
Answer: $a = 2$, $d = 4$, $S_{12} = 288$.

Worked Example 4

Q. Find the sum of all multiples of 7 between 100 and 300.

- First multiple above 100 is 105, last below 300 is 294. AP with $a = 105$, $d = 7$.
- $294 = 105 + (n - 1)7$ gives $n = 28$.
- $S = (28/2)(105 + 294) = 14 \times 399 = 5586$.

KEY POINT
Answer: 5586.

Worked Example 5

Q. The sum of n terms of a sequence is $S_n = 3n^2 + 2n$. Show that it is an AP and find its nth term.

- $T_n = S_n - S_{n-1} = (3n^2 + 2n) - [3(n - 1)^2 + 2(n - 1)]$.
- $= 3n^2 + 2n - 3n^2 + 6n - 3 - 2n + 2 = 6n - 1$.
- T_n is linear in n, so the difference is constant: $d = 6$, and $a = T_1 = 5$.

KEY POINT
Answer: $T_n = 6n - 1$ (an AP: 5, 11, 17, ...).

Worked Example 6

Q. Three numbers in AP have sum 27 and product 693. Find them.

- Let them be $9 - d$, 9, $9 + d$ (sum 27 means middle = 9).
- $9(81 - d^2) = 693$, so $81 - d^2 = 77$, $d^2 = 4$, $d = \pm 2$.

KEY POINT
Answer: 7, 9, 11 (or 11, 9, 7).

Worked Example 7

Q. For the GP 3, 6, 12, ... find the 8th term and the sum of the first 6 terms.

- $a = 3$, $r = 2$. $T_8 = 3 \times 2^7 = 384$.
- $S_6 = 3(2^6 - 1)/(2 - 1) = 3 \times 63 = 189$.

KEY POINT
Answer: $T_8 = 384$; $S_6 = 189$.

Worked Example 8

Q. In a GP the 3rd term is 12 and the 6th term is 96. Find a, r and S_7.

- $ar^2 = 12$ and $ar^5 = 96$. Divide: $r^3 = 8$, so $r = 2$.
- $a = 12/4 = 3$.
- $S_7 = 3(2^7 - 1)/1 = 3 \times 127 = 381$.

KEY POINT
Answer: $a = 3$, $r = 2$, $S_7 = 381$.

Worked Example 9

Q. Which term of the GP 1, 2, 4, 8, ... is 512? Also find the sum of the first 5 terms of 1, 1/2, 1/4, ...

- $2^{n-1} = 512 = 2^9$, so $n - 1 = 9$, $n = 10$.
- Second GP: $a = 1$, $r = 1/2$. $S_5 = (1 - 1/32)/(1 - 1/2) = (31/32) \times 2 = 31/16$.

KEY POINT
Answer: 10th term; $S_5 = 31/16$.

Worked Example 10

Q. Asha saves ₹ 500 in the first month and increases her saving by ₹ 50 every month. How much has she saved in 18 months?

- AP with $a = 500$, $d = 50$, $n = 18$.
- $S_{18} = 9[1000 + 17 \times 50] = 9 \times 1850 = 16{,}650$.

KEY POINT
Answer: ₹ 16,650.

Worked Example 11

Q. A culture has 500 bacteria which double every hour. How many are there after 6 hours?

- GP with $a = 500$ and $r = 2$. After 6 hours it is the 7th term (term 1 is the start).
- $T_7 = 500 \times 2^6 = 32{,}000$.

KEY POINT
Answer: 32,000 bacteria.

BOARD ALERT
Common errors. (1) Using n instead of $n - 1$ in T_n. (2) Using the AP sum formula for a GP. (3) Picking the wrong form of the GP sum: use $(r^n - 1)/(r - 1)$ for $r > 1$ and $(1 - r^n)/(1 - r)$ for $r < 1$; both give the same value. (4) Not rejecting n that is negative or a fraction.

REAL LIFE
A taxi fare that rises by a fixed amount per kilometre is an AP. A loan or investment that grows by a fixed percentage every year is a GP.

HOT QUESTION
The 5th term of an AP is 3 times the 2nd term and the sum of the first 6 terms is 54. Find the AP. Then say what S_n looks like for it.

Mind Map Recap: AP and GP

REVISE IN ONE LOOK

AP: $T_n = a + (n-1)d$; $S_n = (n/2)[2a + (n-1)d]$; test $2b = a + c$.

GP: $T_n = ar^{n-1}$; $S_n = a(r^n-1)/(r-1)$; test $b^2 = ac$.

Two terms given: form two equations. In an AP subtract; in a GP divide.

From S_n: $T_n = S_n - S_{n-1}$.

Three terms in AP: $a-d$, a, $a+d$.

Practice Set

Try each one on paper first. Answers are at the end of the set.

1. Find the 25th term of 4, 9, 14, ...
2. Which term of the AP 5, 11, 17, ... is 107?

3. Find the sum of the first 15 terms of 2, 5, 8, ...
4. The 3rd term of an AP is 11 and the 8th term is 31. Find the AP's first term and common difference.
5. Find the sum of all two-digit numbers divisible by 6.
6. If $S_n = 2n^2 + 5n$, find the 10th term.
7. Three numbers in AP have sum 21 and product 231. Find them.
8. Find the 7th term of the GP 2, 6, 18, ... and the sum of its first 5 terms.
9. The 2nd term of a GP is 6 and the 5th term is 162. Find the GP up to 5 terms.
10. Find the sum of the first 6 terms of 1, −2, 4, ...
11. A ladder has 20 rungs whose lengths decrease uniformly from 80 cm at the bottom to 40 cm at the top. Find the total length of the rungs.
12. The population of a village is 8000 and grows by a factor 1.5 each decade, a GP. Find the population after 3 decades.

Answers

1. $a = 4$, $d = 5$: $4 + 24 \times 5 =$ **124**.

2. $5 + (n - 1)6 = 107$, $n - 1 = 17$: **18th term**.

3. $S = (15/2)[4 + 14 \times 3] = (15/2)(46) =$ **345**.

4. $5d = 20$, $d = 4$, $a = 3$: **a = 3, d = 4**.

5. 12, 18, ... 96: $n = 15$, $S = (15/2)(12 + 96) =$ **810**.

6. $T_{10} = S_{10} - S_9 = 250 - 207 =$ **43**.

7. Middle 7; $7(49 - d^2) = 231$, $d^2 = 16$: **3, 7, 11** (or 11, 7, 3).

8. $T_7 = 2 \times 3^6 =$ **1458**; $S_5 = 2(3^5 - 1)/2 =$ **242**.

9. $r^3 = 27$, $r = 3$, $a = 2$: **2, 6, 18, 54, 162**.

10. $r = -2$: $S_6 = (1 - 64)/(1 + 2) =$ **−21**.

11. AP with $a = 80$, $l = 40$, $n = 20$: $S = 10(120) =$ **1200 cm**.

12. $a = 8000$, $r = 1.5$, after 3 decades is $T_4 = 8000 \times 3.375 =$ **27,000**.

10

CHAPTER · MATHEMATICS

Reflection

A mirror flips a point to the other side, and the rules are just sign changes and one handy formula. Learn them for the axes, the origin and lines parallel to the axes.



IN THIS CHAPTER

- Axes and origin
- Lines x = a and y = a
- Invariant points
- Images of figures

Reflection as a Mirror Image

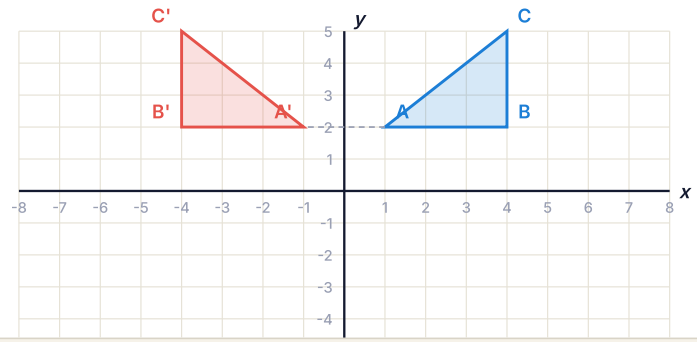
The **reflection** of a point P in a line (the mirror) is a point P' such that the line is the **perpendicular bisector** of PP'. P and P' are the same distance from the mirror, on opposite sides. A figure is reflected by reflecting each of its vertices.

The Reflection Rules

REFLECTION IN	P(X, Y) GOES TO	MIRROR
x-axis (y = 0)	(x, −y)	Horizontal axis
y-axis (x = 0)	(−x, y)	Vertical axis
The origin	(−x, −y)	A point, not a line
Line x = a	(2a − x, y)	Vertical line
Line y = a	(x, 2a − y)	Horizontal line

SIDEKICK TIP

For a mirror x = a, the y-coordinate does not change; only x does. The new x is the mirror position plus (mirror minus old x): x' = a + (a − x) = 2a − x. Same logic for y = a.



Triangle ABC and its image A'B'C' in the y-axis. Each image point is as far behind the mirror as the object is in front.

Invariant Points

A point that does not move under a transformation is **invariant**. Every point on the mirror line is invariant for a line reflection. For reflection in the **origin**, the only invariant point is the origin itself.

KEY POINT

Properties: reflection preserves distances, angles and areas. The image is the same size and shape but laterally inverted (a mirror image), so the order of vertices is reversed in orientation.

Two Reflections in a Row

Reflect in the x-axis and then in the origin: (x, y) → (x, −y) → (−x, y). The result is the same as a single reflection in the **y-axis**. Check combined results this way for quick answers.

Worked Example 1

Q. Find the images of P(3, −5) in (i) the x-axis (ii) the y-axis (iii) the origin.

- x-axis: change the sign of y, giving (3, 5).
- y-axis: change the sign of x, giving (−3, −5).
- Origin: change both signs, giving (−3, 5).

KEY POINT

Answer: (3, 5), (−3, −5) and (−3, 5).

Worked Example 2

Q. Find the image of A(−2, 7) in the line x = 3 and in the line y = −1.

- In x = 3: x' = 2(3) − (−2) = 8, y unchanged: (8, 7).
- In y = −1: y' = 2(−1) − 7 = −9, x unchanged: (−2, −9).

KEY POINT

Answer: (8, 7) and (−2, −9).

Worked Example 3

Q. The point (−4, 6) is reflected in the line x = a and its image is (10, 6). Find a.

- The mirror lies midway between the point and its image.
- a = (−4 + 10)/2 = 3.

KEY POINT

Answer: a = 3.

Worked Example 4

Q. A triangle has vertices A(1, 2), B(4, 2) and C(4, 5). Write the vertices of its image in the y-axis and in the line y = −1.

- y-axis: (x, y) → (−x, y), giving A'(−1, 2), B'(−4, 2), C'(−4, 5).
- Line y = −1: y' = −2 − y, giving A''(1, −4), B''(4, −4), C''(4, −7).

KEY POINT

Answer: Both images listed above (the triangle keeps its size and shape).

Worked Example 5

Q. P(−3, 2) is reflected in the x-axis to Q, and Q is reflected in the origin to R. Find Q and R. Which single reflection maps P to R?

- Q = (−3, −2) (x-axis).
- R = (3, 2) (origin: change both signs).
- P(−3, 2) to R(3, 2): only x has changed sign.

KEY POINT

Answer: Q(−3, −2), R(3, 2); the single reflection is in the y-axis.

Worked Example 6

Q. The image of P(a, b) in the x-axis is (4, −3). Find P and then the image of P in the origin.

- Image in x-axis is (a, −b) = (4, −3), so a = 4 and b = 3: P = (4, 3).
- Image of (4, 3) in the origin is (−4, −3).

KEY POINT

Answer: P = (4, 3); its image in the origin is (−4, −3).

Worked Example 7

Q. Find the length of the segment joining Q(5, −3) to its image in the line y = 1.

- Image: y' = 2(1) − (−3) = 5, so Q' = (5, 5).
- Same x, so distance = 5 − (−3) = 8. (Twice the distance 4 from the point to the mirror.)

KEY POINT

Answer: 8 units.

Worked Example 8

Q. Give two points that are invariant under reflection in the line y = 2 and say whether (0, 0) is invariant under reflection in the origin.

- Any point on y = 2, such as (5, 2) and (−3, 2): y' = 2(2) − 2 = 2, so the point stays.
- Origin: (−0, −0) = (0, 0), so it does not move.

KEY POINT

Answer: (5, 2) and (−3, 2) are invariant; yes, (0, 0) is invariant under reflection in the origin.

BOARD ALERT

Common errors. (1) Changing both signs for an axis reflection. Only one coordinate changes. (2) For x = a, writing (a − x, y) or (x − a, y): the correct image is (2a − x, y). (3) Forgetting that a reflection in the origin is a point reflection, not a line reflection.

REAL LIFE

A lake reflects the shoreline trees upside down. The water line is the mirror: a tree top 8 m above it has an image 8 m below it.

HOT QUESTION

A(−2, 3) is reflected in the line x = 1 to A' and then A' is reflected in the line x = 5 to A''. Find A'' and say how A'' relates to A. What does this suggest about two parallel mirrors?

Mind Map Recap: Reflection

REVISE IN ONE LOOK

x-axis: (x, −y). **y-axis:** (−x, y). **Origin:** (−x, −y).

x = a: (2a − x, y). **y = a:** (x, 2a − y).

Mirror is the perpendicular bisector of PP'.

Invariant points lie on the mirror; for the origin, only (0, 0).

Reflection keeps lengths and angles unchanged.

Practice Set

Try each one on paper first. Answers are at the end of the set.

1. Find the images of A(−6, 4) in the x-axis, the y-axis and the origin.
2. Find the image of B(2, −3) in the line x = −1.
3. Find the image of C(−5, 1) in the line y = 4.
4. A point P is reflected in the y-axis and the image is (7, −2). Find P.
5. The image of (3, 5) in the line x = a is (−1, 5). Find a.
6. Reflect the triangle with vertices (1, 1), (3, 1), (3, 4) in the x-axis. Write the image vertices.
7. Find the image of (−2, 8) in the line y = 3, then reflect the image in the x-axis.
8. If the image of P(a, b) in the origin is (−3, 6), find a and b.
9. Name the single reflection that is equivalent to a reflection in the x-axis followed by a reflection in the y-axis.
10. Find the length of the line joining D(4, 7) to its image in the line x = −2.
11. Write the coordinates of two invariant points for the reflection in the line x = −4.
12. A(3, 2), B(7, 2), C(5, 6) form a triangle. Find the image of the triangle in the line x = 1.

Answers

1. **(−6, −4), (6, 4), (6, −4)**.

2. x' = −2 − 2 = −4: **(−4, −3)**.

3. y' = 8 − 1 = 7: **(−5, 7)**.

4. **(−7, −2)**.

5. a = (3 + (−1))/2 = **1**.

6. **(1, −1), (3, −1), (3, −4)**.

7. Image in y = 3: (−2, −2). In x-axis: **(−2, 2)**.

8. (−a, −b) = (−3, 6): **a = 3, b = −6**.

9. (x, y) → (x, −y) → (−x, −y): **reflection in the origin**.

10. Image: (−4 − 4, 7) = (−8, 7). Distance = 4 − (−8) = **12**.

11. Any two points with x = −4, e.g. **(−4, 0) and (−4, 9)**.

12. x' = 2 − x: **A'(−1, 2), B'(−5, 2), C'(−3, 6)**.

11

CHAPTER · MATHEMATICS

Section and Mid-point Formula

Locate a point that splits a segment in a given ratio, find the middle, find the centroid, and use all three to crack coordinate-geometry puzzles.

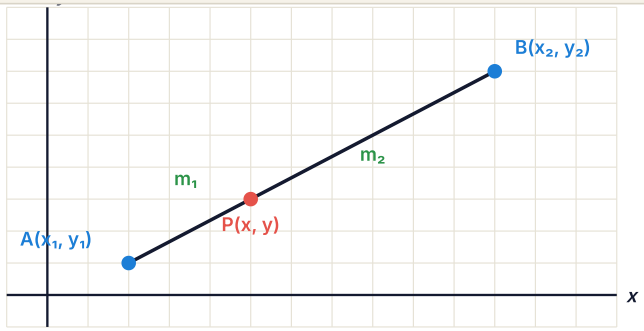


IN THIS CHAPTER

- Section formula
- Mid-point
- Centroid
- Finding ratio
- Parallelograms

The Section Formula

A point P on the segment AB **divides it internally** in the ratio $m_1 : m_2$ if $AP : PB = m_1 : m_2$. The coordinates of P are a weighted average of those of A and B. Your syllabus covers **internal division only**.



P divides AB internally in the ratio $m_1 : m_2$.

$$\text{If } A(x_1, y_1), B(x_2, y_2): \quad P = \left(\frac{m_1x_2 + m_2x_1}{m_1 + m_2}, \frac{m_1y_2 + m_2y_1}{m_1 + m_2} \right)$$

SIDEKICK TIP

Memory trick: **cross-multiply the ratio with the far end**. The part m_1 (nearest A) multiplies the coordinates of B, and m_2 multiplies those of A. This is why a point closer to A has a larger m_2.

The Mid-point Formula

The mid-point is the special case $m_1 = m_2 = 1$ (ratio 1 : 1).

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

The Centroid of a Triangle

The three medians of a triangle meet at the **centroid G**, which divides each median in the ratio 2 : 1 counting from the vertex.

$$G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$$

Finding the Ratio

If P is known, put the ratio as k : 1 (or $m_1 : m_2$), equate one coordinate of the formula to the given value and solve. A line that crosses the x-axis has y = 0 at that point; one that crosses the y-axis has x = 0.

KEY POINT

Parallelogram trick: the diagonals of a parallelogram bisect each other, so the mid-point of AC equals the mid-point of BD. Use it to find a missing vertex.

Worked Example 1

Q. Find the mid-point of the line joining (−4, 7) and (10, −1).

- x = (−4 + 10)/2 = 3.
- y = (7 + (−1))/2 = 3.

KEY POINT

Answer: (3, 3)

Worked Example 2

Q. Find the point P that divides A(2, −3) and B(8, 6) internally in the ratio 1 : 2.

- $m_1 = 1$, $m_2 = 2$. $x = (1 \times 8 + 2 \times 2)/3 = 12/3 = 4$.
- $y = (1 \times 6 + 2 \times (-3))/3 = 0$.

KEY POINT

Answer: P = (4, 0)

Worked Example 3

Q. In what ratio does the x-axis divide the line joining A(3, −4) and B(7, 6)? Find the point of division.

- On the x-axis $y = 0$. Let the ratio be $k : 1$: $y = (6k + (-4))/(k + 1) = 0$, so $6k = 4$, $k = 2/3$, ratio 2 : 3.
- $x = (2 \times 7 + 3 \times 3)/5 = 23/5 = 4.6$.

KEY POINT

Answer: Ratio 2 : 3 at the point (4.6, 0)

Worked Example 4

Q. P(5, 7) lies on the line joining A(−1, 1) and B(8, 10). Find the ratio in which P divides AB.

- Let the ratio be $m : n$. x: $(8m - n)/(m + n) = 5$, so $8m - n = 5m + 5n$, $3m = 6n$, $m : n = 2 : 1$.
- Check with y: $(10 \times 2 + 1 \times 1)/3 = 7$ ✓

KEY POINT

Answer: 2 : 1

Worked Example 5

Q. Find the centroid of the triangle with vertices A(1, 5), B(7, −1) and C(−2, 8).

- $x = (1 + 7 - 2)/3 = 2$.
- $y = (5 - 1 + 8)/3 = 4$.

KEY POINT

Answer: G = (2, 4)

Worked Example 6

Q. Two vertices of a triangle are (3, 1) and (5, −4) and the centroid is (4, 2). Find the third vertex.

- Let the third vertex be (a, b). $(3 + 5 + a)/3 = 4$ gives $a = 4$.
- $(1 - 4 + b)/3 = 2$ gives $b = 9$.

KEY POINT

Answer: (4, 9)

Worked Example 7

Q. M(2, −1) is the mid-point of AB and A is (−4, 5). Find B.

- $(-4 + x)/2 = 2$ so $x = 8$.
- $(5 + y)/2 = -1$ so $y = -7$.

KEY POINT

Answer: B = (8, −7)

Worked Example 8

Q. Points P and Q trisect the segment joining A(1, −2) and B(7, 4). Find P and Q.

- P divides AB in 1 : 2: x = (7 + 2)/3 = 3, y = (4 − 4)/3 = 0, so P = (3, 0).
- Q divides AB in 2 : 1: x = (14 + 1)/3 = 5, y = (8 − 2)/3 = 2, so Q = (5, 2).

KEY POINT
Answer: P(3, 0), Q(5, 2)

Worked Example 9

Q. A(1, 2), B(5, 3), C(8, 7) are three vertices of a parallelogram ABCD in order. Find D.

- Diagonals AC and BD bisect each other. Mid-point of AC = (4.5, 4.5).
- If D = (x, y): (5 + x)/2 = 4.5 so x = 4, and (3 + y)/2 = 4.5 so y = 6.

KEY POINT
Answer: D = (4, 6). Check: AB = (4, 1) = DC.

Worked Example 10

Q. The y-axis divides the line joining A(−3, 5) and B(6, −1). Find the ratio and the point.

- On the y-axis x = 0. With ratio k : 1: (6k − 3)/(k + 1) = 0, k = 1/2, ratio 1 : 2.
- y = (1 × (−1) + 2 × 5)/3 = 3.

KEY POINT
Answer: Ratio 1 : 2 at (0, 3)

BOARD ALERT
Common errors. (1) Writing $m_1x_1 + m_2x_2$ (wrong pairing). m_1 goes with x_2. (2) Dividing by $m_1 - m_2$ or by 2 for a non-1:1 ratio. The divisor is $m_1 + m_2$. (3) Using the mid-point formula for the centroid: centroid divides by 3. (4) Not checking the second coordinate when the ratio is found from the first.

REAL LIFE
A map app finds the mid-point between two friends so you can meet halfway. Splitting a route 2 : 1 to put a rest stop nearer to one end is the section formula.

HOT QUESTION
If G(3, −2) is the centroid of a triangle with vertices (a, 1), (7, b) and (4, −3), find a and b. Then find the mid-point of the side joining (a, 1) and (7, b).

Mind Map Recap: Section and Mid-point

REVISE IN ONE LOOK

Section (internal): $((m_1x_2 + m_2x_1)/(m_1 + m_2), (m_1y_2 + m_2y_1)/(m_1 + m_2))$.

Mid-point: average the coordinates.

Centroid: sum of the x's over 3 and sum of the y's over 3.

Axis crossing: set y = 0 or x = 0 and find the ratio.

Parallelogram: both diagonals have the same mid-point.

Practice Set

Try each one on paper first. Answers are at the end of the set.

1. Find the mid-point of the segment joining (6, −3) and (−2, 9).

2. Find the point dividing (−1, 4) and (5, −2) in the ratio 2 : 1.
3. In what ratio does the point (1, 3) divide the join of (−1, 1) and (4, 6)?
4. Find the centroid of the triangle (2, 3), (−4, 6) and (8, −3).
5. The centroid is (3, 1) and two vertices are (1, 5) and (6, −3). Find the third vertex.
6. The mid-point of AB is (1, 4) and A = (−3, 7). Find B.
7. Find the ratio in which the x-axis divides the line joining (2, −3) and (5, 6) and the point of division.
8. Find the points that trisect the line joining (0, 3) and (9, 12).
9. Three vertices of a parallelogram PQRS taken in order are P(−2, 1), Q(3, 0) and R(6, 4). Find S.
10. Find the ratio in which the y-axis divides the line joining (−4, 2) and (8, 5).
11. P(a, b) divides the line joining (3, 9) and (11, 1) internally in the ratio 3 : 1. Find a and b.
12. If (4, 3) is the mid-point of the segment joining (a, 5) and (7, b), find a and b.

Answers

1. **(2, 3)**.

2. x = (2 × 5 + 1 × (−1))/3 = 3, y = (2 × (−2) + 4)/3 = 0: **(3, 0)**.

3. x: (4m − n)/(m + n) = 1, 3m = 2n: **2 : 3**. (y: (6 × 2 + 3)/5 = 3 ✓)

4. x = 6/3 = 2, y = 6/3 = 2: **(2, 2)**.

5. x: 1 + 6 + a = 9, a = 2; y: 5 − 3 + b = 3, b = 1: **(2, 1)**.

6. **(5, 1)**.

7. y = 0: (6k − 3)/(k + 1) = 0, k = 1/2: ratio **1 : 2**; x = (5 + 4)/3 = 3: **(3, 0)**.

8. **(3, 6) and (6, 9)**.

9. Diagonals PR and QS: mid-point of PR = (2, 2.5); S = (4 − 3, 5 − 0) = **(1, 5)**.

10. (8k − 4)/(k + 1) = 0, k = 1/2: **1 : 2**.

11. x = (3 × 11 + 1 × 3)/4 = 9, y = (3 × 1 + 9)/4 = 3: **a = 9, b = 3**.

12. (a + 7)/2 = 4, (5 + b)/2 = 3: **a = 1, b = 1**.

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CHAPTER · MATHEMATICS

Equation of a Line

Slope is the heart of every line. Build equations from a point and a slope, from two points, and use slopes to test parallel, perpendicular and collinear.



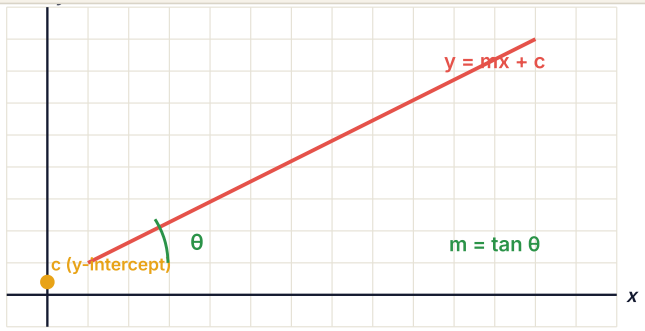
IN THIS CHAPTER

- Slope and tan θ
- y = mx + c
- Two-point form
- Parallel and perpendicular
- Intercepts

Slope of a Line

The **slope (gradient)** m measures how steep a line is: the rise divided by the run. If a line makes an angle θ with the positive direction of the x-axis, then **m = tan θ**.

$$m = \frac{y_2 - y_1}{x_2 - x_1} = \tan\theta$$



The slope m is tan θ, where θ is the angle the line makes with the positive x-axis.

- Line going **up** to the right: m > 0 (θ acute). Line going **down**: m < 0 (θ obtuse).
- Horizontal line (parallel to the x-axis): m = 0. Equation y = k.
- Vertical line (parallel to the y-axis): slope is not defined. Equation x = k.

Equation Forms

FORM	EQUATION	USE WHEN YOU KNOW
Slope-intercept	$y = mx + c$	Slope m and the y-intercept c
Point-slope	$y - y_1 = m(x - x_1)$	Slope m and one point (x_1, y_1)
Two-point	$y - y_1 = \frac{y_2 - y_1}{x_2 - x_1}(x - x_1)$	Two points on the line
General	$ax + by + c = 0$	Convert to $y = mx + c$: $m = -a/b$

KEY POINT

c is the y-intercept: the ordinate of the point where the line meets the y-axis, i.e. the point on the line where x = 0.

Parallel and Perpendicular Lines

$$\text{Parallel: } m_1 = m_2 \qquad \text{Perpendicular: } m_1 \times m_2 = -1$$

- Perpendicular slope is the **negative reciprocal**: if m = 2/3, the perpendicular has slope −3/2.
- Three points are **collinear** if the slope of AB equals the slope of BC.
- A point lies on a line if its coordinates satisfy the equation.

SIDEKICK TIP

To find where a line cuts the axes, put y = 0 for the x-intercept and x = 0 for the y-intercept.

Worked Example 1

Q. Find the slope of the line through (2, −3) and (6, 5). Find also the angle that the line through (1, 2) and (4, 5) makes with the x-axis.

- Slope = (5 − (−3))/(6 − 2) = 8/4 = 2.

- Second line: m = (5 − 2)/(4 − 1) = 1, so tan θ = 1 and θ = 45°.

KEY POINT
Answer: Slope 2; angle 45°.

Worked Example 2

Q. Find the equation of the line with slope −3 passing through (2, 5).

- y − 5 = −3(x − 2)
- y − 5 = −3x + 6, so y = −3x + 11.

KEY POINT
Answer: y = −3x + 11, i.e. 3x + y − 11 = 0.

Worked Example 3

Q. Find the equation of the line through (−1, 4) and (3, −4).

- m = (−4 − 4)/(3 + 1) = −2.
- y − 4 = −2(x + 1) = −2x − 2, so y = −2x + 2.

KEY POINT
Answer: y = −2x + 2 (slope −2, y-intercept 2).

Worked Example 4

Q. Find the slope and y-intercept of 4x − 3y + 12 = 0.

- 3y = 4x + 12, so y = (4/3)x + 4.

KEY POINT
Answer: Slope 4/3, y-intercept 4.

Worked Example 5

Q. Find the line through (3, −2) parallel to 2x + 5y − 7 = 0.

- Slope of the given line = −2/5, so the required slope is −2/5.
- y + 2 = (−2/5)(x − 3). Multiply by 5: 5y + 10 = −2x + 6.

KEY POINT
Answer: 2x + 5y + 4 = 0.

Worked Example 6

Q. Find the line through (4, 1) perpendicular to 3x − 2y + 5 = 0.

- Given slope = 3/2, so the perpendicular slope = −2/3.
- y − 1 = (−2/3)(x − 4). Multiply by 3: 3y − 3 = −2x + 8.

KEY POINT
Answer: 2x + 3y − 11 = 0.

Worked Example 7

Q. Find k if the lines kx + 3y − 5 = 0 and 2x − y + 7 = 0 are (i) perpendicular (ii) parallel.

- Slopes: first = −k/3, second = 2.
- Perpendicular: (−k/3)(2) = −1, so k = 3/2.
- Parallel: −k/3 = 2, so k = −6.

KEY POINT
Answer: (i) k = 3/2 (ii) k = −6.

Worked Example 8

Q. A(1, 2), B(3, 6), C(5, k) are collinear. Find k.

- Slope AB = (6 − 2)/(3 − 1) = 2.
- Slope BC = (k − 6)/(5 − 3) = (k − 6)/2 = 2, so k = 10.

KEY POINT
Answer: k = 10.

Worked Example 9

Q. Find the equation of the perpendicular bisector of the line joining A(2, 1) and B(6, 5).

- Mid-point = (4, 3). Slope AB = 1, so the perpendicular slope = −1.
- y − 3 = −(x − 4), so x + y − 7 = 0.

KEY POINT
Answer: x + y = 7.

Worked Example 10

Q. The line 3x + 4y = 24 meets the axes at P and Q. Find P, Q and the area of the triangle POQ (O is the origin).

- x-intercept (y = 0): x = 8, P = (8, 0). y-intercept (x = 0): y = 6, Q = (0, 6).
- Area = ½ × 8 × 6 = 24.

KEY POINT
Answer: P(8, 0), Q(0, 6); area 24 square units.

BOARD ALERT
Common errors. (1) Taking the slope of ax + by + c = 0 as a/b instead of −a/b. (2) Mixing coordinates in the slope: subtract in the **same order** top and bottom. (3) For perpendicular lines, forgetting to change the sign as well as flipping. (4) Writing the equation of a horizontal line as x = k.

REAL LIFE
A ramp with slope 1/12 rises 1 m for every 12 m of ground. Road signs show slopes as percentages: 8% means m = 0.08.

HOT QUESTION
A line through (−2, 3) makes an angle of 45° with the positive x-axis. Find its equation. Another line is perpendicular to it through the origin: find where the two lines meet.

Mind Map Recap: Equation of a Line

REVISE IN ONE LOOK

$m = (y_2 - y_1)/(x_2 - x_1) = \tan\theta$.

$y = mx + c$; **$y - y_1 = m(x - x_1)$**; two-point form from the slope.

Parallel: $m_1 = m_2$. **Perpendicular:** $m_1 m_2 = -1$.

Collinear: equal slopes for AB and BC.

Intercepts: y = 0 gives x-intercept; x = 0 gives y-intercept.

Practice Set

Try each one on paper first. Answers are at the end of the set.

1. Find the slope of the line through (−2, 3) and (4, −9).

2. Find the equation of the line with slope 3 through (1, −2).
3. Find the equation of the line through (2, 5) and (−2, −3).
4. Find the slope and y-intercept of $5x + 2y - 10 = 0$.
5. Find the line through (2, 1) parallel to $y = 3x - 4$.
6. Find the line through (0, 3) perpendicular to $2x + y = 7$.
7. Find k so that $3x + ky = 4$ and $6x - 2y = 9$ are parallel.
8. Find k so that $kx - 3y + 4 = 0$ and $6x + 2y - 5 = 0$ are perpendicular.
9. Find k if (−1, −3), (2, 3) and (k, 9) are collinear.
10. Find the equation of the line through (0, −4) making an angle of 45° with the positive x-axis.
11. Find the equation of the line that cuts the y-axis at (0, 5) and passes through (−2, −1).
12. Find the equation of the perpendicular bisector of the join of (−1, 2) and (5, −6).

Answers

1. m = −12/6 = **−2**.

2. y + 2 = 3(x − 1): **y = 3x − 5**.

3. m = 8/4 = 2: y − 5 = 2(x − 2): **y = 2x + 1**.

4. y = −(5/2)x + 5: slope **−5/2**, intercept **5**.

5. y − 1 = 3(x − 2): **y = 3x − 5**.

6. Given slope −2, so required slope 1/2: **y = x/2 + 3**.

7. −3/k = 3: **k = −1**.

8. (k/3)(−3) = −1: **k = 1**.

9. Slope of first two = 2; (9 − 3)/(k − 2) = 2: **k = 5**.

10. m = tan 45° = 1: **y = x − 4**.

11. m = (−1 − 5)/(−2 − 0) = 3: **y = 3x + 5**.

12. Mid-point (2, −2), slope of join = −4/3, so perpendicular slope 3/4: y + 2 = (3/4)(x − 2): **3x − 4y − 14 = 0**.

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CHAPTER · MATHEMATICS

Similarity

Same shape, different size. Prove similarity with AA, SSS or SAS, use the Basic Proportionality Theorem, and square the scale factor when areas are involved.



IN THIS CHAPTER

- AA, SSS, SAS
- Basic Proportionality Theorem
- Areas of similar triangles
- Maps and models

Similar Figures

Two figures are **similar** when they have the **same shape** but not necessarily the same size. Think of a photograph and its enlargement: angles stay the same and lengths are all scaled by the same factor.

	CONGRUENT	SIMILAR
Shape	Same	Same
Size	Same	May differ
Corresponding angles	Equal	Equal
Corresponding sides	**Equal**	**Proportional**
Symbol	≡	~

Tests for Similar Triangles

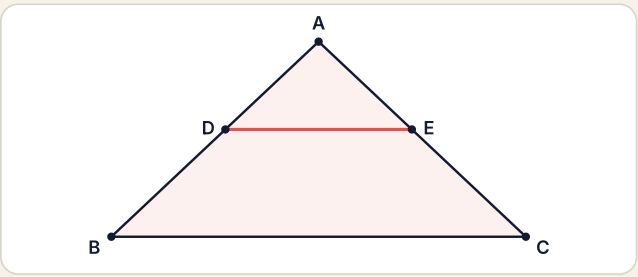
- **AA:** two angles of one triangle equal two angles of the other (the third then matches too).
- **SSS:** all three pairs of corresponding sides are in the same ratio.
- **SAS:** two pairs of sides are in the same ratio and the angles between them are equal.

When ΔABC ~ ΔPQR, the order of the letters matters: A matches P, B matches Q, C matches R, and AB/PQ = BC/QR = CA/RP.

Basic Proportionality Theorem (BPT)

KEY POINT

BPT (Thales): a line drawn parallel to one side of a triangle, meeting the other two sides, divides them in the **same ratio**. **Converse:** if a line divides two sides in the same ratio, it is parallel to the third side.



DE || BC: AD/DB = AE/EC, and triangle ADE is similar to triangle ABC.

$$DE \parallel BC \Leftrightarrow \frac{AD}{DB} = \frac{AE}{EC}$$

Areas of Similar Triangles

If two triangles are similar, the **ratio of their areas equals the square of the ratio of their corresponding sides**.

$$\frac{\text{Area of } \Delta ABC}{\text{Area of } \Delta PQR} = \frac{AB^2}{PQ^2} = \frac{BC^2}{QR^2} = \frac{CA^2}{RP^2}$$

Example idea. Two similar triangles have sides in the ratio 3 : 5. Their areas are in the ratio 9 : 25. If the smaller has area 18 cm², the larger has 18 × 25/9 = 50 cm².

BOARD ALERT

Do not square the ratio of areas. The ratio of **sides** is squared to get the ratio of areas. To go back from areas to sides, take the **square root**.

Maps and Models

- A map with scale 1 : n for lengths has areas in the ratio 1 : n^2.
- A model and the real object are similar: use the scale factor for lengths and its square for areas.
- Convert units at the end: 1 km = 100,000 cm, so 1 km^2 = 10^{10} cm^2.

Worked Example 1

Q. $\Delta ABC \sim \Delta PQR$ with AB = 6, BC = 9 and PQ = 8. Find QR.

- AB/PQ = BC/QR, so 6/8 = 9/QR.
- QR = 9 × 8/6 = 12.

KEY POINT
Answer: QR = 12.

Worked Example 2

Q. In ΔABC, DE || BC with D on AB and E on AC. If AD = 3 cm, DB = 5 cm and AE = 4.5 cm, find EC.

- BPT: AD/DB = AE/EC, so 3/5 = 4.5/EC.
- EC = 4.5 × 5/3 = 7.5.

KEY POINT
Answer: EC = 7.5 cm.

Worked Example 3

Q. Two similar triangles have areas 81 cm^2 and 144 cm^2. A side of the first is 6 cm. Find the corresponding side of the second.

- Ratio of areas 81 : 144, so ratio of sides = $\sqrt{81} : \sqrt{144}$ = 9 : 12 = 3 : 4.
- Side = 6 × 4/3 = 8.

KEY POINT
Answer: 8 cm.

Worked Example 4

Q. In trapezium ABCD, AB || DC and the diagonals meet at O. If AB = 6, CD = 9 and OA = 4, find OC.

- $\Delta AOB \sim \Delta COD$ (alternate angles equal, vertically opposite angles equal: AA).
- AB/CD = OA/OC, so 6/9 = 4/OC and OC = 6.

KEY POINT
Answer: OC = 6.

Worked Example 5

Q. A pole 5 m high casts a shadow 3.2 m long. At the same time a tower casts a shadow 19.2 m long. How tall is the tower?

- Sun's rays make equal angles, so the two right triangles are similar.
- 5/3.2 = h/19.2, so h = 5 × 19.2/3.2 = 30.

KEY POINT
Answer: 30 m.

Worked Example 6

Q. In ΔABC, DE || BC with AD : DB = 2 : 3. If the area of ΔABC is 75 cm^2, find the area of ΔADE and of the trapezium DBCE.

- AD/AB = 2/5. $\Delta ADE \sim \Delta ABC$, so area ratio = $(2/5)^2$ = 4/25.
- Area of ADE = 75 × 4/25 = 12. Trapezium = 75 − 12 = 63.

KEY POINT

Answer: 12 cm² and 63 cm².

Worked Example 7

Q. On a map of scale 1 : 50,000 a road is 6 cm long and a lake covers 8 cm². Find the actual road length and the lake's actual area.

- Road: 6 × 50,000 = 300,000 cm = 3 km.
- Area scale = $50{,}000^2 = 2.5 \times 10^9$. Lake = $8 \times 2.5 \times 10^9 = 2 \times 10^{10}$ cm² = 2 km².

KEY POINT

Answer: 3 km and 2 km².

Worked Example 8

Q. A model of a building has scale 1 : 200. The actual floor area is 5000 m². Find the area of the floor on the model and the actual height if the model is 15 cm high.

- Height: 15 × 200 = 3000 cm = 30 m.
- Area on model = $5000/200^2 = 5000/40{,}000 = 0.125$ m² = 1250 cm².

KEY POINT

Answer: Actual height 30 m; model floor 1250 cm².

Worked Example 9

Q. In ΔABC right-angled at B, BD ⊥ AC. If AD = 4 and DC = 9, find BD and show that $AB^2 + BC^2 = AC^2$.

- ΔADB ~ ΔBDC, so AD/BD = BD/DC and $BD^2 = 4 \times 9 = 36$, BD = 6.
- $AB^2 = AD \times AC = 4 \times 13 = 52$ and $BC^2 = DC \times AC = 9 \times 13 = 117$.
- $AB^2 + BC^2 = 169 = 13^2$.

KEY POINT

Answer: BD = 6; verified.

BOARD ALERT

Common errors. (1) Matching the wrong corresponding sides: write the similarity with letters in order and use the same order for sides. (2) Applying BPT as AD/AB = AE/EC (mixed parts and wholes). Use parts to parts: AD/DB = AE/EC, or wholes to wholes: AD/AB = AE/AC. (3) Forgetting to square for areas.

REAL LIFE

Architects test designs with scale models. If the model is 1/100 of the real size, its surface (and so the paint needed) is 1/10,000 of the real amount.

HOT QUESTION

In ΔABC, D and E are on AB and AC with DE || BC. If AD = x, DB = x − 2, AE = x + 2 and EC = x − 1, find x and verify the ratios.

Mind Map Recap: Similarity

REVISE IN ONE LOOK

Similar: equal angles, proportional sides. Tests: **AA, SSS, SAS**.

BPT: DE || BC ⇔ AD/DB = AE/EC.

Areas: ratio of areas = (ratio of sides)².

Maps and models: lengths use scale k, areas use k^2.

Right triangle altitude to hypotenuse: $BD^2 = AD \times DC$.

Practice Set

Try each one on paper first. Answers are at the end of the set.

1. $\Delta ABC \sim \Delta DEF$ with AB = 4, BC = 6, CA = 8 and DE = 6. Find EF and FD.
2. In ΔABC, DE $\parallel$ BC. If AD = 2.4, DB = 3.6 and AC = 10, find AE.
3. The areas of two similar triangles are 49 cm² and 100 cm². A side of the first is 14 cm. Find the corresponding side of the second.
4. Two similar triangles have corresponding sides in the ratio 2 : 3. If the smaller triangle's area is 20 cm², find the larger one's area.
5. A girl 1.8 m tall casts a shadow 2.4 m. At the same time a tree's shadow is 14 m. Find the height of the tree.
6. In ΔABC, DE $\parallel$ BC and area of ΔADE : area of trapezium DBCE = 1 : 3. If BC = 12 cm, find DE.
7. On a map of scale 1 : 25,000, a park has an area of 256 cm². Find its actual area in km².
8. Are the triangles with sides 5, 12, 13 and 15, 36, 39 similar? Give the reason.
9. ABCD is a trapezium with AB $\parallel$ DC and the diagonals meet at O. If AB = 8, DC = 12 and OB = 5, find OD.
10. A plot is shown on a plan of scale 1 : 500 with an area of 20 cm². Find the actual area in m².
11. ΔABC has DE $\parallel$ BC with AD = x, DB = x − 2, AE = x + 2 and EC = x − 1. Find x.
12. In ΔABC, $\angle B = 90°$ and BD $\perp$ AC. If AD = 3 and DC = 12, find BD, AB and BC.

Answers

1. Ratio 6/4 = 3/2: EF = 6 × 3/2 = **9**, FD = 8 × 3/2 = **12**.

2. AD/AB = 2.4/6 = 2/5, so AE = 2/5 × 10 = **4**.

3. Sides ratio 7 : 10: **20 cm**.

4. Area ratio 4 : 9: 20 × 9/4 = **45 cm²**.

5. 1.8/2.4 = h/14: h = **10.5 m**.

6. Area ADE : ABC = 1 : 4, so DE : BC = 1 : 2: DE = **6 cm**.

7. 256 × (25,000)² = 1.6×10^{11} cm² = **16 km²**.

8. Ratios 15/5 = 36/12 = 39/13 = 3: **similar by SSS**.

9. OB/OD = AB/DC = 2/3: OD = 5 × 3/2 = **7.5**.

10. 20 × 250,000 cm² = 5,000,000 cm² = **500 m²**.

11. $x/(x-2) = (x+2)/(x-1)$: $x^2 - x = x^2 - 4$: **x = 4**.

12. $BD^2 = 3 \times 12 = 36$: **BD = 6**; $AB^2 = 3 \times 15 = 45$: **AB = 3√5**; $BC^2 = 12 \times 15 = 180$: **BC = 6√5**.

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CHAPTER · MATHEMATICS

Loci

A locus is simply the path of a point that obeys a rule. Learn the three standard loci, how to combine them and how to describe each in words.



IN THIS CHAPTER

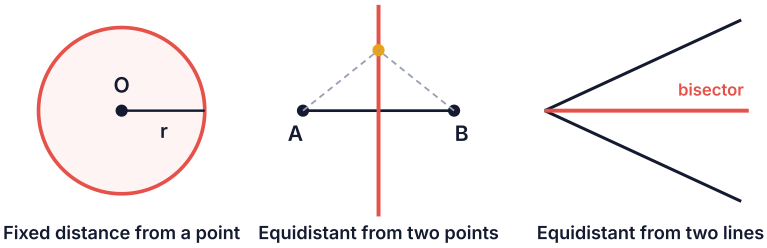
- Circle locus
- Perpendicular bisector
- Angle bisectors
- Combining loci

What a Locus Is

A **locus** (plural: loci) is the set of all the points that satisfy a given condition, and no other points. It is the **path** traced by a point that moves according to a rule. We draw it, then describe it in words.

To find a locus: (1) mark a few points that obey the rule, (2) mark more until a pattern appears, (3) join them, and (4) describe the shape in words with its measurements.

The Three Basic Loci



Fixed distance from a point | Equidistant from two points | Equidistant from two lines

CONDITION	LOCUS	HOW TO DRAW IT
At a fixed distance r from a fixed point O	A **circle** with centre O and radius r	Compasses, set to r
Equidistant from two fixed points A and B	The **perpendicular bisector** of AB	Two arcs of equal radius from A and B; join their crossing points
Equidistant from two intersecting lines	The two **angle bisectors** of the lines (they are perpendicular to each other)	Bisect each of the angles formed

More Loci Built from These

- **At a fixed distance d from a line:** two lines parallel to it, one on each side, each d away.
- **Equidistant from two parallel lines:** a parallel line midway between them.
- **Centre of a circle touching two parallel lines** d apart: the mid-parallel; the radius is d/2.
- **Circumcentre of a triangle** (equidistant from all three vertices): where the perpendicular bisectors of the sides meet.
- **Incentre of a triangle** (equidistant from all three sides): where the angle bisectors meet.

KEY POINT

Most exam questions combine two loci. The answer is the **point(s) where the two loci cross**. Draw each locus lightly, then mark the intersection clearly.

Constructions You Must Be Able to Do

1. Perpendicular bisector of AB
Open compasses to more than half of AB. Draw arcs from A and B above and below the line. Join the two crossing points.

2. Bisector of angle PQR
Draw an arc from Q cutting both arms. From each cutting point draw arcs of equal radius to cross. Join Q to the crossing.

3. Parallel line at distance d
At two points on the line draw perpendiculars, mark d on each and join the marks.

4. Circle through three points
Perpendicular bisectors of two chords meet at the centre.

Worked Example 1

Q. State the locus of a point P that is 3 cm from a fixed point O, and find the area enclosed.

- Fixed distance from a fixed point: a circle with centre O and radius 3 cm.
- Area = $\pi r^2 = \pi \times 9 = 28.3\ cm^2$ (taking $\pi = 3.14$).

KEY POINT
Answer: A circle of radius 3 cm about O; area about 28.3 cm^2.

Worked Example 2

Q. A and B are 6 cm apart. Find the distance from A of the point P on the perpendicular bisector of AB that is 4 cm from the line AB.

- The bisector passes through the mid-point M, with AM = 3 cm and PM = 4 cm, at right angles to AB.
- $PA^2 = AM^2 + PM^2 = 9 + 16 = 25$.

KEY POINT
Answer: PA = 5 cm (and PB = 5 cm, as it must be).

Worked Example 3

Q. Points are 3 cm from a point A and also 2.5 cm from a line l, where A is 4 cm from l. How many such points are there?

- Locus 1: circle, centre A, radius 3 cm. Locus 2: two lines parallel to l, 2.5 cm on each side.
- The parallel on A's side is 4 − 2.5 = 1.5 cm from A, which is less than 3, so it cuts the circle at 2 points.
- The other parallel is 4 + 2.5 = 6.5 cm from A, more than 3 cm, so it misses the circle.

KEY POINT
Answer: Exactly 2 points.

Worked Example 4

Q. Find the radius of the circle passing through all three vertices of a triangle with sides 6 cm, 8 cm and 10 cm.

- Since $6^2 + 8^2 = 10^2$, the triangle is right-angled.
- The circumcentre (intersection of perpendicular bisectors) lies at the mid-point of the hypotenuse.
- Radius = 10/2 = 5.

KEY POINT
Answer: 5 cm.

Worked Example 5

Q. Describe the locus of the centre of a circle that touches two parallel lines 8 cm apart, and give the radius.

- The centre is equidistant from both lines, so it lies on the parallel line midway between them.
- Distance from each line = 4 cm = radius.

KEY POINT
Answer: A line parallel to both, 4 cm from each; radius 4 cm.

Worked Example 6

Q. In ΔABC a point P is equidistant from AB and AC and also from B and C. Describe how to find P.

- Equidistant from AB and AC: P lies on the bisector of angle A.
- Equidistant from B and C: P lies on the perpendicular bisector of BC.
- P is where these two lines intersect.

KEY POINT
Answer: Intersection of the bisector of $\angle A$ and the perpendicular bisector of BC.

Worked Example 7

Q. A goat is tied to a stake by a rope 7 m long and can graze in a circle. How much area can it reach ($\pi = 22/7$)?

- Locus of the goat's farthest point: a circle of radius 7 m.

- Area = (22/7) × 7 × 7 = 154 m².

KEY POINT
Answer: 154 m².

Worked Example 8

Q. Find the locus of points 3 cm from a straight line l and give the distance between the loci.

- Two lines parallel to l, one on each side, each 3 cm from l.
- Distance between them = 3 + 3 = 6 cm.

KEY POINT
Answer: Two parallels 3 cm from l; they are 6 cm apart.

BOARD ALERT
Common errors. (1) Giving only one parallel line for "at a fixed distance from a line". There are two. (2) For two intersecting lines, giving one bisector only. There are two, at right angles. (3) Leaving construction arcs off: examiners want to see them. (4) Answering with a drawing only. Always **state the locus in words**.

REAL LIFE
A sprinkler that waters a circle, a radio mast with a fixed range, and the safe line between two ships that must stay equally far from both are all loci.

HOT QUESTION
Three villages form a right triangle with sides 9 km, 12 km and 15 km. A hospital must be equidistant from all three. Where should it be built and how far from each village?

Mind Map Recap: Loci

REVISE IN ONE LOOK

Fixed distance from a point: circle.

Equidistant from two points: perpendicular bisector.

Equidistant from two lines: angle bisector(s); parallel lines: midway line.

Fixed distance from a line: two parallel lines.

Combine loci: answer is the intersection. Always write the locus in words.

Practice Set

Try each one on paper first. Answers are at the end of the set.

1. What is the locus of a point at a distance of 5 cm from a fixed point P? What is the circumference of this locus (π = 3.14)?
2. State the locus of points equidistant from two points A and B.
3. A and B are 10 cm apart. A point P equidistant from them is 12 cm from the line AB. Find PA.
4. State the locus of the centre of a circle that passes through two fixed points A and B.
5. What is the locus of points equidistant from two intersecting lines?
6. What is the locus of the centre of a wheel rolling on level ground whose radius is 35 cm?
7. Points are 4 cm from a line m and 5 cm from a point P on m. How many are there and how far are they along m from P?
8. Two parallel lines are 6 cm apart. Describe the locus of points equidistant from both and the locus of points 2 cm from one line.
9. Find the circumradius of a triangle with sides 5 cm, 12 cm and 13 cm.
10. Name the point where the angle bisectors of a triangle meet and what it is equidistant from.
11. A point moves so that it is always 2 cm from a fixed line segment of length 6 cm. Describe the locus (a stadium shape) and give the total length of the two straight parts.

12. Find the locus of the centre of a circle that touches both of two lines that meet at O making an angle of 60°.

Answers

1. A circle of radius 5 cm about P; circumference = 2 × 3.14 × 5 = **31.4 cm**.

2. The **perpendicular bisector** of AB.

3. AM = 5, PM = 12: PA = √(25 + 144) = **13 cm**.

4. The **perpendicular bisector** of AB.

5. The **two angle bisectors** (they are perpendicular to each other).

6. A **line parallel to the ground at a height 35 cm** above it.

7. 2 points (one on each side of m): each is √(25 − 16) = **3 cm** along m from P.

8. Equidistant: a **parallel line midway** (3 cm from each). 2 cm from one line: **two parallels**, 2 cm either side of it.

9. $5^2 + 12^2 = 13^2$: right triangle, so R = 13/2 = **6.5 cm**.

10. The **incentre**; it is equidistant from all three sides.

11. Two parallel segments of length 6 cm at distance 2 cm on each side, joined by semicircles of radius 2 cm at the ends. Straight parts total **12 cm**.

12. The **bisector of the 60° angle** (each point of it is equidistant from both lines). For circles lying in the adjacent 120° region, the second bisector, perpendicular to the first.

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CHAPTER · MATHEMATICS

Circles

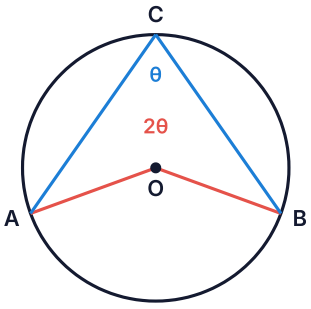
Angles at the centre, cyclic quadrilaterals, tangents and chord-product rules. Learn each theorem, its name and when to quote it.



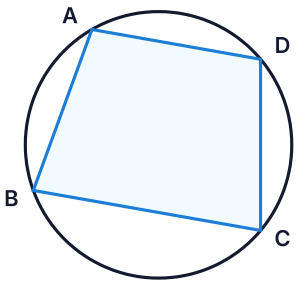
IN THIS CHAPTER

- Angle properties
- Cyclic quadrilaterals
- Tangent properties
- Chord and secant lengths
- Constructions

Angles in a Circle



Angle at centre = 2 × angle at circumference



Cyclic quadrilateral: A + C = 180°

THEOREM	STATEMENT
Angle at the centre	The angle an arc subtends at the centre is **double** the angle it subtends at any point on the remaining part of the circle.
Same segment	Angles in the same segment of a circle are **equal**.
Semicircle	The angle in a semicircle is a **right angle**.

SIDEKICK TIP

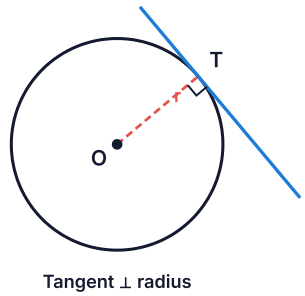
In a circle problem, hunt first for a **diameter** (right angle), then for angles on the **same arc** (equal), then for **centre and circumference** (double).

Cyclic Quadrilaterals

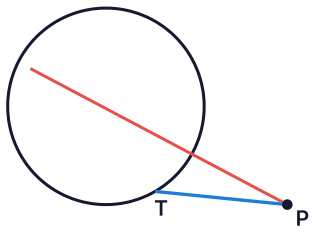
A quadrilateral with all four vertices on a circle is **cyclic**.

- **Opposite angles are supplementary:** ∠A + ∠C = 180° and ∠B + ∠D = 180°.
- **Exterior angle** (made by extending one side) = the **interior opposite angle**.

Tangents and Secants



Tangent ⊥ radius



$PT^2 = PA \times PB$

- A tangent at a point is **perpendicular to the radius** through that point.
- From an external point, the **two tangents are equal in length**.
- If two circles touch, the **point of contact lies on the line joining their centres**. Distance between centres = $r_1 + r_2$ (external) or $r_1 - r_2$ (internal).
- **Alternate segment theorem:** the angle between a tangent and a chord through the point of contact equals the angle in the alternate segment.
- **Chords:** a perpendicular from the centre bisects a chord; equal chords are equidistant from the centre.

Lengths Theorems

Chords AB and CD meeting at P (inside or outside): $PA \times PB = PC \times PD$

Tangent PT and secant PAB from an external point P: $PT^2 = PA \times PB$

Constructions: Tangents and Circles on Triangles

1. Tangents from an external point P
Join P to the centre O. Find the mid-point M of OP. With M as centre and MO as radius draw a circle meeting the given circle at A and B. PA and PB are the tangents.

2. Circumscribed circle of a triangle
Draw perpendicular bisectors of two sides. Their meeting point is the centre; radius is its distance to a vertex.

3. Inscribed circle of a triangle
Bisect two angles. The meeting point is the centre; drop a perpendicular to a side for the radius.

4. Circle in or around a regular hexagon
Both circles have the same centre as the hexagon. Radius of the circumcircle equals the side.

Worked Example 1

Q. An arc subtends an angle of 110° at the centre of a circle. Find the angle it subtends at a point on the remaining part of the circle.

- Angle at the centre = 2 × angle at the circumference.
- 110° ÷ 2 = 55°.

KEY POINT
Answer: 55°.

Worked Example 2

Q. AB is a diameter of a circle and C lies on the circle. If ∠CAB = 38°, find ∠ABC.

- ∠ACB = 90° (angle in a semicircle).
- ∠ABC = 180° − 90° − 38° = 52°.

KEY POINT
Answer: 52°.

Worked Example 3

Q. ABCD is a cyclic quadrilateral with ∠A = 95° and ∠B = 70°. Find ∠C and ∠D. If BC is extended to E, find ∠DCE.

- ∠C = 180° − 95° = 85° and ∠D = 180° − 70° = 110°.
- Exterior angle DCE = interior opposite angle A = 95°. (Check: 180° − 85° = 95°.)

KEY POINT
Answer: ∠C = 85°, ∠D = 110°, ∠DCE = 95°.

Worked Example 4

Q. A tangent PT is drawn from a point P at 13 cm from the centre O of a circle. If PT = 12 cm, find the radius.

- OT ⊥ PT, so ΔOTP is right-angled at T.
- $OT^2 = OP^2 - PT^2 = 169 - 144 = 25$.

KEY POINT
Answer: Radius = 5 cm.

Worked Example 5

Q. Chords AB and CD of a circle meet at P inside the circle. If AP = 6, PB = 4 and CP = 3, find PD.

- PA × PB = PC × PD, so 6 × 4 = 3 × PD.
- PD = 24/3 = 8.

KEY POINT
Answer: PD = 8.

Worked Example 6

Q. From a point P outside a circle a tangent PT and a secant PAB are drawn. If PA = 4 and AB = 5, find PT.

- PB = PA + AB = 9.
- PT^2 = PA × PB = 4 × 9 = 36.

KEY POINT
Answer: PT = 6.

Worked Example 7

Q. A tangent at A and chord AB make an angle of 62°. Find the angle in the alternate segment, and ∠ACB for a point C there.

- Angle between tangent and chord = angle in the alternate segment.
- So ∠ACB = 62°.

KEY POINT
Answer: 62°.

Worked Example 8

Q. A chord of length 16 cm lies in a circle of radius 10 cm. Find its distance from the centre.

- The perpendicular from the centre bisects the chord: half-chord = 8.
- $d^2 = 10^2 - 8^2 = 36$.

KEY POINT
Answer: 6 cm.

Worked Example 9

Q. Two circles of radii 5 cm and 3 cm touch each other. Find the distance between their centres if they touch (i) externally (ii) internally.

- (i) 5 + 3 = 8 cm.
- (ii) 5 − 3 = 2 cm.

KEY POINT
Answer: 8 cm and 2 cm.

Worked Example 10

Q. Tangents PA and PB are drawn to a circle with centre O and ∠AOB = 120°. Find ∠APB.

- ∠OAP = ∠OBP = 90° (tangent perpendicular to radius).
- In quadrilateral OAPB: ∠APB = 360° − 90° − 90° − 120° = 60°.

KEY POINT
Answer: ∠APB = 60°.

BOARD ALERT
Common errors. (1) Using the angle at the centre theorem with a major arc and a minor arc mixed up. (2) Forgetting that the semicircle angle needs the chord to be a **diameter**. (3) In PT^2 = PA × PB, using AB instead of the whole secant PB. (4) Quoting theorems without naming them: write the reason beside each step.

REAL LIFE

A security camera sees the same angle at all points along an arc of a circle through two doors: that is the “same segment” theorem at work.

HOT QUESTION

PA and PB are tangents from P to a circle with centre O. If ∠APB = 50°, find ∠OAB. Then explain why OAPB is a cyclic quadrilateral.

Mind Map Recap: Circles

REVISE IN ONE LOOK

Angle at centre = 2 × angle at circumference. Same segment angles are equal. Semicircle = 90°.

Cyclic quadrilateral: opposite angles add to 180°; exterior angle = interior opposite.

Tangent ⊥ radius; tangents from a point are equal; alternate segment theorem.

Chords: PA × PB = PC × PD. Tangent-secant: $PT^2 = PA \times PB$.

Touching circles: centres and contact point are collinear.

Practice Set

Try each one on paper first. Answers are at the end of the set.

1. An arc subtends 84° at the centre. Find the angle at the circumference on the major arc.
2. In a cyclic quadrilateral ABCD, ∠A = 3x and ∠C = 5x − 20°. Find x and ∠A.
3. AB is a diameter and ∠ABC = 41°. Find ∠BAC.
4. The tangent from a point 17 cm from the centre is 15 cm long. Find the radius.
5. Two chords PQ and RS intersect inside a circle at X. PX = 8, XQ = 3 and RX = 4. Find XS.
6. A tangent PT = 10 cm, and the secant PAB has PA = 5 cm. Find AB.
7. Tangents from P touch a circle at A and B and ∠APB = 70°. Find ∠AOB.
8. The angle between a tangent and chord is 48°. Find the angle in the alternate segment.
9. Radii 8 cm and 3 cm; the circles touch externally. How far apart are the centres? And if one lies inside the other touching?
10. A chord 24 cm long is 5 cm from the centre. Find the radius.
11. In a cyclic quadrilateral, an exterior angle is 105°. Find the opposite interior angle.
12. Find the length of a tangent from a point 25 cm from the centre of a circle of radius 7 cm.

Answers

1. 42°.

2. 3x + 5x − 20 = 180: x = 25; ∠A = **75°**.

3. ∠ACB = 90°, so ∠BAC = 180 − 90 − 41 = **49°**.

4. $r^2 = 289 - 225 = 64$: **8 cm**.

5. 8 × 3 = 4 × XS: **XS = 6**.

6. $10^2 = 5 \times PB$: PB = 20, AB = **15 cm**.

7. ∠AOB = 180° − 70° = **110°**.

8. **48°**.

9. External: **11 cm**; internal: **5 cm**.

10. Half-chord 12: $r^2 = 144 + 25 = 169$: **13 cm**.

11. Interior opposite = **105°**.

12. $\sqrt{(625 - 49)} = \sqrt{576}$ = **24 cm**.

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CHAPTER · MATHEMATICS

Constructions

Ruler and compasses only. Draw tangents from an outside point, circles around and inside triangles, and a regular hexagon, and check each by calculation.



IN THIS CHAPTER

- Tangents from a point
- Circumscribed circle
- Inscribed circle
- Regular hexagon

The Ground Rules

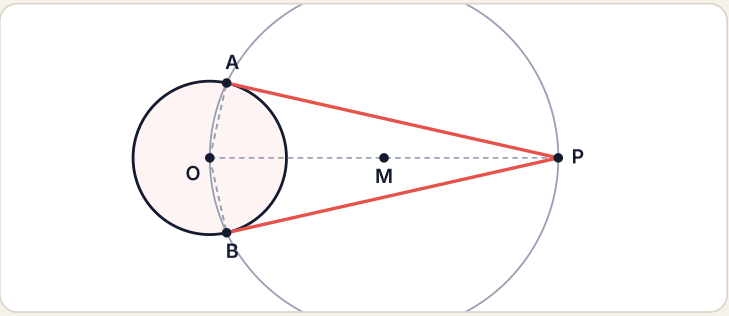
In a construction you use only a **ruler (for drawing straight lines and measuring) and a pair of compasses**, plus a sharp pencil. Keep every construction arc visible. Examiners award marks for the arcs, the neat final figure and a short written statement.

- Use a hard pencil (2H) for construction lines, a softer one for the final figure.
- Do not rub out the arcs.
- Label every point and write the measurement asked for (such as the radius or the tangent length) after measuring.
- Check your result: the tangent should look like it just touches the circle, and the right angle should look like 90°.

Quick Recall of Basic Constructions

TO CONSTRUCT	IDEA
Perpendicular bisector of AB	Arcs of equal radius (more than half of AB) from A and B; join the two crossing points
Bisector of an angle	Arc cutting both arms; two equal arcs from the cut points; join the vertex to their crossing
Triangle, three sides given	Draw the longest base; arcs from each end with the other two lengths
Triangle, two sides and the included angle	Draw one side, make the angle with a protractor or compasses, mark the second side
Angle of 60°	Arc from the point; arc of the same radius from where it meets the line; join

Tangents to a Circle from an External Point



Tangents PA and PB from an external point P: the circle on OP as diameter meets the given circle at A and B (angle OAP = 90 degrees).

1. Draw
Draw the circle with centre O and mark the external point P.

2. Join and bisect
Join OP and construct its perpendicular bisector to find the mid-point M.

3. Second circle
With centre M and radius MO (= MP) draw a circle cutting the first circle at A and B.

4. Tangents
Join PA and PB. These are the two tangents.

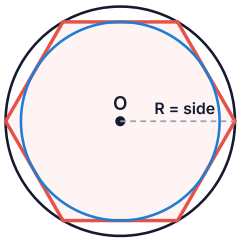
KEY POINT

Why it works: OP is a diameter of the second circle, so ∠OAP is an angle in a semicircle and equals 90°. A line perpendicular to a radius at its end is a tangent. The two tangent lengths are equal: PA = PB.

Circumscribed and Inscribed Circles of a Triangle

- **Circumscribed circle** (passes through all vertices): draw perpendicular bisectors of two sides. The crossing point is the **circumcentre**; the circle has radius equal to its distance to any vertex.
- **Inscribed circle** (touches all three sides): bisect two angles. The crossing point is the **incentre**. Drop a perpendicular from it to a side; this is the radius.

Circles and a Regular Hexagon



Regular hexagon: circumradius = side; inradius = side × √3 / 2.

1. Draw the circle
Draw a circle of radius equal to the hexagon side. Mark a point A on it.

2. Step round
With the same radius step off arcs round the circle: A, B, C, D, E, F. Six steps return to A.

3. Join
Join the six points in order. This is the regular hexagon, and the circle is its circumscribed circle.

4. Inscribed circle
Bisect two sides perpendicularly; the centre is O (the same centre) and the radius is the distance from O to a side mid-point.

SIDEKICK TIP
In a regular hexagon the six triangles meeting at the centre are equilateral. So circumradius = side, and inradius = (√3/2) × side. A side of 4 cm gives R = 4 cm and r = 3.46 cm.

Worked Example 1

Q. Draw a circle of radius 3 cm. From a point P 7 cm from its centre construct the two tangents and measure their length. Check by calculation.

- Follow the four steps: OP = 7 cm, M at 3.5 cm from O, second circle radius 3.5 cm.
- Calculation: $OT^2 + PT^2 = OP^2$, so PT = √(49 − 9) = √40 = 6.32 cm.

KEY POINT
Answer: Each tangent measures about 6.32 cm.

Worked Example 2

Q. Construct a triangle with sides 5 cm, 6 cm and 7 cm and draw its circumscribed circle. Find the radius by calculation to check your drawing.

- Draw BC = 7 cm, then arcs 5 cm (from C) and 6 cm (from B) to meet at A.
- Draw perpendicular bisectors of AB and BC; they meet at the circumcentre O. Draw the circle through A, B, C.
- Check: s = 9, area = √(9 × 4 × 3 × 2) = √216 = 14.70. R = abc/(4 × area) = 210/58.79.

KEY POINT
Answer: Circumradius about 3.57 cm.

Worked Example 3

Q. For the same triangle (5, 6, 7) construct the inscribed circle and find its radius.

- Bisect two angles. Their meeting point I is the incentre.
- Perpendicular from I to BC gives the radius.
- Check: r = area/s = 14.70/9 = 1.63 cm.

KEY POINT
Answer: Inradius about 1.63 cm.

Worked Example 4

Q. Construct a regular hexagon of side 4 cm and draw its circumscribed and inscribed circles.

- Draw a circle of radius 4 cm and step off six chords of 4 cm.
- Circumcircle is the drawn circle (R = 4 cm).
- Inradius = 4 × √3/2 = 3.46 cm; check by measuring from O to a side mid-point.

KEY POINT
Answer: R = 4 cm; r = 3.46 cm.

Worked Example 5

Q. Two tangents are drawn from P (7 cm from the centre) to a circle of radius 3 cm. Find ∠APB by calculation.

- In the right triangle OAP: sin(∠APO) = OA/OP = 3/7, so ∠APO = 25.4°.
- OP bisects ∠APB, so ∠APB = 2 × 25.4 = 50.8°.

KEY POINT
Answer: ∠APB is about 50.8° (measure it to check).

Worked Example 6

Q. Construct triangle ABC with AB = 6 cm, BC = 5 cm and ∠B = 60°. Draw its incircle and state its radius.

- Draw AB = 6, make ∠B = 60° and mark BC = 5; join AC. Measure AC (calculation: $AC^2 = 36 + 25 - 30 = 31$, so AC = 5.57 cm).
- Bisect ∠A and ∠B; their meeting point is the incentre.
- Check: area = ½ × 6 × 5 × sin 60° = 12.99; s = 8.28; r = area/s = 1.57 cm.

KEY POINT
Answer: Inradius about 1.57 cm.

BOARD ALERT
Common errors. (1) Rubbing out arcs. (2) Using different compass settings for the two arcs that find a bisector. (3) Drawing the tangent from P without actually passing through the points of intersection A and B. (4) Not stating the final measurement or conclusion.

REAL LIFE
Belt-and-pulley drawings and the design of a nut or a hexagonal tile all use the constructions of tangents and of regular hexagons.

HOT QUESTION
A circle of radius 4 cm and a point P 9 cm from its centre: before constructing, find the length of the tangents. Then explain, without measuring, why the angle between the tangents is less than 60°.

Mind Map Recap: Constructions

REVISE IN ONE LOOK

Tangents from P: bisect OP at M, circle on OP as diameter, join P to the two crossing points.

Circumcircle: perpendicular bisectors of two sides. **Incircle:** two angle bisectors, perpendicular to a side.

Regular hexagon: side = radius of its circumcircle; step off six chords.

Keep arcs, label points, write the measurement.

Check lengths by calculation: tangent = $\sqrt{(d^2 - r^2)}$; r = area/s; R = abc/4×area.

Practice Set

Try each one on paper first. Answers are at the end of the set.

1. Draw a circle of radius 2.5 cm. From a point 6.5 cm from the centre draw the tangents. Find the tangent length by calculation.
2. Construct triangle ABC with BC = 6 cm, AB = 5 cm, AC = 7 cm and draw its circumscribed circle. Give the radius.
3. For the triangle in Q2, draw the inscribed circle and give its radius.
4. Construct a right triangle with sides 6 cm, 8 cm, 10 cm and its circumcircle. Where is its centre and what is the radius?
5. For the 6-8-10 triangle, find the radius of the incircle.
6. Construct a regular hexagon of side 3.5 cm and draw the circle through its vertices. Give the radius.
7. Find the inradius of a regular hexagon of side 6 cm.
8. Draw a circle of radius 3 cm. Draw a tangent at a point T on it and measure the angle between the tangent and the radius OT.
9. From a point 8 cm from the centre of a 4 cm circle, draw the two tangents. Find $\angle APB$.
10. Construct an equilateral triangle of side 6 cm and draw its circumcircle. Find the radius.
11. Construct an equilateral triangle of side 6 cm and draw its incircle. Find the radius.
12. A tangent PT of length 9 cm is drawn to a circle of radius 12 cm. How far is P from the centre?

Answers

1. PT = $\sqrt{(42.25 - 6.25)}$ = **6 cm**.

2. s = 9, area = $\sqrt{(9 \times 3 \times 4 \times 2)}$ = 14.70; R = 210/(4 × 14.70) = **3.57 cm**.

3. r = area/s = 1.63 = **1.63 cm**.

4. The centre is the **mid-point of the hypotenuse**; R = **5 cm**.

5. r = area/s = 24/12 = **2 cm**.

6. Radius = **3.5 cm**.

7. r = 6 × $\sqrt{3}/2$ = $3\sqrt{3}$ = **5.20 cm**.

8. **90°**.

9. $\sin(\angle APO) = 4/8 = 1/2$, $\angle APO = 30°$: $\angle APB$ = **60°**.

10. R = side/$\sqrt{3}$ = $6/\sqrt{3}$ = $2\sqrt{3}$ = **3.46 cm**.

11. r = side/$(2\sqrt{3})$ = **1.73 cm**.

12. OP = $\sqrt{(81 + 144)}$ = **15 cm**.

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CHAPTER · MATHEMATICS

Mensuration: Cylinder, Cone and Sphere

Surface areas and volumes of three solids, combined solids, hollow pipes, and melting one solid into another. Learn the formulae and when to add or subtract.



IN THIS CHAPTER

- Cylinder
- Cone
- Sphere and hemisphere
- Combination of solids
- Melting and recasting

Formulae for the Three Solids

SOLID	CURVED SURFACE AREA (CSA)	TOTAL SURFACE AREA (TSA)	VOLUME
Cylinder (radius r, height h)	$2\pi rh$	$2\pi r(h + r)$	$\pi r^2 h$
Cone (slant height l, $l^2 = r^2 + h^2$)	πrl	$\pi r(l + r)$	$\frac{1}{3}\pi r^2 h$
Sphere	$4\pi r^2$	$4\pi r^2$	$(4/3)\pi r^3$
Hemisphere	$2\pi r^2$	$3\pi r^2$	$\frac{2}{3}\pi r^3$

SIDEKICK TIP

A hemisphere has a flat circular face in addition to its curved surface: TSA = $2\pi r^2 + \pi r^2 = 3\pi r^2$. A sphere has no flat face, so TSA = CSA.

Using π

Use the value of π that the question gives (22/7 or 3.14). If none is given and the radius is a multiple of 7, use 22/7. Keep π in your working until the last step when you can.

Four Kinds of Problems

1. Direct
Substitute into the formula. Match the question to CSA, TSA or volume.

2. Combination of solids
Break the object into standard solids. Add volumes. For surface area add only the surfaces you can see; the joined faces vanish.

3. Hollow solids
Volume of material = outer volume − inner volume. For a pipe: $\pi h(R^2 - r^2)$.

4. Melting and recasting
The volume stays the same. Volume of the old solid = (number of new solids) × (volume of one).

KEY POINT

Units: lengths in cm or m, areas in cm^2 or m^2, volumes in cm^3 or m^3. 1 m^3 = 1000 litres and 1 litre = 1000 cm^3. Convert all lengths to one unit before you start.

BOARD ALERT

Frustum problems are **not included** in the syllabus. Skip them.

Worked Example 1

Q. A cylinder has radius 7 cm and height 10 cm. Find its curved surface area, total surface area and volume (π = 22/7).

- CSA = $2\pi rh = 2 \times 22/7 \times 7 \times 10 = 440\ cm^2$.
- TSA = $2\pi r(h + r) = 2 \times 22/7 \times 7 \times 17 = 748\ cm^2$.
- Volume = $\pi r^2 h = 22/7 \times 49 \times 10 = 1540\ cm^3$.

KEY POINT

Answer: CSA 440 cm^2, TSA 748 cm^2, volume 1540 cm^3.

Worked Example 2

Q. A cone has base radius 6 cm and height 8 cm. Find its slant height, curved surface area and volume (π = 3.14).

- $l = \sqrt{(6^2 + 8^2)} = 10$ cm.
- CSA = $\pi rl = 3.14 \times 6 \times 10 = 188.4\ cm^2$.
- Volume = $\frac{1}{3} \times 3.14 \times 36 \times 8 = 301.44\ cm^3$.

KEY POINT

Answer: l = 10 cm, CSA 188.4 cm², volume 301.44 cm³.

Worked Example 3

Q. Find the surface area and volume of a sphere of radius 10.5 cm (π = 22/7).

- Surface area = $4\pi r^2 = 4 \times 22/7 \times 10.5 \times 10.5 = 1386$ cm².
- Volume = $(4/3)\pi r^3 = 4/3 \times 22/7 \times 1157.625 = 4851$ cm³.

KEY POINT

Answer: 1386 cm² and 4851 cm³.

Worked Example 4

Q. A solid metal sphere of radius 9 cm is melted and recast into small cones of radius 3 cm and height 6 cm. How many cones are formed?

- Volume of sphere = $(4/3)\pi(9)^3 = 972\pi$.
- Volume of one cone = $(1/3)\pi(9)(6) = 18\pi$.
- Number = $972\pi/18\pi = 54$.

KEY POINT

Answer: 54 cones.

Worked Example 5

Q. A toy is a cone on top of a hemisphere, both of radius 3.5 cm. The total height of the toy is 12.5 cm. Find its volume (π = 22/7).

- Cone height = 12.5 − 3.5 = 9 cm.
- Hemisphere: $(2/3)(22/7)(3.5)^3 = 89.83$ cm³.
- Cone: $(1/3)(22/7)(3.5)^2(9) = 115.50$ cm³.
- Total = 205.33 cm³.

KEY POINT

Answer: About 205.33 cm³.

Worked Example 6

Q. A metal pipe is 21 cm long with outer radius 5 cm and inner radius 4 cm. Find the volume of metal (π = 22/7).

- Volume = $\pi h(R^2 - r^2) = 22/7 \times 21 \times (25 - 16)$.
- $= 66 \times 9 = 594$.

KEY POINT

Answer: 594 cm³.

Worked Example 7

Q. A cylindrical tank has radius 3.5 m and height 4 m. Find the cost of painting its curved surface at ₹ 15 per m². How many litres does it hold?

- CSA = $2 \times 22/7 \times 3.5 \times 4 = 88$ m². Cost = $88 \times 15 =$ ₹ 1,320.
- Volume = $22/7 \times 12.25 \times 4 = 154$ m³ = 154,000 litres.

KEY POINT

Answer: ₹ 1,320 and 154,000 litres.

Worked Example 8

Q. A cylindrical vessel of radius 6 cm contains water. A solid sphere of radius 3 cm is dropped in and sinks. By how much does the water level rise?

- Volume of sphere = $(4/3)\pi(27) = 36\pi$.
- This equals the extra cylinder volume: $\pi(36)h = 36\pi$, so h = 1.

KEY POINT

Answer: The level rises by 1 cm.

Worked Example 9

Q. A tent is a cylinder of radius 7 m and height 3 m surmounted by a cone of the same radius and slant height 12.5 m. Find the canvas needed for its curved surfaces (π = 22/7).

- Cylinder CSA = 2 × 22/7 × 7 × 3 = 132 m².
- Cone CSA = 22/7 × 7 × 12.5 = 275 m².
- Canvas = 132 + 275 = 407 m² (no base is needed).

KEY POINT

Answer: 407 m².

BOARD ALERT

Common errors. (1) Using height instead of slant height in πrl. (2) Giving TSA when CSA is asked (or the reverse). (3) Forgetting to convert cm to m or the volume to litres. (4) Counting joined faces twice in a combination of solids.

REAL LIFE

Water tanks, ice-cream cones and cricket balls are cylinders, cones and spheres. Recasting is how a goldsmith turns one gold sphere into many tiny beads.

HOT QUESTION

A solid cylinder of radius r and height 2r is melted and recast into a sphere. Find the radius of the sphere in terms of r.

Mind Map Recap: Mensuration

REVISE IN ONE LOOK

Cylinder: $2\pi rh$, $2\pi r(h+r)$, $\pi r^2 h$.

Cone: πrl, $\pi r(l+r)$, $\frac{1}{3}\pi r^2 h$; $l^2 = r^2 + h^2$.

Sphere: $4\pi r^2$ and $(4/3)\pi r^3$. **Hemisphere:** $2\pi r^2$, $3\pi r^2$, $\frac{2}{3}\pi r^3$.

Combination: add volumes; add visible surfaces only. Hollow: outer minus inner.

Recast: volume unchanged; number = big volume / small volume.

Practice Set

Try each one on paper first. Answers are at the end of the set.

1. A cylinder has radius 14 cm and height 20 cm. Find CSA, TSA and volume (π = 22/7).
2. A cone has radius 5 cm and height 12 cm. Find its slant height, TSA and volume (π = 3.14).
3. Find the volume and surface area of a sphere of radius 21 cm (π = 22/7).
4. Find the TSA and volume of a hemisphere of radius 7 cm (π = 22/7).
5. A solid sphere of radius 6 cm is melted and cast into small spheres of radius 1.5 cm. How many are made?
6. A cylindrical drum of radius 14 cm is filled with water to a height of 21 cm. How many litres? (π = 22/7)
7. Find the cost of painting the outside of a closed cylindrical tin of radius 7 cm and height 15 cm at ₹ 0.50 per 10 cm². (π = 22/7)
8. A metal pipe 14 cm long has outer radius 3.5 cm and inner radius 2.5 cm. Find the volume of metal. (π = 22/7)
9. A solid cone of radius 4 cm and height 9 cm is melted into a cylinder of radius 3 cm. Find the height of the cylinder.
10. A toy is a hemisphere of radius 3 cm surmounted on a cone of height 4 cm and radius 3 cm. Find its total surface area (π = 3.14).
11. Water flows through a pipe of radius 3.5 cm at 2 m/s. Find the volume of water in litres that flows out in 1 minute (π = 22/7).
12. A cone has curved surface area 550 cm² and slant height 25 cm. Find its base radius (π = 22/7).

Answers

1. CSA **1760 cm²**, TSA **2992 cm²**, volume **12320 cm³**.

2. l = **13 cm**; TSA = 3.14 × 5 × 18 = **282.60 cm²**; volume = **314 cm³**.

3. Volume **38808 cm³**; surface area **5544 cm²**.

4. TSA = 3 × 22/7 × 49 = **462 cm²**; volume = **718.67 cm³**.

5. $(6/1.5)^3$ = **64**.

6. Volume = 22/7 × 196 × 21 = 12936 cm^3 = **12.936 litres**.

7. TSA = 2 × 22/7 × 7 × 22 = 968 cm^2; cost = **₹ 48.40**.

8. 22/7 × 14 × (12.25 − 6.25) = **264 cm³**.

9. (1/3)(16)(9) = 48 = 9h: **h = 5.33 cm** (16/3).

10. l = 5; cone CSA = 3.14 × 3 × 5 = 47.1; hemisphere CSA = 2 × 3.14 × 9 = 56.52; total **103.62 cm²**.

11. Length in 1 min = 120 m = 12,000 cm; volume = 22/7 × 12.25 × 12000 = 462000 cm^3 = **462 litres**.

12. 22/7 × r × 25 = 550: **r = 7 cm**.

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CHAPTER · MATHEMATICS

Trigonometry

Three ratios, three identities. Use them to find values from one given ratio, to evaluate expressions and to prove identities step by step.



IN THIS CHAPTER

- Trig ratios
- Standard angles
- Complementary angles
- Identities
- Proofs

The Three Ratios

In a right triangle, with respect to an acute angle θ, call the side opposite to it **perpendicular (p)**, the side next to it **base (b)** and the longest side **hypotenuse (h)**.

$$\sin\theta = \frac{p}{h} \quad \cos\theta = \frac{b}{h} \quad \tan\theta = \frac{p}{b} = \frac{\sin\theta}{\cos\theta}$$

Their reciprocals: cosec θ = 1/sin θ, sec θ = 1/cos θ, cot θ = 1/tan θ = cos θ/sin θ.

Standard Values

θ	0°	30°	45°	60°	90°
sin θ	0	1/2	1/√2	√3/2	1
cos θ	1	√3/2	1/√2	1/2	0
tan θ	0	1/√3	1	√3	not defined



SIDEKICK TIP

Sines go 0, ½, 1/√2, √3/2, 1 (write as √0/2, √1/2, √2/2, √3/2, √4/2). Cosines are the same list backwards.

Complementary Angles

$$\sin(90° - A) = \cos A \quad \cos(90° - A) = \sin A \quad \tan(90° - A) = \cot A$$

The Three Identities

These hold for every acute angle A (and in fact for all angles where the terms are defined). They come from Pythagoras: $p^2 + b^2 = h^2$. Divide by h^2 to get the first, by b^2 to get the second, and by p^2 to get the third.

$$\sin^2 A + \cos^2 A = 1$$

$$1 + \tan^2 A = \sec^2 A$$

$$1 + \cot^2 A = \operatorname{cosec}^2 A$$

Useful rearrangements: $\sin^2 A = 1 - \cos^2 A$; $\sec^2 A - \tan^2 A = 1$; $\operatorname{cosec}^2 A - \cot^2 A = 1$; $(\sec A - \tan A)(\sec A + \tan A) = 1$.

How to Prove an Identity

1. Pick the harder side
Start with the more complicated side. Do not move terms across the equals sign.

2. Convert
If stuck, rewrite everything in terms of sin and cos.

3. Use identities
Replace $1 - \sin^2 A$ by $\cos^2 A$, and so on.

4. Simplify and match
Stop when you reach the other side. Write 'LHS = RHS'.

Worked Example 1

Q. If sin θ = 5/13 and θ is acute, find cos θ and tan θ.

- $\cos^2\theta = 1 - 25/169 = 144/169$, so cos θ = 12/13.

- $\tan\theta = \sin\theta/\cos\theta = 5/12$.

KEY POINT
Answer: $\cos\theta = 12/13$, $\tan\theta = 5/12$.

Worked Example 2

Q. If $\tan\theta = 3/4$, find $\sin\theta + \cos\theta$ (θ acute).

- Take a triangle with $p = 3$, $b = 4$, so $h = 5$.
- $\sin\theta = 3/5$, $\cos\theta = 4/5$.

KEY POINT
Answer: 7/5.

Worked Example 3

Q. Prove that $(\sec A - \tan A)(\sec A + \tan A) = 1$.

- LHS = $\sec^2 A - \tan^2 A$ (difference of squares).
- Since $1 + \tan^2 A = \sec^2 A$, this is 1.

KEY POINT
Answer: LHS = RHS.

Worked Example 4

Q. Prove that $\frac{\sin A}{1 + \cos A} + \frac{1 + \cos A}{\sin A} = 2 \operatorname{cosec} A$.

- Common denominator: $[\sin^2 A + (1 + \cos A)^2] / [\sin A (1 + \cos A)]$.
- Numerator = $\sin^2 A + 1 + 2\cos A + \cos^2 A = 2 + 2\cos A = 2(1 + \cos A)$.
- So the whole = $2(1 + \cos A)/[\sin A(1 + \cos A)] = 2/\sin A = 2 \operatorname{cosec} A$.

KEY POINT
Answer: LHS = RHS.

Worked Example 5

Q. Prove that $\frac{1 - \sin\theta}{1 + \sin\theta} = (\sec\theta - \tan\theta)^2$.

- RHS = $(1/\cos\theta - \sin\theta/\cos\theta)^2 = (1 - \sin\theta)^2/\cos^2\theta$.
- $\cos^2\theta = (1 - \sin\theta)(1 + \sin\theta)$, so RHS = $(1 - \sin\theta)/(1 + \sin\theta)$.

KEY POINT
Answer: RHS = LHS.

Worked Example 6

Q. Prove that $\cot^2 A - \cos^2 A = \cot^2 A \cos^2 A$.

- $\cot^2 A - \cos^2 A = \cos^2 A/\sin^2 A - \cos^2 A = \cos^2 A (1/\sin^2 A - 1)$.
- $= \cos^2 A (1 - \sin^2 A)/\sin^2 A = \cos^2 A \times \cos^2 A/\sin^2 A = \cos^2 A \cot^2 A$.

KEY POINT
Answer: LHS = RHS.

Worked Example 7

Q. Evaluate: $(\sin 35° / \cos 55°) + (\cos 20° / \sin 70°) - 2\cos 60°$.

- $\cos 55° = \sin 35°$ (as $55 = 90 - 35$), so the first ratio is 1.
- $\sin 70° = \cos 20°$, so the second ratio is 1.
- $2\cos 60° = 1$.

KEY POINT

Answer: 1 + 1 − 1 = 1.

Worked Example 8

Q. If $\sec\theta + \tan\theta = 3$, find $\sec\theta - \tan\theta$ and then $\sin\theta$.

- $(\sec\theta - \tan\theta)(\sec\theta + \tan\theta) = 1$, so $\sec\theta - \tan\theta = 1/3$.
- Add: $2\sec\theta = 3 + 1/3 = 10/3$, $\sec\theta = 5/3$, $\cos\theta = 3/5$.
- Subtract: $2\tan\theta = 3 - 1/3 = 8/3$, $\tan\theta = 4/3$. $\sin\theta = \tan\theta\cos\theta = 4/5$.

KEY POINT

Answer: $\sec\theta - \tan\theta = 1/3$; $\sin\theta = 4/5$.

Worked Example 9

Q. Evaluate $4(\sin^4 30° + \cos^4 60°) - 3(\cos^2 45° - \sin^2 90°)$.

- $\sin 30° = \cos 60° = 1/2$, so $\sin^4 30° + \cos^4 60° = 1/16 + 1/16 = 1/8$; times 4 = 1/2.
- $\cos^2 45° = 1/2$, $\sin^2 90° = 1$: $1/2 - 1 = -1/2$; times $-3 = 3/2$.
- Sum = 1/2 + 3/2 = 2.

KEY POINT

Answer: 2.

Worked Example 10

Q. If $x = a\cos\theta$ and $y = b\sin\theta$, eliminate θ.

- $\cos\theta = x/a$ and $\sin\theta = y/b$.
- Use $\sin^2\theta + \cos^2\theta = 1$.

KEY POINT

Answer: $x^2/a^2 + y^2/b^2 = 1$.

BOARD ALERT

Common errors. (1) Writing $\sin^2 A$ as $\sin A^2$ or "sin A + sin A". (2) Cancelling sin from sin A + cos A. (3) Proving an identity by cross-multiplying both sides (this assumes what you must show). (4) Using tan 90° as a number: it is not defined.

REAL LIFE

Surveyors use these ratios to find inaccessible lengths and to check slopes; the identities guarantee that their calculators and tables agree.

HOT QUESTION

Prove that $(\sin A + \operatorname{cosec} A)^2 + (\cos A + \sec A)^2 = 7 + \tan^2 A + \cot^2 A$.

Mind Map Recap: Trigonometry

REVISE IN ONE LOOK

sin = p/h, cos = b/h, tan = p/b.

Standard angles: 0, 30, 45, 60, 90. Complementary: $\sin(90-A) = \cos A$.

$\sin^2 + \cos^2 = 1$; $1 + \tan^2 = \sec^2$; $1 + \cot^2 = \operatorname{cosec}^2$.

Prove: start on the complicated side; convert to sin and cos; use identities.

Given one ratio: draw the triangle and use Pythagoras.

Practice Set

Try each one on paper first. Answers are at the end of the set.

1. If $\cos\theta = 8/17$ (acute), find $\sin\theta$ and $\cot\theta$.
2. If $\tan A = 5/12$, find $\sin A$ and $\sec A$.
3. Prove that $(1 - \cos^2 A)(1 + \cot^2 A) = 1$.
4. Prove that $\dfrac{\cos A}{1 - \sin A} = \dfrac{1 + \sin A}{\cos A}$.
5. Evaluate $\sin^2 25° + \sin^2 65° + \tan^2 45°$.
6. Evaluate $(\tan 20°/\cot 70°) + (\sin 50°/\cos 40°) - 2\sin 30°$.
7. Prove that $\sec^4 A - \sec^2 A = \tan^4 A + \tan^2 A$.
8. If $\sin\theta + \cos\theta = \sqrt{2}\cos\theta$, show that $\tan\theta = \sqrt{2} - 1$.
9. Prove that $\dfrac{1}{\sec A - \tan A} - \dfrac{1}{\cos A} = \tan A$.
10. Evaluate $2\sin^2 30° - 3\cos^2 45° + \tan^2 60°$.
11. If $\operatorname{cosec}\theta - \cot\theta = 1/2$, find $\operatorname{cosec}\theta + \cot\theta$ and hence $\sin\theta$.
12. Find the value of $\sin 60° \cos 30° + \cos 60° \sin 30°$.

Answers

1. $\sin\theta = 15/17$; $\cot\theta = $ **8/15**.

2. $h = 13$: $\sin A = $ **5/13**, $\sec A = $ **13/12**.

3. $(\sin^2 A)(\operatorname{cosec}^2 A) = $ **1**.

4. Multiply across: $\cos^2 A = (1 + \sin A)(1 - \sin A) = 1 - \sin^2 A$. True.

5. $\sin 65° = \cos 25°$, so $\sin^2 25° + \cos^2 25° + 1 = $ **2**.

6. $\tan 20°/\tan 20° = 1$; $\sin 50°/\sin 50° = 1$; $2 \times 1/2 = 1$: **1**.

7. $\sec^2 A(\sec^2 A - 1) = \sec^2 A \tan^2 A = (1 + \tan^2 A)\tan^2 A = $ RHS.

8. Divide by $\cos\theta$: $\tan\theta + 1 = \sqrt{2}$, so $\tan\theta = \sqrt{2} - 1$.

9. $1/(\sec A - \tan A) = \sec A + \tan A$; minus $\sec A$ leaves **tan A**.

10. $2 \times 1/4 - 3 \times 1/2 + 3 = 1/2 - 3/2 + 3 = $ **2**.

11. Product = 1, so $\operatorname{cosec}\theta + \cot\theta = $ **2**; $\operatorname{cosec}\theta = 5/4$, $\sin\theta = $ **4/5**.

12. $3/4 + 1/4 = $ **1**.

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CHAPTER · MATHEMATICS

Heights and Distances

Turn a tower, a ladder or a kite string into a right triangle and let trigonometry do the measuring. Angles of elevation and depression, with one or two triangles.



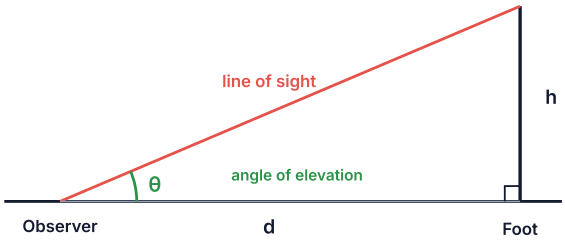
IN THIS CHAPTER

- Elevation and depression
- Choosing the ratio
- Two observation points
- Using the tables

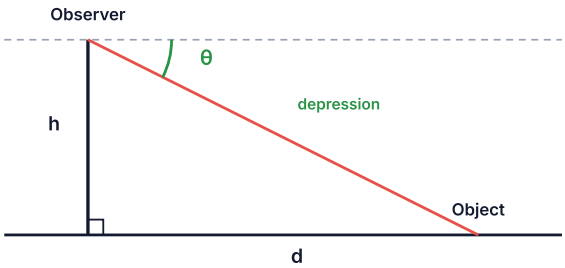
Two Angles, One Idea

A tall object, a point on the ground and the line of sight form a **right-angled triangle**. Trigonometry then links the angle with the height and distance.

- **Angle of elevation:** the angle **above the horizontal** through which you raise your eyes to look at an object that is higher than you.
- **Angle of depression:** the angle **below the horizontal** through which you lower your eyes to see an object lower than you.
- The angle of depression of an object from A equals the angle of elevation of A from the object (alternate angles between parallels).



Angle of elevation θ: tan θ = h / d.



Angle of depression θ (below the horizontal) equals the angle of elevation from the object: tan θ = h / d.

The Method

1. Sketch
Draw the situation as a right triangle. Put the angle at the right place. Mark known and unknown sides.

2. Choose the ratio
Height and distance together: use tan. Height and line of sight: sin. Distance and line of sight: cos.

3. Write and solve
For example tan 60° = h/20 gives h = 20 tan 60°. Use √3 = 1.732 or read the value from the tables.

4. Answer
Give the answer with the unit, and round as asked.

θ	15°	20°	25°	30°	35°	40°	45°	50°	55°	60°	65°	70°	75°
tan θ	0.2679	0.3640	0.4663	0.5774	0.7002	0.8391	1.0000	1.1918	1.4281	1.7321	2.1445	2.7475	3.7321
sin θ	0.2588	0.3420	0.4226	0.5000	0.5736	0.6428	0.7071	0.7660	0.8192	0.8660	0.9063	0.9397	0.9659
cos θ	0.9659	0.9397	0.9063	0.8660	0.8192	0.7660	0.7071	0.6428	0.5736	0.5000	0.4226	0.3420	0.2588

SIDEKICK TIP
The syllabus allows trigonometric tables. Values like those above are what the table gives (to four decimal places). Use √3 = 1.732 and √2 = 1.414 when the angle is 30°, 45° or 60°.

KEY POINT

Cases with **more than two right triangles** are excluded. Most questions use two triangles: two observation points, or a building with a flagstaff.

Worked Example 1

Q. From a point 20 m from the foot of a tower the angle of elevation of the top is 60°. Find the height of the tower.

- tan 60° = h/20.
- h = 20 × 1.732 = 34.64 m.

KEY POINT

Answer: About 34.64 m.

Worked Example 2

Q. A kite is flying at the end of a 100 m string making 30° with the ground. Find its height, taking the string as straight.

- sin 30° = h/100.
- h = 100 × 1/2 = 50.

KEY POINT

Answer: 50 m.

Worked Example 3

Q. From the top of a 30 m building the angle of depression of a car on the road is 45°. How far is the car from the foot of the building?

- tan 45° = 30/d, and tan 45° = 1.
- d = 30.

KEY POINT

Answer: 30 m.

Worked Example 4

Q. From a point on the ground 40 m from the foot of a tower, the angles of elevation of the bottom and the top of a flagstaff on the tower are 30° and 45°. Find the height of the tower and of the flagstaff.

- Tower: tan 30° = h/40, so h = 40/√3 = 23.09 m.
- Tower + flagstaff: tan 45° = H/40, so H = 40 m.
- Flagstaff = 40 − 23.09 = 16.91 m.

KEY POINT

Answer: Tower 23.09 m; flagstaff 16.91 m.

Worked Example 5

Q. A tower is observed from two points A and B in line with its foot, on the same side, 20 m apart. The elevations are 30° at A and 60° at B. Find the height of the tower.

- Let B be x m from the foot. h = x tan 60° = x√3.
- From A: h = (x + 20) tan 30° = (x + 20)/√3.
- x√3 = (x + 20)/√3, so 3x = x + 20, x = 10.
- h = 10√3 = 17.32 m.

KEY POINT

Answer: About 17.32 m.

Worked Example 6

Q. A 50 m tall tower stands between two points P and Q on opposite sides, in line with it. The elevations of the top are 30° from P and 45° from Q. Find PQ.

- From P: 50/PO = tan 30°, so PO = 50√3 = 86.60 m.
- From Q: QO = 50/tan 45° = 50 m.
- PQ = 86.60 + 50 = 136.60 m.

KEY POINT
Answer: About 136.60 m.

Worked Example 7

Q. A 10 m ladder leans against a wall making an angle of 60° with the ground. How high up the wall does it reach and how far is its foot from the wall?

- Height = 10 sin 60° = 10 × √3/2 = 8.66 m.
- Foot distance = 10 cos 60° = 5 m.

KEY POINT
Answer: Height 8.66 m; foot 5 m from the wall.

Worked Example 8

Q. From the top of a tower 60 m high, the angles of depression of the top and the bottom of a building are 30° and 60°. Find the height of the building.

- Horizontal distance d: tan 60° = 60/d, so d = 60/√3 = 20√3 = 34.64 m.
- Drop from the tower's top to the building's top: d tan 30° = 20√3 × (1/√3) = 20 m.
- Height of building = 60 − 20 = 40 m.

KEY POINT
Answer: 40 m.

Worked Example 9

Q. The angle of elevation of the top of a pole from a point 50 m away is 35°. Find the height using the table (tan 35° = 0.7002).

- h = 50 × tan 35° = 50 × 0.7002 = 35.010 m.

KEY POINT
Answer: About 35.01 m.

Worked Example 10

Q. The shadow of a 6 m pole is 6√3 m long. Find the angle of elevation of the Sun.

- tan θ = 6/(6√3) = 1/√3, so θ = 30°.

KEY POINT
Answer: 30°.

BOARD ALERT
Common errors. (1) Placing the angle of elevation inside the triangle at the wrong vertex. (2) Taking the angle of depression as the angle at the object's top. It is measured from the horizontal at the observer. (3) Ignoring the observer's height when told. (4) Using sin or cos when only the height and distance are involved.

REAL LIFE
Lifeguards, pilots and surveyors measure these angles with an instrument called a clinometer; the math is just the right triangle you drew.

HOT QUESTION
From a window 12 m above the ground the angle of elevation of the top of a tower is 30°, and the angle of depression of its foot is 45°. Find the tower's height.

Mind Map Recap: Heights and Distances

REVISE IN ONE LOOK

Elevation: above the horizontal. Depression: below the horizontal.

tan θ = height / distance. sin uses the line of sight.

Two triangles: share the common height and form two equations, or find lengths separately.

Use √3 = 1.732, √2 = 1.414 or the table.

Always sketch, mark the right angle, and give units.

Practice Set

Try each one on paper first. Answers are at the end of the set.

1. The angle of elevation of the top of a tower from a point 45 m from its foot is 45°. Find the height.
2. A 12 m long ladder reaches a window 6√3 m above the ground. Find the angle the ladder makes with the ground.
3. From a point 30 m from a building the angle of elevation of its top is 30°. Find the height of the building.
4. From the top of a 75 m cliff the angle of depression of a boat is 30°. Find the boat's distance from the cliff foot.
5. The angle of elevation of the top of a pillar is 40° from a point 25 m from the pillar. Find its height (tan 40° = 0.8391).
6. Two points A and B are on a line through the foot of a pole, on the same side, 12 m apart. The elevations of the pole top are 45° and 60° at A and B. Find the pole's height.
7. A man standing on a 15 m high platform sees the top of a building at 30° elevation and its foot at 60° depression. Find the height of the building.
8. A vertical flagpole stands on a building. From a point 30 m from the building the elevations of the bottom and top of the pole are 30° and 45°. Find the length of the flagpole.
9. The shadow of a vertical tower on level ground increases by 10 m when the Sun's elevation falls from 60° to 30°. Find the height of the tower.
10. From a point on a road the angle of elevation of the top of a 20 m pole is 60°. Find the distance from the point to the foot of the pole.
11. A kite string of length 200 m makes 60° with the ground. Find the kite's height.
12. Two buildings are 60 m apart. From the top of the first (height 80 m) the angle of depression of the top of the second is 30°. Find the height of the second.

Answers

1. **45 m**.

2. sin θ = 6√3/12 = √3/2: **60°**.

3. 30/√3 = 10√3 = **17.32 m**.

4. 75√3 = **129.90 m**.

5. 25 × 0.8391 = **20.98 m**.

6. Let B be x m away: h = x√3 and h = x + 12. x(√3 − 1) = 12; x = 16.39; h = **28.39 m**.

7. Distance = 15/√3 = 5√3; rise = 5√3 × 1/√3 = 5; height = 15 + 5 = **20 m**.

8. Top 30 m; bottom 30/√3 = 17.32 m; pole = **12.68 m**.

9. Shadows: h/tan 30° − h/tan 60° = h(√3 − 1/√3) = 2h/√3 = 10, so h = 5√3 = **8.66 m**.

10. 20/√3 = **11.55 m**.

11. 200 sin 60° = 100√3 = **173.21 m**.

12. Drop = 60 tan 30° = 20√3 = 34.64; height = 80 − 34.64 = **45.36 m**.

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CHAPTER · MATHEMATICS

Statistics

Summarise data with the mean (three ways), median and mode, show it with histograms and ogives, and read quartiles and the interquartile range off the graph.



IN THIS CHAPTER

- Mean: three methods
- Median and mode
- Histogram
- Ogive
- Quartiles and IQR

Measures of Central Tendency

A single number that represents a whole data set is a measure of central tendency. Three are in your syllabus.

MEASURE	MEANING	RAW OR ARRAYED DATA
Mean	Sum of all values divided by the number of values	$\bar{x} = \Sigma x / n$
Median	The middle value when data is arranged in order	n odd: the middle value. n even: the average of the two middle values
Mode	The value that occurs **most often**	Highest frequency; a data set may have no mode or more than one

Grouped Data: Three Methods for the Mean

For a frequency table the class mark x is the mid-point of a class: (lower + upper)/2. Pick whichever method makes the arithmetic lightest; all three give the same answer.

METHOD	FORMULA	WHEN IT IS HANDY
Direct	$\bar{x} = \Sigma fx / \Sigma f$	Small numbers
Short-cut (assumed mean A)	$\bar{x} = A + \Sigma fd / \Sigma f$, where $d = x - A$	Large class marks; choose A near the middle
Step-deviation	$\bar{x} = A + (\Sigma ft / \Sigma f) \times i$, where $t = (x - A)/i$	Equal class width i; makes numbers tiny

Median and Mode of Grouped Data

$$\text{Median} = L + \frac{(n/2) - cf}{f} \times h$$

L = lower limit of the **median class** (the class where the cumulative frequency first reaches n/2), cf = cumulative frequency of the class before it, f = frequency of the median class, h = class width.

$$\text{Mode} = L + \frac{f_1 - f_0}{2f_1 - f_0 - f_2} \times h$$

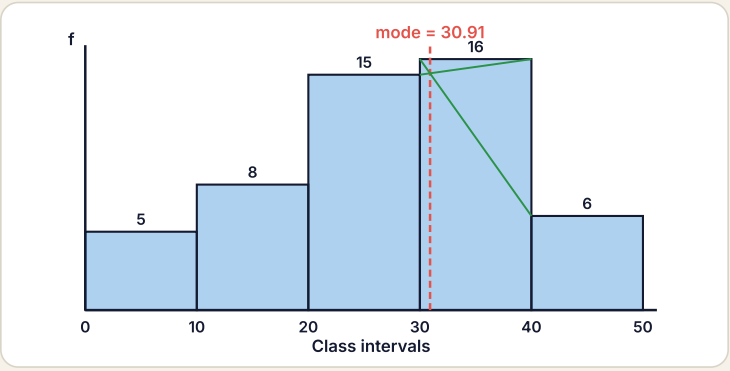
The **modal class** has the highest frequency; f_1 is its frequency, f_0 and f_2 are those of the classes before and after it.



SIDEKICK TIP

For discontinuous classes such as 10–19, 20–29, make the classes continuous first: subtract 0.5 from each lower limit and add 0.5 to each upper limit (9.5–19.5, 19.5–29.5, ...). This matters for median and mode.

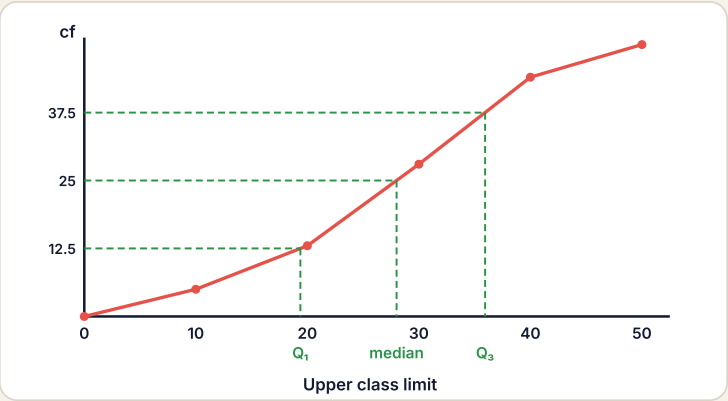
Histograms and Ogives



Histogram: bars touch because classes are continuous. Height = frequency (equal class widths).

Mode from a histogram: in the tallest bar draw a line from its top left corner to the top left corner of the next bar, and from its top right corner to the top right corner of the previous bar, crossing each other. Drop a perpendicular from the crossing point to the x-axis. That value is the mode.

Less than ogive: plot cumulative frequency against the **upper** class limits (add the point (lowest lower limit, 0)) and join the points with a smooth curve.



Less than ogive for the same data. From n/4, n/2 and 3n/4 draw across to the curve, then down to the axis.

Quartiles and the Interquartile Range

- **Lower quartile Q_1**: value at n/4. **Median Q_2**: value at n/2. **Upper quartile Q_3**: value at 3n/4.
- **Interquartile range = $Q_3 - Q_1$**. It measures the spread of the middle half of the data.
- The same interpolation formula as the median works with n/4 or 3n/4 in place of n/2.

BOARD ALERT

Read the ogive carefully. The vertical axis is cumulative frequency, the horizontal axis is the variable. To find a quartile, start at the cumulative frequency (n/4 or 3n/4), go across to the curve, then down.

Worked Example 1

Q. Find the mean, median and mode of 12, 15, 18, 15, 20, 25, 15, 22, 19.

- Sum = 161, n = 9, mean = 161/9 = 17.89.
- Arranged: 12, 15, 15, 15, 18, 19, 20, 22, 25. The 5th value is 18.
- 15 occurs three times.

KEY POINT

Answer: Mean 17.89; median 18; mode 15.

Worked Example 2

Q. Find the mean of the data below by the direct method.
Class: 0–10, 10–20, 20–30, 30–40, 40–50; frequency: 5, 8, 15, 16, 6.

- Class marks 5, 15, 25, 35, 45. fx = 25, 120, 375, 560, 270.
- Σf = 50, Σfx = 1350.
- Mean = 1350/50 = 27.

KEY POINT

Answer: Mean = 27.

Worked Example 3

Q. Marks: 10–20, 20–30, 30–40, 40–50, 50–60 with frequencies 4, 9, 15, 8, 4. Find the mean by the short-cut method.

- Class marks x = 15, 25, 35, 45, 55. Take A = 35, so d = −20, −10, 0, 10, 20.
- fd = −80, −90, 0, 80, 80; Σfd = −10; Σf = 40.
- Mean = 35 + (−10/40) = 34.75.

KEY POINT

Answer: Mean = 34.75.

Worked Example 4

Q. For the same data find the mean by the step-deviation method.

- A = 35, i = 10, t = (x − 35)/10 = −2, −1, 0, 1, 2.
- ft = −8, −9, 0, 8, 8; Σft = −1.
- Mean = 35 + (−1/40) × 10 = 34.75 (same as before).

KEY POINT

Answer: Mean = 34.75.

Worked Example 5

Q. For the data of Example 2 find the median and the mode.

- Cumulative frequencies: 5, 13, 28, 44, 50. n/2 = 25, so the median class is 20–30.
- Median = 20 + (25 − 13)/15 × 10 = 20 + 8 = 28.
- Modal class 30–40 (f = 16): Mode = 30 + (16 − 15)/(32 − 15 − 6) × 10 = 30 + 10/11 = 30.91.

KEY POINT

Answer: Median 28; mode 30.91.

Worked Example 6

Q. Find the median of: 10–19, 20–29, 30–39, 40–49, 50–59 with frequencies 3, 7, 12, 5, 3.

- Make classes continuous: 9.5–19.5, 19.5–29.5, 29.5–39.5, ... (width 10).
- n = 30, n/2 = 15. cf: 3, 10, 22, 27, 30. Median class 29.5–39.5, cf before = 10, f = 12.
- Median = 29.5 + (15 − 10)/12 × 10 = 29.5 + 4.17 = 33.67.

KEY POINT

Answer: Median = 33.67.

Worked Example 7

Q. The mean of the data is 49. Find the missing frequency x.
Class: 0–20, 20–40, 40–60, 60–80, 80–100; f: 6, 8, x, 10, 4.

- Class marks 10, 30, 50, 70, 90. Σfx = 60 + 240 + 50x + 700 + 360 = 1360 + 50x. Σf = 28 + x.
- (1360 + 50x)/(28 + x) = 49, so 1360 + 50x = 1372 + 49x.
- x = 12.

KEY POINT

Answer: x = 12.

Worked Example 8

Q. From the ogive of the Example 2 data, find Q_1, the median, Q_3 and the interquartile range.

- n = 50: n/4 = 12.5, n/2 = 25, 3n/4 = 37.5.
- Q_1 lies in 10–20: 10 + (12.5 − 5)/8 × 10 = 19.375.
- Median = 28.
- Q_3 lies in 30–40: 30 + (37.5 − 28)/16 × 10 = 35.9375.
- IQR = 35.9375 − 19.375 = 16.5625.

KEY POINT

Answer: $Q_1 \approx 19.4$, median 28, $Q_3 \approx 35.9$, IQR ≈ 16.6. Reading from the graph you get values close to these.

BOARD ALERT

Common errors. (1) Plotting the ogive against lower limits. Use upper limits. (2) Using the frequency of the class instead of the cumulative frequency before it (cf). (3) Taking the class mark as the upper limit. (4) Using the mode formula with the wrong neighbouring frequencies. (5) Not making discontinuous classes continuous.

REAL LIFE

Marks sheets, rainfall records and heights of a class all become useful when summarised by a mean, a median and a spread. The IQR tells you whether the middle half of a class is tightly bunched or widely spread.

HOT QUESTION

The mean of 10 numbers is 24. When one number is removed the mean of the remaining numbers is 23. Which number was removed? What happens to the median if the removed number was the largest?

Mind Map Recap: Statistics

REVISE IN ONE LOOK

Mean: direct $\Sigma fx/\Sigma f$; short-cut $A + \Sigma fd/\Sigma f$; step-deviation $A + (\Sigma ft/\Sigma f)i$.

Median $= L + ((n/2 - cf)/f)h$. **Mode** $= L + ((f_1 - f_0)/(2f_1 - f_0 - f_2))h$.

Histogram: bars for continuous classes; mode by the criss-cross method. Ogive: cf against upper limits.

Q_1 at $n/4$, Q_2 at $n/2$, Q_3 at $3n/4$; IQR $= Q_3 - Q_1$.

Discontinuous classes: make them continuous (±0.5).

Practice Set

Try each one on paper first. Answers are at the end of the set.

1. Find the mean, median and mode of 8, 11, 14, 11, 17, 20, 11, 9.
2. Find the mean of 4, 10, 18, 12, 6 as frequencies for classes 0–20, 20–40, 40–60, 60–80, 80–100, by the direct method.
3. For the data in Q2 find the median.
4. For the data in Q2 find the mode.
5. Classes 5–15, 15–25, 25–35, 35–45, 45–55 with frequencies 7, 11, 16, 9, 7. Find the mean by the step-deviation method (take A = 30).
6. For the data in Q5 find the median.
7. For the data in Q5 find the modal value.
8. The mean of the data is 34. Find p: classes 10–20, 20–30, 30–40, 40–50, 50–60 with frequencies 4, 7, p, 6, 3.
9. Make 20–29, 30–39, 40–49, 50–59 with frequencies 6, 11, 9, 4 continuous and find the median.
10. For the data in Q2 find Q_1, Q_3 and the interquartile range.
11. The mean of 5 numbers is 18. When a sixth number is added the mean becomes 20. Find the sixth number.
12. Draw (describe) a less than ogive for Q5's data: give the cumulative frequencies and the points to plot.

Answers

1. Sum 101, n = 8: mean **12.625**; arranged 8, 9, 11, 11, 11, 14, 17, 20: median (11 + 11)/2 = **11**; mode **11**.

2. n = 50; mids 10, 30, 50, 70, 90; $\Sigma fx = 2620$; mean = **52.40**.

3. cf: 4, 14, 32, 44, 50; n/2 = 25 lies in 40–60: $40 + (25 - 14)/18 \times 20 =$ **52.22**.

4. Modal class 40–60: $40 + (18 - 10)/(36 - 10 - 12) \times 20 = 40 + 8/14 \times 20 =$ **51.43**.

5. $t = (x - 30)/10 = -2, -1, 0, 1, 2$: $\Sigma ft = -14 - 11 + 0 + 9 + 14 = -2$; mean $= 30 + (-2/50) \times 10 =$ **29.60**.

6. n = 50, cf: 7, 18, 34, 43, 50. n/2 = 25 in 25–35: $25 + (25 - 18)/16 \times 10 =$ **29.38**.

7. Modal class 25–35: $25 + (16 - 11)/(32 - 11 - 9) \times 10 =$ **29.17**.

8. **p = 10**.

9. n = 30, n/2 = 15, cf: 6, 17, 26, 30: class 29.5–39.5: $29.5 + (15 - 6)/11 \times 10 =$ **37.68**.

10. Q_1 (12.5) in 20–40: $20 + (12.5 - 4)/10 \times 20 =$ **37**; Q_3 (37.5) in 40–60: $40 + (37.5 - 14)/18 \times 20 =$ **69.17**; IQR = **32.17**.

11. Total of six = 120, total of five = 90: sixth number = **30**.

12. Plot (5, 0), (15, 7), (25, 18), (35, 34), (45, 43), (55, 50), joined by a smooth curve: upper limits against cf **7, 18, 34, 43, 50**.

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CHAPTER · MATHEMATICS

Probability

How likely is it? Count the favourable outcomes, divide by the total, and use sample spaces for coins, dice, cards and bags.



IN THIS CHAPTER

- Sample space
- P(E)
- Complementary events
- Coins, dice and cards

Random Experiments and Sample Space

A **random experiment** has several possible outcomes and you cannot be sure which one will occur. The set of all possible outcomes is the **sample space S**. An **event** is a part of the sample space.

EXPERIMENT	SAMPLE SPACE	NUMBER OF OUTCOMES
Toss one coin	{H, T}	2
Toss two coins	{HH, HT, TH, TT}	4
Roll one die	{1, 2, 3, 4, 5, 6}	6
Roll two dice	36 ordered pairs, (1,1) to (6,6)	36
Draw a card from a pack	52 cards (4 suits × 13)	52

Probability of an Event

$$P(E) = \frac{\text{number of outcomes favourable to E}}{\text{total number of possible outcomes}}$$

- $0 \le P(E) \le 1$.
- **Impossible event:** P = 0. **Sure (certain) event:** P = 1.
- **Complementary event** (not E): P(not E) = 1 − P(E).
- The probabilities of all the outcomes of an experiment add up to 1.
- Outcomes should be **equally likely**: a fair coin, an unbiased die, a well-shuffled pack.



KEY POINT

Only **single events** are in your syllabus: one coin, one die, one card or one draw from a bag, or two coins/dice thrown together, but not combined events with "and/or" rules.

A Pack of 52 Cards

ITEM	COUNT
Suits: spades and clubs (black), hearts and diamonds (red)	13 each, 26 red, 26 black
Face cards (J, Q, K)	12 (3 per suit)
Kings / Queens / Jacks / Aces	4 of each
Court cards (J, Q, K and Ace) are sometimes used; read the question	—



SIDEKICK TIP

Write the sample size first. Then count the favourable cases by listing them. Only then write the fraction and simplify it.

Worked Example 1

Q. A fair die is rolled once. Find the probability of getting (i) a prime number (ii) a number greater than 4 (iii) a multiple of 7.

- Total = 6. Primes: 2, 3, 5, so P = 3/6 = 1/2.
- Greater than 4: 5, 6, so P = 2/6 = 1/3.
- No multiple of 7 on a die: P = 0.



KEY POINT

Answer: 1/2, 1/3 and 0.

Worked Example 2

Q. Two coins are tossed together. Find P(two heads), P(exactly one head) and P(at least one head).

- Sample space: HH, HT, TH, TT (4).
- P(HH) = 1/4. Exactly one head: HT, TH, so 2/4 = 1/2.
- At least one head: HH, HT, TH, so 3/4.

KEY POINT
Answer: 1/4, 1/2 and 3/4.

Worked Example 3

Q. Two dice are thrown together. Find the probability that (i) the sum is 8 (ii) the sum is at least 10 (iii) both show the same number.

- Total outcomes = 36.
- (i) Sum 8: (2,6), (3,5), (4,4), (5,3), (6,2) = 5, so 5/36.
- (ii) Sum 10, 11, 12: 3 + 2 + 1 = 6, so 6/36 = 1/6.
- (iii) Doublets: 6, so 6/36 = 1/6.

KEY POINT
Answer: 5/36, 1/6 and 1/6.

Worked Example 4

Q. A card is drawn from a well-shuffled pack of 52 cards. Find the probability of getting (i) a king (ii) a red face card (iii) a black card that is not an ace.

- (i) 4 kings: 4/52 = 1/13.
- (ii) Red face cards: 3 in hearts + 3 in diamonds = 6, so 6/52 = 3/26.
- (iii) 26 black cards minus 2 black aces = 24, so 24/52 = 6/13.

KEY POINT
Answer: 1/13, 3/26 and 6/13.

Worked Example 5

Q. A bag has 5 red, 8 white and 7 green balls. One ball is drawn at random. Find the probability that it is (i) white (ii) not green (iii) red or green.

- Total = 20.
- (i) 8/20 = 2/5.
- (ii) Not green = 1 − 7/20 = 13/20 (or 5 + 8 = 13 balls).
- (iii) Red or green = 5 + 7 = 12, so 12/20 = 3/5.

KEY POINT
Answer: 2/5, 13/20 and 3/5.

Worked Example 6

Q. A letter is chosen at random from the word MATHEMATICS. Find the probability that it is (i) a vowel (ii) the letter M (iii) not T.

- The word has 11 letters: M, A, T, H, E, M, A, T, I, C, S.
- (i) Vowels A, E, A, I = 4: 4/11.
- (ii) M occurs twice: 2/11.
- (iii) T occurs twice, so not T = 9/11.

KEY POINT
Answer: 4/11, 2/11 and 9/11.

Worked Example 7

Q. A bag contains 6 red balls and x blue balls. If P(blue) = 3/5, find x. What is P(red)?

- P(blue) = x/(6 + x) = 3/5, so 5x = 18 + 3x.
- x = 9. Then P(red) = 6/15 = 2/5 (or 1 − 3/5).

KEY POINT
Answer: x = 9; P(red) = 2/5.

Worked Example 8

Q. Cards numbered 1 to 25 are shuffled and one is drawn. Find P(multiple of 4) and P(a number divisible by both 2 and 3).

- Multiples of 4: 4, 8, 12, 16, 20, 24 = 6, so 6/25.
- Divisible by 6: 6, 12, 18, 24 = 4, so 4/25.

KEY POINT
Answer: 6/25 and 4/25.

Worked Example 9

Q. In a class of 40 students, 18 are girls. A student is chosen at random to be the monitor. Find the probability that the monitor is a boy. If 2 more girls join, what is the new P(girl)?

- Boys = 22, P(boy) = 22/40 = 11/20.
- Now girls = 20 and total = 42: P(girl) = 20/42 = 10/21.

KEY POINT
Answer: 11/20; then 10/21.

BOARD ALERT
Common errors. (1) Counting (1, 2) and (2, 1) as one outcome with two dice: they are different. (2) Forgetting that the total changes after adding or removing items. (3) Writing probability as a ratio such as 3 : 5 or a value above 1. (4) Not simplifying the fraction.

REAL LIFE
A weather forecast of "30% chance of rain", the toss at the start of a cricket match and the way insurance premiums are set all rest on probability.

HOT QUESTION
A bag has 3 red and 5 blue balls. How many red balls must be added so that P(red) becomes 1/2? Then explain why the new answer for P(blue) is not simply the old value.

Mind Map Recap: Probability

REVISE IN ONE LOOK

P(E) = favourable / total, with equally likely outcomes.

0 ≤ P ≤ 1; sure event 1; impossible event 0; P(not E) = 1 − P(E).

Two coins: 4 outcomes. Two dice: 36. Cards: 52 (26 red, 26 black, 12 face cards).

List favourable outcomes, then write the fraction in simplest form.

Adding or removing items changes the total as well.

Practice Set

Try each one on paper first. Answers are at the end of the set.

1. A die is rolled. Find the probability of getting an even number greater than 2.
2. Three coins are tossed together. Find the probability of getting exactly two heads.
3. Two dice are thrown. Find the probability that the sum is 7.

4. Two dice are thrown. Find the probability that the product of the numbers is 12.
5. A card is drawn from a pack of 52. Find the probability that it is a heart that is also a face card.
6. A card is drawn from a pack. Find P(not a spade).
7. A bag has 4 red, 6 blue and 5 yellow balls. Find P(blue) and P(not yellow).
8. A bag has 9 white balls and x black balls. If P(black) = 2/5, find x.
9. A number is chosen from 1 to 40. Find P(a prime number).
10. A letter is chosen at random from the word PROBABILITY. Find P(letter B).
11. In a lottery there are 8 prizes and 32 blanks. Find the probability of winning a prize with one ticket.
12. A bag has 5 red and 3 green balls. How many red balls must be added so that P(red) = 3/4?

Answers

1. Even numbers greater than 2: 4, 6: **1/3**.

2. Outcomes 8; exactly two heads: HHT, HTH, THH = 3: **3/8**.

3. Sum 7: 6 ways: **1/6**.

4. Pairs (2,6), (3,4), (4,3), (6,2) = 4: **1/9**.

5. Hearts that are face cards: 3: **3/52**.

6. 39 non-spades: **3/4**.

7. Total 15: P(blue) = **2/5**; not yellow = 10/15 = **2/3**.

8. x/(9 + x) = 2/5: 5x = 18 + 2x: **x = 6**.

9. Primes up to 40: 2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37 (12): **3/10**.

10. PROBABILITY has 11 letters, B occurs twice: **2/11**.

11. 1/5.

12. (5 + x)/(8 + x) = 3/4: 20 + 4x = 24 + 3x: **x = 4**.

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CHAPTER · MATHEMATICS

Internal Assessment: Assignment Guide

Twenty marks come from assignments marked by your teacher and an external examiner. See how the marks are awarded and start with ready-made project ideas, each with a worked head start.



IN THIS CHAPTER

- Marking criteria
- Layout of an assignment
- Project ideas
- Worked head starts

How the 20 Marks Work

Internal Assessment carries **20 marks**, alongside the 80-mark written paper. You complete **at least two assignments** set or approved by your teacher. Your subject teacher (internal examiner) awards 10 marks and an external examiner awards 10 marks, each assessing independently.

CRITERION	WHAT IT LOOKS FOR	MARKS
Preparation	A well-defined problem, an approach that suits it, proper techniques	4
Concepts	Correct use of mathematical ideas and methods, and a good range of techniques	4
Computation	Careful and accurate work with correct constructions, measurements and units	4
Presentation	Clear conclusions, correct mathematical language, tidy tables, diagrams and graphs	4
Understanding	Your own contribution, and being able to explain or apply it in new situations	4



KEY POINT

Each criterion is graded into levels (the top level for careful, original and well-presented work; lower levels for routine or limited work). The five criteria total 20 marks. Teachers combine them; do not worry about the arithmetic, worry about the quality.

A Five-Part Layout That Works

1. Question and aim
State in one or two lines what you are trying to find out.

2. Method and data
Say how you collected or generated data. Use a table with units.

3. Working
Show formulae, calculations and at least one clean graph or diagram.

4. Conclusion
Answer the question, and say how sure you are.

5. Reflection
What would you do differently? Where else does the idea apply?

Project Ideas from the Syllabus, with a Head Start

1. Which rectangles have area equal to perimeter?

With whole-number sides l and w, $l \times w = 2(l + w)$ rearranges to $(l - 2)(w - 2) = 4$. The pairs of factors of 4 give $l - 2$ and $w - 2$ as (1, 4), (2, 2), so the rectangles are **3 × 6** (area = perimeter = 18) and **4 × 4** (area = perimeter = 16). Check: $3 \times 6 = 18$ and $2(3 + 6) = 18$; $4 \times 4 = 16$ and $2(4 + 4) = 16$.

2. The Tower of Hanoi

With n discs the least number of moves is $\mathbf{2^n - 1}$. For n = 1, 2, 3, 4, 5 that is 1, 3, 7, 15, 31. Tabulate your own counts and look for the pattern: each new disc doubles the old number and adds 1. This is a geometric-style progression, so link it to the AP and GP chapter.

3. Swapping red and blue marbles

With n marbles of each colour the number of moves needed is $\mathbf{n^2 + 2n}$: 3, 8, 15, 24, 35 for n = 1 to 5. Test the rule yourself with coins on paper.

4. Biggest open box from a square sheet

From a 10 cm square, cut a square of side x from each corner and fold up the sides. Volume = $x(10 - 2x)^2$. Try x = 1, 1.5, 1.6, 1.7, 2 and plot a graph of V against x. The maximum is about **74.07 cm³ when x ≈ 1.67 cm** (that is, 10/6). Compare your table with this.

5. Find the centre of a circle with only a 60–30 set-square

An angle in a semicircle is 90°. Place the set-square's right-angle corner on the circle and mark where its two edges cut the circle. Those two points are the ends of a **diameter**. Repeat at another position for a second diameter. The two diameters meet at the centre.

6. A cryptarithm

In TO + GO = OUT each letter is a different digit. The solution is T = 2, O = 1, G = 8, U = 0 because 21 + 81 = 102. Create your own and test that it has a single solution.

7. Survey projects

Mode of travel to school, favourite sports, prices of essential goods: collect at least 30 responses, make a frequency table, draw a bar graph or histogram and find the mode. **Bank accounts:** compare interest rates of savings and recurring deposit accounts from three banks. **Shares:** follow two companies for a week from a newspaper and record the market value and dividend. **Home budget:** list monthly income and expenses, draw a pie chart and compute percentages.

8. Ink-drop and marble-volume experiments

Ink drop: release a drop from heights of 20, 30, 40, 50 cm onto plain paper, measure the diameter of the splash and plot height against diameter. **Marbles in a box:** fill a box with same-size marbles, then pour in water to measure the empty space. Do this for four marble sizes and compare the fraction of space left.



SIDEKICK TIP

Examiners reward your **own** idea. Change one number or condition in a suggested project (a different sheet size, different heights, a different survey question) and explain what changes. That shows understanding.



BOARD ALERT

Common errors. (1) Copying a classmate's data or write-up. (2) Presenting results without units or without a title on a graph. (3) A conclusion that does not answer the original question. (4) Leaving everything to the last week.



HOT QUESTION

Choose one project above and extend it: for the open box, try a rectangle 12 cm by 8 cm. What is the best cut-out size? Describe how you would find it and why a graph helps.

Checklist for a Full-Marks Assignment

REVISE IN ONE LOOK

A clear aim and a tidy data table with units.

At least one graph or diagram, labelled, drawn to scale.

Correct formulae, neat calculations and a stated conclusion.

Your own twist and a short reflection.

Neat handwriting, page numbers and a cover page with your name and class.

Done reading?
Now go break the board.



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